Multivariate Statistics



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These lecture notes are based on the *Multivariate Statistics* course given at EPFL for the academic year 2024/2025 by Prof. Victor Panaretos. In these notes, any Proposition/Lemma/Example/Proof/... with a $(*)$ are taken from the exercise sheets. The figures are from the course slides.

Chapter 1

Linear Algebra Recap

In this section, we aim to recapitulate some concepts and results from Linear Algebra that will be essential for the remainder of this course. We set Q to be a $n \times p$ real matrix.

1.1 Subspaces, Partitions, Spectra and Projections

Definition 1.1.1. *The **range** (or column space) of Q is the set spanned by its columns:*

$$\mathcal{R}(Q) := \{Q\beta : \beta \in \mathbb{R}^p\} \subseteq \mathbb{R}^n. \tag{1.1}$$

Definition 1.1.2. *The **kernel** (or null space) of Q is the subspace defined as:*

$$\mathcal{N}(Q) := \{x \in \mathbb{R}^p : Qx = 0\}. \tag{1.2}$$

Definition 1.1.3. *The **orthogonal complement** of $\mathcal{R}(Q)$ is the subspace defined as:*

$$\mathcal{R}^{\perp}(Q) := \{y \in \mathbb{R}^n : y^T Qx = 0, \forall x \in \mathbb{R}^p\} = \{y \in \mathbb{R}^n : y^T v = 0, \forall v \in \mathcal{R}(Q)\}. \tag{1.3}$$

The orthogonal complement may be defined for arbitrary subspaces by using the second inequality.

Theorem 1.1.4 (Singular Value Decomposition - SVD)**.** *Any $n \times p$ real matrix Q may be factorized as $Q_{n\times p} = U_{n\times n}\Sigma_{n\times p}V^T_{p\times p}$, where U and V^T are orthogonal with columns called respectively the **left/right singular vectors**, and Σ is diagonal with non-negative real entries called **singular values**.*

We have some interesting consequences. For instance (i) the left singular vectors corresponding to non-zero singular values form an orthonormal basis for $\mathcal{R}(Q)$, (ii) the left singular vectors corresponding to zero singular values form an orthonormal basis of $\mathcal{R}^{\perp}(Q)$, and (iii) writing $\{u_i\}_{i=1}^n$ for the left singular vectors and $\{v_j\}_{j=1}^n$ for the right singular vectors, the SVD can also be expressed as $Q_{n\times p} = \sum_{j=1}^{\text{rank}(Q)} \sigma_j u_j v_j^T$.

Proof. Since the statement is invariant to transposition, assume wlog that $n \geq p$. We will prove the statement by induction on p. Assume that $p = 1$ so that Q is a column vector. Then the statement holds true trivially by taking simply $V^T = V = \mathbb{I}$, $\Sigma = \left(\|Q\|, 0_{1\times(n-1)}\right)^T$, $U = (u_1 ... u_n)$, where $u_1 = Q/\|Q\|$ and $(u_2, ..., u_n)$ an orthonormal basis for $\text{span}^{\perp}(u_1)$. Thus the statement is true for all $n \geq p$ when $p = 1$, consisting of the base case for the induction. For the inductive step, assume that the statement is true for some $p > 1$ and all $n \geq p$. Let us prove that it is also true for $p+1$ and all $n \geq p+1$. Let $\mathbb{S}^{p+1} = \{x \in \mathbb{R}^{p+1} :$

$\|x\| = 1\}$ and $q(x) = \|Qx\|$. Since $q(\cdot)$ is continuous and $\mathbb{S}^{p+1}$ is compact, we have that $q(x)$ is bounded over $\mathbb{S}^{p+1}$ and attains its bounds. So there exists $v_1 \in \mathbb{S}^{p+1}$ such that $q(v_1) = \max_{x\in\mathbb{S}^{p+1}} q(x) = \sigma_1 < \infty$, and let $v_1 \in \mathbb{S}^{p+1}$ be a maximiser of $q(x)$, i.e. such that $q(v_1) = \max_{x\in\mathbb{S}^{p+1}} q(x)$. Now, define $u_1 = \sigma_1^{-1} Q v_1$ so that $\|u_1\| = 1$. Given any orthonormal basis $\{u_j\}_{j=2}^n$ for $\text{span}^{\perp}(u_1)$ and $\{v_j\}_{j=2}^p$ for $\text{span}^{\perp}(v_1)$ define U and V to be orthogonal matrices $U = (u_1 \;\; u_2 ... u_n) = (u_1 \;\; U_1)$ and $V = (v_1 \;\; v_2 ... v_n) = (v_1 \;\; V_1)$. Using block matrix multiplication, we see that:

$$U^T_{n\times n} Q_{n\times(p+1)} V_{(p+1)\times(p+1)} = \begin{pmatrix} u_1^T \\ U_1^T \end{pmatrix} Q \begin{pmatrix} v_1 & V_1 \end{pmatrix} = \begin{pmatrix} u_1^T Q v_1 & u_1^T Q V_1 \\ U_1^T Q v_1 & U_1^T Q V_1 \end{pmatrix} = \begin{pmatrix} \sigma_1 & \theta^T_{1\times p} \\ 0_{(n-1)\times 1} & Z_{(n-1)\times p} \end{pmatrix}.$$

Now we claim that $\theta = 0$, and to see this we first observe that,

$$\sigma_1 = \max_{x\in\mathbb{S}^{p+1}} \|Qx\| = \max_{x\in\mathbb{S}^{p+1}} \|U^T Q x\| = \max_{x\in\mathbb{S}^{p+1}} \|U^T Q V x\|. \tag{1.4}$$

Now, let us consider the following norm:

$$\left\| U^T Q V \begin{pmatrix} \sigma_1 \\ \theta \end{pmatrix} \right\| = \left\| \begin{pmatrix} \sigma_1 & \theta^T \\ 0 & Z \end{pmatrix} \begin{pmatrix} \sigma_1 \\ \theta \end{pmatrix} \right\| = \left\| \begin{pmatrix} \sigma_1^2 + \theta^T\theta \\ Z\theta \end{pmatrix} \right\| = \sqrt{(\sigma_1^2 + \theta^T\theta)^2 + \|Z\theta\|^2} \geq \sigma_1^2 + \theta^T\theta$$
$$= (\sigma_1^2 + \theta^T\theta)^{\frac{1}{2}} \left\| \begin{pmatrix} \sigma_1 \\ \theta \end{pmatrix} \right\|.$$

Dividing across by $\|(\sigma_1 \;\; \theta)^T\|$, we see that we must necessarily have,

$$(\sigma_1^2 + \theta^T\theta)^{\frac{1}{2}} \leq \max_{x\in\mathbb{S}^{p+1}} \|U^T Q V x\| = \sigma_1, \tag{1.5}$$

and so it must be that $\theta^T\theta = 0$. We then conclude that,

$$U^T Q V = \begin{pmatrix} \sigma_1 & 0_{1\times p} \\ 0_{(n-1)\times 1} & Z \end{pmatrix} \Longrightarrow Q = U \begin{pmatrix} \sigma_1 & 0_{1\times p} \\ 0_{(n-1)\times 1} & Z \end{pmatrix} V^T. \tag{1.6}$$

However, Z is an $(n-1)\times p$ matrix, and since $n \geq p+1$ it holds that $n-1 \geq p$. So by our inductive hypothesis, $Z_{(n-1)\times p} = W_{(n-1)\times(n-1)} \Omega_{(n-1)\times p} R^T_{p\times p}$, where W,R are orthogonal and Ω is diagonal. Thus:

$$\begin{aligned} Q_{n\times p} &= U_{n\times n} \begin{pmatrix} \sigma_1 & 0_{1\times p} \\ 0_{(n-1)\times 1} & W\Omega R^T \end{pmatrix} V^T_{p\times p} \\ &= \underbrace{U \begin{pmatrix} 1 & 0_{1\times(n-1)} \\ 0_{(n-1)\times 1} & W_{(n-1)\times(n-1)} \end{pmatrix}}_{\text{orthogonal}} \underbrace{\begin{pmatrix} \sigma_1 & 0_{1\times p} \\ 0_{(n-1)\times 1} & \Omega_{(n-1)\times p} \end{pmatrix}}_{\text{diagonal}} \underbrace{\begin{pmatrix} 1 & 0_{1\times p} \\ 0_{p\times 1} & R^T_{p\times p} \end{pmatrix} V^T}_{\text{orthogonal}}. \end{aligned}$$

□

Theorem 1.1.5 (Spectral Theorem)**.** *A $p\times p$ matrix A is symmetric if and only if there exists a $p\times p$ orthogonal matrix U and a real diagonal matrix Λ such that $A = U\Lambda U^T$.*

In particular, (i) the orthonormal columns of $U = (u_1 \;\; ... \;\; u_p)$ are **eigenvectors** of A, i.e. $Au_j = \lambda_j u_j$ and where $\Lambda = \text{diag}(\lambda_1, ..., \lambda_p)$ is the matrix of (real) **eigenvalues** of A. Furthermore, (ii) the rank of A is the number of non-zero eigenvalues, (iii) if the eigenvalues are distinct then the eigenvectors are unique (up to re-ordering and sign flips), and (iv) the spectral representation may also be expressed as,

$$A_{p\times p} = \sum_{j=1}^{\text{rank(A)}} \lambda_j u_j u_j^T. \tag{1.7}$$

Proof. If $A = 0$ the statement holds trivially, thus consider $A = A^T \neq 0$. First note that the SVD of A guarantees the existence of a singular vector pair (u, v) with non-zero singular value σ so that,

$$A(v+u) = Av + Au = Av + A^T u = \sigma u + \sigma v = \sigma(u+v). \tag{1.8}$$

As a result, $w = (u+v)/\|u+v\|$ is a unit eigenvector of A with real eigenvalue σ. Now, the theorem is obviously true for 1×1 matrices (i.e. scalars). So we can use induction. Assume any non-zero $p \times p$ symmetric matrix satisfies the statement. Let $A = A^T \neq 0$ be $(p+1) \times (p+1)$. By (1.8), A has at least one unit eigenvector $w \in \mathbb{R}^p$ with real eigenvalue $\sigma \neq 0$. Let also $W = (w \;\; R)$ where R has p orthonormal columns spanning $\text{span}^{\perp}(w)$. Then:

$$W^T A W = \begin{pmatrix} w^T \\ R^T \end{pmatrix} A \begin{pmatrix} w & R \end{pmatrix} = \begin{pmatrix} w^T Aw & w^T AR \\ R^T Aw & R^T AR \end{pmatrix} = \begin{pmatrix} \sigma & (Aw)^T R \\ R^T Aw & R^T AR \end{pmatrix} = \begin{pmatrix} \sigma & 0_{1\times p} \\ 0_{p\times 1} & R^T AR \end{pmatrix}$$
$$= \begin{pmatrix} \sigma & 0_{1\times p} \\ 0_{p\times 1} & B \end{pmatrix},$$

where $B = R^T AR$ is a symmetric $p \times p$ matrix. Now, since B is symmetric, we have that $B = V \Omega V^T$ for $V_{p\times p}$ orthogonal and $\Omega_{p\times p}$ diagonal by our induction hypothesis. Thus in summary:

$$A = W \begin{pmatrix} \sigma & 0_{1\times p} \\ 0_{p\times 1} & B \end{pmatrix} W^T = \underbrace{W \begin{pmatrix} 1 & 0_{1\times p} \\ 0_{p\times 1} & V_{p\times p} \end{pmatrix}}_{\text{orthogonal}} \underbrace{\begin{pmatrix} \sigma & 0_{1\times p} \\ 0_{p\times 1} & \Omega_{p\times p} \end{pmatrix}}_{\text{diagonal}} \underbrace{\begin{pmatrix} 1 & 0_{1\times p} \\ 0_{p\times 1} & V^T_{p\times p} \end{pmatrix} W^T}_{\text{orthogonal}} \equiv U \Lambda U^T.$$

□

Combining the SVD and the spectral theorem, we remark three properties: (i) the left singular vectors of Q are eigenvectors of $A = QQ^T$, (ii) the right singular vectors of Q are eigenvectors of $A = Q^T Q$, and (iii) the squared singular values of Q are eigenvalues of both QQ^T and $Q^T Q$.

Definition 1.1.6. *A matrix Q is said to be **idempotent** if $Q^2 = Q$.*

Definition 1.1.7. *An **orthogonal projection** (and thus a projection) onto a subspace $\mathcal{V}$ is a symmetric idempotent matrix H such that $\mathcal{R}(H) = \mathcal{V}$.*

Proposition 1.1.8. *The only possible eigenvalues of a projection matrix are 0 and 1.*

Proof. $(*)$ Any eigenvector v must satisfy $Qv = \lambda v$ for some $\lambda \in \mathbb{R}$, and since any projection is idempotent we have that:

$$\lambda^2 v = Q^2 v = Qv = \lambda v \Longrightarrow \lambda(\lambda - 1)v = 0. \tag{1.9}$$

Since $v \neq 0$ it must be that $\lambda = 0$ or $\lambda = 1$. □

Proposition 1.1.9. *Let $\mathcal{V}$ be a subspace and H be a projection onto $\mathcal{V}$. Then $\mathbb{I} - H$ is the projection matrix onto $\mathcal{V}^{\perp}$.*

Proof. First note that $(\mathbb{I}-H)^T = \mathbb{I}-H^T = \mathbb{I}-H$ since H is symmetric, and $(\mathbb{I}-H)^2 = \mathbb{I}-2H+H^2 = \mathbb{I}-H$. Thus, $\mathbb{I}-H$ is a projection matrix. It remains to identify the column space of $\mathbb{I}-H$. Let $H = U\Lambda U^T$ be the spectral decomposition of H. Then, $\mathbb{I}-H = UU^T - U\Lambda U^T = U(\mathbb{I}-\Lambda)U^T$. Hence, the column space of $\mathbb{I}-H$ is spanned by the eigenvectors of H corresponding to the zero eigenvalues of H, which coincides with $\mathcal{R}^{\perp}(H) = \mathcal{V}^{\perp}$. □

Proposition 1.1.10. *Let $\mathcal{V}$ be a subspace and H be a projection onto $\mathcal{V}$. Then $Hy = y \iff y \in \mathcal{V}$.*

Proof. If $y \in \mathcal{V} \equiv \mathcal{R}(H)$, then $y = Hu$ for some u and so $Hy = HHu = Hu = y$. Conversely, if $y = Hy$ then $y \in \mathcal{R}(H) \equiv \mathcal{V}$ by default (being of the form Hu for $u = y$). □

Proposition 1.1.11. *If P and Q are projection matrices onto a subspace $\mathcal{V}$, then $P = Q$.*

Proof. $(*)$ • First note that any vector v may be decomposed uniquely as $v = u + w$ where $u \in \mathcal{R}(Q)$ and $w \in \ker(Q)$. Indeed, $u = Qv$ is the projection of v onto the subspace defined by the range of Q, while $w = (\mathbb{I}-Q)v$ is the projection of v onto the orthogonal complement of the range of Q corresponding to its null space. Now, suppose there exists another pair of vectors u' and w' such that $v = u' + w'$ with $u' = Qv$ and $w' = (\mathbb{I}-Q)v$. Then, $Q(u'-u) = 0$ implying that $u'-u \in \ker(Q)$ while $u, u' \in \mathcal{R}(Q)$. As a result it must be that $u = u'$, and in a similar way $w = w'$.
• Thus, if we decompose u as $v + w$ where $v \in \mathcal{V}$ and $w \in \mathcal{V}^{\perp}$ then since P and Q are projections onto $\mathcal{V}$ we have that $Qv = Pv = v$ while $Qw = Pw = 0$, and we are done. □

Proposition 1.1.12. *Let $\mathcal{V}$ be a subspace of $\mathbb{R}^n$ and H a projection onto $\mathcal{V}$. Then $\|x - Hx\| \le \|x - v\|$ for all $v \in \mathcal{V}$.*

Proof. We use the Pythagorean Theorem and Proposition (1.1.9) to find:

$$\|x-v\|^2 = \|x-Hx+Hx-v\|^2 = \|x-Hx\|^2 + \|Hx-v\|^2 = \underbrace{\|(\mathbb{I}-H)x\|^2}_{\in\mathcal{V}^{\perp}} + \underbrace{\|Hx-v\|^2}_{\in\mathcal{V}} \ge \|x-Hx\|^2.$$

□

1.2 The Pseudoinverse and Positive Definite Matrices

In this section we define the Moore-Penrose pseudoinverse matrix, and present some properties of positive definite matrices.

Definition 1.2.1. *Let Q be an $n \times p$ matrix with SVD,*

$$Q_{n\times p} = U_{n\times n}\Sigma_{n\times p}V^T_{p\times p} = U\begin{pmatrix}\Omega_{r\times r} & 0_{p-r} \\ 0_{n-r} & 0_{(n-r)\times(p-r)}\end{pmatrix}V^T, \tag{1.10}$$

*where we assume wlog that $n \ge p$ so that $r := rank(Q) \le p$ and Ω is diagonal with non-zero entries. The **pseudoinverse** of Q is the $p \times n$ matrix $Q^{\dagger}$ defined as:*

$$Q^\dagger := \left(U \begin{pmatrix} \Omega_{r\times r}^{-1} & 0_{p-r} \\ 0_{n-r} & 0_{(n-r)\times(p-r)} \end{pmatrix} V^T \right)^T = V \begin{pmatrix} \Omega_{r\times r}^{-1} & 0_{n-r} \\ 0_{p-r} & 0_{(p-r)\times(n-r)} \end{pmatrix} U^T \tag{1.11}$$

.

Intuitively, $Q^\dagger$ acts as an inverse of Q on $\mathcal{R}(Q) \subset \mathbb{R}^n$. Its action on $\mathbb{R}^n$ is to first project onto $\mathcal{R}(Q)$ and then acts as the inverse of Q on that range. Furthermore note that when Q is symmetric then so does $Q^\dagger$ and the expressions become:

$$Q = U \begin{pmatrix} \Omega_{r\times r} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} U^T \quad \text{and,} \quad Q^\dagger = U \begin{pmatrix} \Omega_{r\times r}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} U^T. \tag{1.12}$$

Furthermore, the pseudoinverse is the **unique** matrix satisfying (i) $Q^\dagger Q$ is the projection onto $\mathcal{R}(Q^T)$ and (ii) $QQ^\dagger$ is the projection onto $\mathcal{R}(Q)$; so that when Q is symmetric $QQ^\dagger = Q^\dagger Q$ by uniqueness of projections. Note that, as immediate corollaries, we have $QQ^\dagger Q = Q$ and $Q^\dagger QQ^\dagger = Q^\dagger$.

Definition 1.2.2 (Non-negative matrix - Quadratic form definition)**.** *A $p \times p$ real symmetric matrix Ω is called **non-negative definite**, written as $\Omega \succeq 0$, if and only if $x^T \Omega x \geq 0$ for all $x \in \mathbb{R}^p$. If $x^T \Omega x > 0$ for all $x \in \mathbb{R}^p \setminus \{0\}$ then $\Omega \succ 0$ is **positive definite**.*

An equivalent definition is given by the Spectral Theorem.

Definition 1.2.3 (Non-negative matrix - Spectral definition)**.** *A $p \times p$ real symmetric matrix Ω is called **non-negative definite**, written as $\Omega \succeq 0$, if and only if the eigenvalues of Ω are non-negative. If the eigenvalues of Ω are strictly positive then $\Omega \succ 0$ is **positive definite**.*

We now list a series of properties all related to the non-negative definite property of some matrix Q.

Proposition 1.2.4. *$Q \succeq 0$ if and only if $Q = \sum_{j=1}^m q_j q_j^T$ for some vectors $\{q_j\}_j \in \mathbb{R}^p$ with $m \leq p$.*

Proof. $(*)$ $(\Longrightarrow)$ Follows trivially from the Spectral Theorem.

$(\Longleftarrow)$ Consider $Q = \sum_{i=1}^m q_i q_i^T$ for some vectors $q_1, ..., q_m$. Since the sum of positive semi-definite matrices is also positive semi-definite, then it suffices to show that the matrix $q_1 q_1^T$ is positive semi-definite. Clearly it is symmetric. Furthermore, for any vector v one has $v^T(q_1 q_1^T)v = (v^T q_1)(v^T q_1)^T = (v^T q_1)^2 \geq 0$ and thus proving the claim. □

Proposition 1.2.5. *For $P, Q \succeq 0$ we have the following two implications:*
(1) $Q - P \succeq 0 \Longrightarrow \mathcal{R}(P) \subset \mathcal{R}(Q)$.
(2) $\mathcal{R}(P) \subset \mathcal{R}(Q) \Longrightarrow cQ - P \succeq 0$ for some $c > 0$.

Proof. $(*)$ We consider each point separately.
(1) We use the fact that for positive semi-definite matrices we have $\mathcal{R}(A) \subset \mathcal{R}(A+B)$. Then we write $Q = P + (Q - P)$ where both $P, (Q-P) \succeq 0$ and so that $\mathcal{R}(P) \supset \mathcal{R}(P + (Q-P)) = \mathcal{R}(Q)$, as required.

(2) It suffices to show that there exists $c > 0$ such that for all $v \in \mathcal{R}(P)$ we have $c \cdot v^T Q v \geq v^T P v$. If it is finite, an obvious choice would be to choose,

$$\sup_{v\in\mathcal{R}(P)}\frac{v^TPv}{v^TQv}\leq\sup_{v\in\mathcal{R}(Q)}\frac{v^TPv}{v^TQv}, \tag{1.13}$$

where the upper-bound comes from the assumed inclusion $\mathcal{R}(P)\subset\mathcal{R}(Q)$. Finally, the above supremum may be upper-bounded by the strictly positive quantity $\lambda_{\max}(P)/\lambda_{\min,+}(Q)<\infty$ where $\lambda_{\max}(P)$ denotes the maximal eigenvalue of P and $\lambda_{\min,+}(Q)$ is the smallest non-zero eigenvalue of Q. □

Proposition 1.2.6. *Let $Q\in\mathbb{R}^{p\times p}$ be symmetric. Then the following are equivalent:*
(1) $x^TQx\geq 0$ for all $x\in\mathbb{R}^p$.
(2) If $Qv=\lambda v$ for some $v\in\mathbb{R}^p$ and $\lambda\in\mathbb{R}$, then $\lambda\geq 0$.

Proof. $(*)$ Let $Q\in\mathbb{R}^{p\times p}$ be symmetric.
• (1) $\Longrightarrow$ (2) If $Qv=\lambda v$ for some $v\neq 0$ then $v^TQv=v^T(\lambda v)=\lambda v^Tv\geq 0$. Since $v^Tv>0$ it follows that $\lambda\geq 0$.

• (2) $\Longrightarrow$ (1) Since Q is symmetric, it admits an orthonormal eigenbasis $\{v_i\}$ with eigenvalues $\lambda_i\geq 0$. Any $x\in\mathbb{R}^p$ may be written as $x=\sum_i\alpha_iv_i$ thus $x^TQx=\sum_i\lambda_i\alpha_i^2\geq 0$.

□

Proposition 1.2.7. *Let $Q\in\mathbb{R}^{p\times p}$ such that $Q\succeq 0$ and $v_1,...,v_m\in\mathbb{R}^p$. Then:*

$$v_1,...,v_m\in\mathcal{R}(Q)\Longleftrightarrow\mathcal{R}\left(\sum_{i=1}^m v_iv_i^T\right)\subseteq\mathcal{R}(Q). \tag{1.14}$$

Proof. $(*)$ $(\Longleftarrow)$ If A,B are positive semi-definite then $\mathcal{R}(A)\subset\mathcal{R}(A+B)$ and so for all $i=\{1,...,m\}$, $v_i\in\mathcal{R}(v_iv_i^T)\subset\mathcal{R}\left(\sum_{i=1}^m v_iv_i^T\right)\subset\mathcal{R}(Q)$. Thus $v_i\in\mathcal{R}(Q)$ as required.

$(\Longrightarrow)$ Let $x\in\mathcal{R}(\sum_{i=1}^m v_iv_i^T)$, i.e. $x=\left(\sum_{i=1}^m v_iv_i^T\right)y$ for some y. However, since each $v_i\in\mathcal{R}(Q)$ we have that $v_i=Qw_i$ so that we may write:

$$x=\left(\sum_{i=1}^m v_iv_i^T\right)y=\left(\sum_{i=1}^m Qw_iw_i^TQ\right)y=Q\left(\sum_{i=1}^m w_iw_i^TQy\right)\equiv Qz, \tag{1.15}$$

which shows that $x\in\mathcal{R}(Q)$. □

Remark 1.2.8. *We also have that $Q\succeq 0$ if and only if there exists $A\succeq 0$ such that $Q=A^2$. We call such an A the* ***square root*** *of Q and denote it by $A=\sqrt{Q}$ or $A=Q^{\frac{1}{2}}$.*

Remark 1.2.9. *We have that any projection P satisfies $P\succeq 0$.*

If we have that $B-A\succeq 0$ then we write $B\succeq A$. Non-negative definite matrices are partially ordered with respect to $\succeq$, and this is called the **Loewner order**.

1.3 Vectorization and the Kronecker Matrix Product

In this final section of the linear algebra recap, we present the concept of vectorization (i.e. transformations of matrices into vectors) and the Kronecker matrix product (i.e. transformations of vectors/matrices into larger vectors/matrices).

The space $\mathbb{R}^{n\times p}$ of $n\times p$ matrices forms a real vector space of dimension np. Indeed, this space is isometrically isomorphic with $\mathbb{R}^{np}$, where the isomorphism is given by the vec operation $\text{vec}:\mathbb{R}^{n\times p}\mapsto\mathbb{R}^{np}$ whose (linear) action is to stack the matrix columns into a tall np-vector:

$$\text{vec}\left\{\begin{pmatrix} v_1 & \dots & v_p\end{pmatrix}\right\} = \begin{pmatrix} v_1 \\ \vdots \\ v_p\end{pmatrix}, \qquad v_i\in\mathbb{R}^n. \tag{1.16}$$

Abstractly, $\mathbb{R}^{n\times p}$ is a real vector space whose elements are p-vectors with coordinates that are themselves elements of $\mathbb{R}^n$. Now, the canonical basis of $\mathbb{R}^{n\times p}$ can directly be seen to be the collection $E_{ij}:=v_iu_j^T$ for $\{v_i\}$ the canonical basis of $\mathbb{R}^n$ and $\{u_j\}$ the canonical basis of $\mathbb{R}^p$. A little bit of algebra also shows that:

$$\langle A,B\rangle_{\mathbb{R}^{n\times p}} := \langle \text{vec}(A),\text{vec}(B)\rangle_{\mathbb{R}^{np}} = \text{vec}(A)^T\text{vec}(B) = \text{trace}(A^TB). \tag{1.17}$$

A linear transformation on $\mathbb{R}^{n\times p}\to\mathbb{R}^{n\times p}$ will be a matrix transformation $\mathbb{R}^{np}\to\mathbb{R}^{np}$, hence an $np\times np$ matrix. In the vectorized perspective, a rank-1 transformation is $\text{vec}(U)\text{vec}(V)^T$ which maps $\text{vec}(A)$ to $\text{vec}(U)\text{vec}(V)^T\text{vec}(A)=\text{trace}(V^TA)\text{vec}(U)$ which can now be re-expressed in matrix form as $A\mapsto\text{trace}(V^TA)U$. Thus by the SVD, a linear $f:\mathbb{R}^{n\times p}\to\mathbb{R}^{n\times p}$ is represented as:

$$f(A) = \sum_{i=1}^{\text{rank}(f)} \sigma_i \text{trace}(V_i^TA)U_i, \tag{1.18}$$

and is therefore such that the vectorization of $f(A)$ is given by:

$$\text{vec}(f(A)) = \sum_{i=1}^{\text{rank}(f)} \sigma_i \text{vec}(U_i)\text{vec}(V_i)^T\text{vec}(A), \tag{1.19}$$

for $\sigma_i>0$ and $\{U_i\}$, $\{V_j\}$ orthonormal bases of $\mathbb{R}^{n\times p}$.

Definition 1.3.1. *The **Kronecker matrix product** between two matrices A and B is defined as:*

$$A\otimes B := \begin{pmatrix} a_{11}B & \dots & a_{1p}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{np}B\end{pmatrix}. \tag{1.20}$$

Then, from the above definition, it is clear that $\text{vec}(U)\text{vec}(V)^T\equiv\text{vec}(U)\otimes\text{vec}(V)^T$ and thus by the SVD, any linear map on $\mathbb{R}^{np}$ may be written (non-uniquely) as:

$$\sum_{i=1}^{np}(A_i)_{np\times 1}\otimes(B_i)_{1\times np}. \tag{1.21}$$

Proposition 1.3.2. *For the matrices A,X,B of appropriate dimensions, we have the following identity:*

$$\text{vec}(AXB) = (B^T\otimes A)\text{vec}(X). \tag{1.22}$$

Finally, as a result of this proposition, one can show that for (non-unique) matrices A_i and B_i, any linear map acts on $\mathbb{R}^{p\times p}$ as:

$$X\mapsto\sum_{i=1}^{np}(A_i)_{n\times n}X_{n\times p}(B_i)_{p\times p}. \tag{1.23}$$

Chapter 2

Random Vectors and Matrices

We now start the study of random vectors and random matrices that will be serve as the central objects for the theory of Multivariate Statistics. We first recall basic results and definitions ranging from joint distributions to conditional independence, before generalizing them to multi-dimension.

2.1 Basics of Random Variables and Random Vectors

• A **random vector** $X = (X_1, ..., X_p)^T$ is a finite collection of jointly distributed real random variables arranged as the coordinates of a vector. The point is that we may want to make probabilistic statements on the joint behaviour of all of these random variables. Thus, we define the **joint distribution function** of a random vector X as:

$$F_X(x_1, ..., x_p) := \mathbb{P}(X_1 \leq x_1, ..., X_p \leq x_p). \tag{2.1}$$

Correspondingly, one defines the **joint frequency function** (or **probability mass function**) if the $\{X_i\}_i$ are all discrete, given by:

$$f_X(x_1, ..., x_p) := \mathbb{P}(X_1 = x_1, ..., X_p = x_p), \tag{2.2}$$

and the associated continuous case with the **joint density function**, if there exists $f_X : \mathbb{R}^p \to [0, +\infty)$ such that:

$$F_X(x_1, ..., x_p) = \int_{-\infty}^{x_1} ... \int_{-\infty}^{x_p} f_X(u_1, ..., u_p) du_1 ... du_p. \tag{2.3}$$

Note that in the case where f_X is continuous at the point x one has:

$$f_X(x_1, ..., x_p) = \frac{\partial^p}{\partial x_1 ... \partial x_p} F_X(x_1, ..., x_p). \tag{2.4}$$

• Now, suppose that given a joint density one is interested only in a single random variable. Then, given the joint distribution of the random vector $X = (X_1, ..., X_p)^T$, we can isolate the distribution of a single coordinate X_i. In the discrete case, this is given by the **marginal frequency function** of X_i:

$$f_{X_i}(x_i) := \mathbb{P}(X_i = x_i) = \sum_{x_1} ... \sum_{x_{i-1}} \sum_{x_{i+1}} ... \sum_{x_p} f_X(x_1, ..., x_{i-1}, x_i, x_{i+1}, ...x_p). \tag{2.5}$$

In the continuous case, it is given by the **marginal density function** of X_i:

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} ... \int_{-\infty}^{\infty} f_X(y_1, ..., y_{i-1}, x_i, y_{i+1}, ..., y_p) dy_1 ... dy_{i-1} dy_{i+1} ... dy_p. \tag{2.6}$$

More generally, we may define the joint frequency/density of a random vector X formed by a subset of the coordinates of X, say the first k. Thus in the discrete case this is given by:

$$f_{X_1,...,X_k}(x_1,...,x_k) = \sum_{x_{k+1}} ... \sum_{x_p} f_X(x_1,...,x_k,x_{k+1},...,x_p), \tag{2.7}$$

and in the continuous case we have:

$$f_{X_1,...,X_k}(x_1,...,x_k) = \int_{-\infty}^{\infty} ... \int_{-\infty}^{\infty} f_X(x_1,...,x_k,x_{k+1},...,x_p)dx_{x+1}...dx_p. \tag{2.8}$$

Remark 2.1.1. *Marginals do **not** uniquely determine the joint distribution.*

• In some instances, we may wish to make probabilistic statements about the potential outcomes of one random variable, if we already know the outcome of another. If $X = (X_1,...,X_p)^T$ is a continuous/discrete random vector, we define the **conditional probability density/frequency function** of $(X_1,...,X_k)^T$ given $\{X_{k+1} = x_{k+1},...,X_p = x_p\}$ as:

$$f_{X_1,...,X_k|X_{k+1},...,X_p}(x_1,...,x_k|x_{k+1},...,x_p) = \frac{f_{X_1,...,X_p}(x_1,...,x_k,x_{k+1},...,x_p)}{f_{X_{k+1},...,X_p}(x_{k+1},...,x_p)}, \tag{2.9}$$

provided that $f_{X_{k+1},...,X_p}(x_{k+1},...,x_p) > 0$.

• The random variables $X_1,...,X_p$ are called **independent** if and only if for all $x_1,...,x_p \in \mathbb{R}$:

$$F_{X_1,...,X_p}(x_1,...,x_p) = \prod_{i=1}^{p} F_{X_i}(x_i). \tag{2.10}$$

Equivalently, $X_1,...,X_p$ are independent if and only if for all $x_1,...,x_p \in \mathbb{R}$:

$$f_{X_1,...,X_p}(x_1,...,x_p) = \prod_{i=1}^{p} f_{X_i}(x_i). \tag{2.11}$$

Note that when random variables are independent, conditional distributions reduce to the corresponding marginal distributions. Knowing the value of one of the random variables gives us no information about the distribution of the rest.

The random vector X in $\mathbb{R}^p$ is called **conditionally independent of the random vector Y given the random vector** Z written as $X \perp_Z Y$ or $\{X \perp Y\}|Z$ if and only if, for all $x_1,...,x_p \in \mathbb{R}$:

$$F_{X_1,...,X_p|Y}(x_1,...,x_p) = F_{X_1,...,X_p|Z}(x_1,...,x_p). \tag{2.12}$$

Equivalently, if and only if, for all $x_1,...,x_p \in \mathbb{R}$:

$$f_{X_1,...,X_p|Y}(x_1,...,x_p) = f_{X_1,...,X_p|Z}(x_1,...,x_p). \tag{2.13}$$

Knowing Y in addition to knowing Z gives us no more information about X. As a result, if X is conditionally independent of Y given Z then $F_{X,Y|Z} = F_{X|Y,Z}F_{Y|Z} = F_{X|Z}F_{Y|Z}$. And thus:

$$X \perp_Z Y \iff Y \perp_Z X. \tag{2.14}$$

• Let $X = (X_1,...,X_p)^T$ be a random vector in $\mathbb{R}^p$ with joint density function $f_X(x_1,...,x_p)$. For any $g : \mathbb{R}^p \to \mathbb{R}$ we define the **expectation of** g to be:

$$\mathbb{E}\big(g(X_1,...,X_p)\big) := \int_{-\infty}^{\infty} ... \int_{-\infty}^{\infty} g(x_1,...,x_p) f_X(x_1,...,x_p) dx_1...dx_p. \tag{2.15}$$

Similarly in the discrete case:

$$\mathbb{E}\big(g(X_1,...,X_p)\big) := \sum_{x_1 \in \mathcal{X}_1} ... \sum_{x_p \in \mathcal{X}_p} g(x_1,...,x_p) f_X(x_1,...,x_p). \tag{2.16}$$

The corresponding **mean vector** of a random vector $X = (X_1,...,X_p)$ is defined as $\mathbb{E}(X) = \big(\mathbb{E}(X_1) \ \ \dots \ \ \mathbb{E}(X_p)\big)^T$.

2.2 Random Matrices, Covariance Matrices and Linear Transformations

We now turn to a more global framework by working with random matrices, where we will encounter the covariance matrix and how linear transformations affects some results. In this section, we consider a random $n \times p$ matrix,

$$X = \begin{pmatrix} X_{11} & \dots & X_{1p} \\ \vdots & & \vdots \\ X_{n1} & \dots & X_{np} \end{pmatrix}, \tag{2.17}$$

which is simply a matrix whose n rows are random vectors in $\mathbb{R}^p$. Equivalently, it is a finite collection of np random variables arranged as the entries of an $n \times p$ matrix. Notions of joint densities/frequencies follow immediately, and the notion of expectation is just a matrix of expectations.

Lemma 2.2.1. *Given a random matrix $X_{n\times p}$ and deterministic matrices $A_{m\times n}$ and $B_{p\times q}$, the following hold:*
(i) $\mathbb{E}(X^T) = \big(\mathbb{E}(X)\big)^T$.
(ii) When $n = p$, $\mathbb{E}(\text{trace}(X)) = \text{trace}(\mathbb{E}(X))$.
(iii) $\mathbb{E}(AXB) = A\mathbb{E}(X)B$.

Now, recall that the **covariance** of a random variable X_1 with another random variable X_2 expresses the degree of linear dependency between the two. If $\mathbb{E}(X_i^2) < \infty$, it is defined as:

$$\text{Cov}(X_1, X_2) := \mathbb{E}\big[(X_1 - \mathbb{E}(X_1))(X_2 - \mathbb{E}(X_2))\big]. \tag{2.18}$$

Definition 2.2.2. *The **covariance matrix** $\Sigma = \{\Sigma_{ij}\}$ of a random vector $X = (X_1,...,X_p)^T$ is a $p \times p$ symmetric matrix with entries,*

$$\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}\big[(X_i - \mathbb{E}(X_i))(X_j - \mathbb{E}(X_j))\big], \quad 1 \le i \le j \le p. \tag{2.19}$$

In other words, the covariance matrix encodes the variances in the diagonal and the pairwise covariances in the off-diagonal of the coordinates of X. Often, the notations $\Sigma_{ii} \equiv \sigma_i^2$ and $\Sigma_{ij} \equiv \sigma_{ij}$ for $i \neq j$ are used, where $\sigma_i := \sqrt{\text{Var}(X_i)}$ is the **standard deviation** of X_i. It can also be easily checked that,

$$\Sigma = \mathbb{E}\big[(X - \mathbb{E}(X))(X - \mathbb{E}(X))^T\big] = \mathbb{E}(XX^T) - \mathbb{E}(X)\mathbb{E}(X)^T. \tag{2.20}$$

Now, we know that for Gaussians the covariance being zero is equivalent to the independence of the random variables. But this is not true in general, and we now look at two examples where this fact does not hold.

Example 2.2.3 ($Cov(X,Y)=0 \not\Longrightarrow$ independence)**.** *Let $X \sim Unif[-\pi,\pi]$ and define $Y=\cos(X)$. Clearly X and Y are <u>not</u> independent, however their covariance is zero. Let us now show this concretely. First we calculate $\mathbb{P}(Y>0)=\frac{1}{2}$ but $\mathbb{P}(Y>0|X\in(-\pi,-2))=1$. Despite this, we have that:*

$$\mathrm{Cov}(X,Y)=\mathbb{E}(XY)-\mathbb{E}(X)\mathbb{E}(Y)=\int_{-\pi}^{\pi} x\cos(x)\frac{1}{2\pi}dx=0. \tag{2.21}$$

This is due to the fact that some non-linear dependencies cannot be detected by covariance.

Example 2.2.4 ($\mathrm{Cov}(X,Y)=0 \not\Longrightarrow$ independence)**.** *Let X and Y have joint density:*

$$f_{XY}(x,y)=\begin{cases}\frac{1}{\pi}, & \text{if } x^2+y^2\leq 1\\ 0 & \text{otherwise}\end{cases}. \tag{2.22}$$

Note that $\mathbb{E}(X)=\mathbb{E}(Y)=0$ by symmetry. Hence $\mathrm{Cov}(X,Y)=\mathbb{E}(XY)$, however:

$$\mathbb{E}(XY)=\int\int_{x^2+y^2\leq 1} xy\frac{1}{\pi}dxdy=\int\int_{x^2+y^2\leq 1,y\geq 0} xy\frac{1}{\pi}dxdy+\int\int_{x^2+y^2\leq 1,y<0} xy\frac{1}{\pi}dxdy. \tag{2.23}$$

The two terms are equal by symmetry, and moreover:

$$\int\int_{x^2+y^2\leq 1,y\geq 0} xy\frac{1}{\pi}dxdy=\frac{1}{\pi}\int_{-1}^{1} x\int_{0}^{1-x^2} ydydx=\frac{1}{\pi}\int_{-1}^{1} x\frac{(1-x^2)^2}{2}dx=0. \tag{2.24}$$

Thus, the covariance is zero but X and Y are clearly dependent since knowing X restricts the possible values of Y.

Lemma 2.2.5. *Let X be a random $p\times 1$ vector such that $\mathbb{E}\|X\|^2<\infty$ and with covariance Σ. Given a real matrix $A_{q\times p}$, the covariance of the $q\times 1$ random vector AX is $A\Sigma A^T$.*

Proof. $(*)$ This is just a simple calculation. Let $\mu=\mathbb{E}(X)$ then,

$$\mathbb{E}(Y)=\mathbb{E}(AX)=A\mathbb{E}(X)=A\mu. \tag{2.25}$$

Thus we obtain:

$$\mathrm{Cov}(Y)=\mathbb{E}\big[(A(X-\mu))(A(X-\mu))^T\big]=A\mathrm{Cov}(X)A^T=A\Sigma A^T. \tag{2.26}$$

□

Corollary 2.2.6. *Let Y be a random $d\times 1$ vector such that $\mathbb{E}\|Y\|^2<\infty$. Let $\beta,\gamma\in\mathbb{R}^d$ be fixed vectors. If Ω denotes the covariance matrix of Y, then:*
(i) $\mathrm{Var}(\beta^TY)=\beta^T\Omega\beta$.
(ii) $\mathrm{Cov}(\beta^TY,\gamma^TY)=\gamma^T\Omega\beta$.

We now state an interesting result relating non-negative matrices with covariance matrices.

Proposition 2.2.7. *Let Ω be a real symmetric matrix. Then Ω is non-negative definite if and only if Ω is the covariance matrix of some random variable X.*

Proof. $(*)$ If $\Omega = \mathbb{E}(XX^T)$ then $u^T\Omega u = \mathbb{E}(u^TXX^Tu) = \mathbb{E}\big[(X^Tu)^T(X^Tu)\big] = \mathbb{E}\|X^Tu\|^2 \geq 0$. For the other direction, it suffices to take Z as a random vector with covariance the identity matrix, and then pick $X = \sqrt{\Omega}Z$, where the square root exists since $\Omega \succeq 0$. □

2.3 Cross-Covariance Matrix and the Support of a Random Vector

Let X and Y be centred random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

Definition 2.3.1. *The **cross-covariance** between X and Y is the $n \times m$ matrix,*

$$\Sigma_{XY} = \mathrm{Cov}(X,Y) := \mathbb{E}\big[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))^T\big] = \mathbb{E}(XY^T) - \mathbb{E}(X)\mathbb{E}(Y)^T. \tag{2.27}$$

Remark that this is not symmetric since $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)^T$, and so will not be non-negative definite in general. Now, if we concatenate into an $(n+m)$-dimensional random vector $Z = (X^T \ Y^T)^T$, and use block notation, we see that:

$$\Sigma_Z = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}. \tag{2.28}$$

In other words, an $n \times m$ matrix is a cross-covariance if and only if it can be represented as the off-diagonal block of some $(n+m) \times (n+m)$ covariance matrix.

Definition 2.3.2. *The **support** of a random vector X in $\mathbb{R}^p$ is defined as:*

$$\mathrm{supp}(X) := \{x \in \mathbb{R}^p : \mathbb{P}[\|X - x\| < \epsilon] > 0, \forall \epsilon > 0\}. \tag{2.29}$$

Intuitively, the support is the region of $\mathbb{R}^p$ that X can reach. In fact, $\mathrm{supp}(X)$ is a closed set since it is indeed the smallest closed set F such that $\mathbb{P}(X \in F) = 1$. Even more interestingly, the covariance provides some information on the support.

Lemma 2.3.3. *Let X be a random vector in $\mathbb{R}^p$ with mean μ_X and covariance Σ_X. Then, recalling that $\Sigma_X\Sigma_X^\dagger = H_X$ is the projection onto $\mathcal{R}(\Sigma_X)$:*
(i) $\mathrm{supp}(X) \subseteq \mathcal{R}(\Sigma_X) + \mu_X$.
(ii) $(\Sigma_X\Sigma_X^\dagger)(X - \mu_X) = X - \mu_X$ *a.s.*
(iii) $(\Sigma_X\Sigma_X^\dagger)\Sigma_{XY} = \Sigma_{XY}$ *for any random vector Y with finite second moment.*

Proof. We prove this in turns.

- (1) $\iff$ (2) Recall that $H_Xu = u$ if and only if $u \in \mathcal{R}(H_X)$. Thus (2) is equivalent to stating that $\mathbb{P}(X - \mu_X \in \mathcal{R}(H_X)) = 1$. Since $\mathcal{R}(H_X) = \mathcal{R}(\Sigma_X)$, and observing that $\mathcal{R}(H_X) + \mu_X$ is closed, the last statement is equivalent to (1).

- (2) Write $X - \mu_X = H_X(X - \mu_X) + (\mathbb{I} - H_X)(X - \mu_X)$ and remark that,

$$\mathrm{Cov}\big((\mathbb{I} - H_X)(X - \mu_X)\big) = (\mathbb{I} - H_X)\Sigma_X(\mathbb{I} - H_X) = (\Sigma_X - \Sigma_X\Sigma_X^\dagger\Sigma_X)(\mathbb{I} - H_X) = 0.$$

As a result, $(\mathbb{I} - H_X)(X - \mu_X) = \mathbb{E}[(\mathbb{I} - H_X)(X - \mu_X)]$ a.s. but $\mathbb{E}[(\mathbb{I} - H_X)(X - \mu_X)] = (\mathbb{I} - H_X)\mathbb{E}[(X - \mu_X)] = 0$. In summary, $X - \mu_X = H_X(X - \mu_X)$ a.s., as required.

- (3) We simply observe the following:

$$H_X\Sigma_{XY} = H_X\mathbb{E}\big[(X-\mu_X)(Y-\mu_Y)^T\big] = \mathbb{E}\big[H_X(X-\mu_X)(Y-\mu_Y)^T\big] = \mathbb{E}\big[(X-\mu_X)(Y-\mu_Y)^T\big] = \Sigma_{XY},$$

where we have used result (2) in the second to last equality.

□

Lemma 2.3.4. *Let Y be a random vector in $\mathbb{R}^p$ with covariance Σ_Y. Then, $\mathrm{Cov}\big((\Sigma_Y^\dagger)^{\frac{1}{2}}Y\big) = H_Y$, where H_Y is the projection onto $\mathcal{R}(\Sigma_Y)$. Consequently, when Σ_Y^{-1} exists, we have $\mathrm{Cov}\big(\Sigma_Y^{-\frac{1}{2}}Y\big) = \mathbb{I}_{p\times p}$.*

Proof. We calculate:

$$\mathrm{Cov}\big((\Sigma_Y^\dagger)^{\frac{1}{2}}Y\big) = (\Sigma_Y^\dagger)^{\frac{1}{2}}\Sigma_Y(\Sigma_Y^\dagger)^{\frac{1}{2}} = (\Sigma_Y^\dagger)^{\frac{1}{2}}\Sigma_Y^{\frac{1}{2}}\Sigma_Y^{\frac{1}{2}}(\Sigma_Y^\dagger)^{\frac{1}{2}}. \tag{2.30}$$

By the definition of the pseudoinverse and the fact that $\Sigma_Y \succeq 0$,

$$(\Sigma_Y^\dagger)^{\frac{1}{2}}\Sigma_Y^{\frac{1}{2}} = \big(\Sigma_Y^{\frac{1}{2}}\big)^\dagger\Sigma_Y^{\frac{1}{2}}. \tag{2.31}$$

Now, the RHS is the projection onto $\mathcal{R}\big(\Sigma_Y^{\frac{1}{2}}\big)$. Finally, $\mathcal{R}(\Sigma_Y) = \mathcal{R}\big(\Sigma_Y^{\frac{1}{2}}\big)$ due to the Spectral Theorem, and the proof is complete. □

Lemma 2.3.5 (Matrix Correlation Inequality)**.** *Let $Z = (X^T \;\; Y^T)^T$ be comprised of two centred random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, with covariance,*

$$\Sigma_Z = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}. \tag{2.32}$$

Then, $\Sigma_X - \Sigma_{XY}\Sigma_Y^\dagger\Sigma_{XY}^T \succeq 0$. Furthermore, if Σ_Z is non-singular, then Σ_Y is also non-singular and, $\Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T \succ 0$.

Note that the matrix $\Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T$ is called the **Schur complement of Σ_Y in Σ_Z**. Also, when $\Sigma_Z \succ 0$, the last inequality may be re-written as $\Sigma_X^{-1}\Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T \prec \mathbb{I}$, explaining the term *correlation inequality*.

Proof. Define the zero-mean random vector $\epsilon = X - \Sigma_{XY}\Sigma_Y^\dagger Y$ and we notice that:

$$\begin{aligned} 0 \preceq \Sigma_\epsilon = \mathbb{E}(\epsilon\epsilon^T) &= \mathbb{E}(XX^T) + \Sigma_{XY}\underbrace{\Sigma_Y^\dagger\mathbb{E}(YY^T)\Sigma_Y^\dagger}_{=\Sigma_Y^\dagger\Sigma_Y\Sigma_Y^\dagger=\Sigma_Y^\dagger}\Sigma_{XY}^T - \underbrace{\mathbb{E}(XY^T)}_{=\Sigma_{XY}}\Sigma_Y^\dagger\Sigma_{XY}^T - \Sigma_{XY}\Sigma_Y^\dagger\underbrace{\mathbb{E}(YX^T)}_{=\Sigma_{XY}^T} \\ &= \Sigma_X - \Sigma_{XY}\Sigma_Y^\dagger\Sigma_{XY}^T. \end{aligned}$$

For the second part, we will argue by contradiction. Assume that $\Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T$ is singular. This means that there exists an $x \in \mathbb{R}^n \setminus \{0\}$ such that $x^T\Sigma_X x - x^T\Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T x = 0$. Now define the $(n+m)\times 1$ vector,

$$u = \begin{pmatrix} x \\ -\Sigma_Y^{-1}\Sigma_{XY}^T x \end{pmatrix}. \tag{2.33}$$

Since $x \neq 0$, it also holds that $u \neq 0$. Now observe that,

$$u^T \Sigma_Z u = \begin{pmatrix} x^T & -x^T \Sigma_{XY} \Sigma_Y^{-1} \end{pmatrix} \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} \begin{pmatrix} x \\ -\Sigma_Y^{-1} \Sigma_{XY}^T x \end{pmatrix} = x^T \Sigma_X x - x^T \Sigma_{XY} \Sigma_Y^{-1} \Sigma_{XY}^T x = 0.$$

Since $u \neq 0$, this contradicts the assumption that Σ_Z is non-singular. □

2.4 Important Random Matrices and the Moment Generating Function

In this section, we present the precision matrix, the correlation matrix and the cross-correlation matrix. We end on the definition of the moment generating function of a random vector.

Definition 2.4.1. *Let X be a random vector in $\mathbb{R}^p$ with non-singular covariance Σ_X. The **precision matrix** of X is defined as $\Theta_X := \Sigma_X^{-1}$.*

As a direct application of Lemma (2.3.5) we have the following result for a 2×2 block precision matrix.

Lemma 2.4.2. *Let Σ be an $(n+m) \times (n+m)$ non-singular covariance,*

$$\Sigma = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}. \tag{2.34}$$

Then, $\Sigma_X - \Sigma_{XY} \Sigma_Y^{-1} \Sigma_{XY}^T \succ 0$ and $\Sigma_Y - \Sigma_{XY}^T \Sigma_X^{-1} \Sigma_{XY} \succ 0$ where we have the following expression for the precision matrix:

$$\Theta := \Sigma^{-1} = \begin{pmatrix} \left(\Sigma_X - \Sigma_{XY} \Sigma_Y^{-1} \Sigma_{XY}^T\right)^{-1} & -\left(\Sigma_X - \Sigma_{XY} \Sigma_Y^{-1} \Sigma_{XY}^T\right)^{-1} \Sigma_{XY} \Sigma_Y^{-1} \\ -\Sigma_Y^{-1} \Sigma_{XY}^T \left(\Sigma_X - \Sigma_{XY} \Sigma_Y^{-1} \Sigma_{XY}^T\right)^{-1} & \left(\Sigma_Y - \Sigma_{XY}^T \Sigma_X^{-1} \Sigma_{XY}\right)^{-1} \end{pmatrix}. \tag{2.35}$$

We now move on to correlations. Recall that the **correlation** between X_1 and X_2 is defined as,

$$\operatorname{corr}(X_1, X_2) := \frac{\operatorname{Cov}(X_1, X_2)}{\sqrt{\operatorname{Var}(X_1) \operatorname{Var}(X_2)}}. \tag{2.36}$$

This quantity conveys equivalent dependence information to covariance. A first advantage of this quantity is that it is invariant to changes of scales, and also that it can be understood in absolute terms since it ranges in $[-1, 1]$. This last point is a result of the correlation inequality (Cauchy-Schwarz) $|\operatorname{corr}(X_1, X_2)| \leq \sqrt{\operatorname{Var}(X_1) \operatorname{Var}(X_2)}$.

Definition 2.4.3. *The **correlation matrix** $R = \{\rho_{ij}\}$ of a random vector $X = (X_1, ..., X_p)^T$ is a $p \times p$ symmetric matrix with entries $\rho_{ij} = \operatorname{corr}(X_i, X_j)$ for $1 \leq i \leq j \leq p$.*

Note that the correlation matrix is well-defined whenever $\operatorname{Var}(X_i) > 0$ for all $1 \leq i \leq n$, i.e. none of the coordinates are degenerate random variables. Equivalently, the correlation matrix is the covariance matrix of the standardised vector $X = (X_1/\sigma_1, ..., X_p/\sigma_p)^T$, where $\sigma_i^2 = \operatorname{Var}(X_i)$. Thus recalling that $\operatorname{Cov}(AX) = A \Sigma A^T$ we have that:

$$R = \begin{pmatrix} \operatorname{Var}(X_1) & & 0 \\ & \ddots & \\ 0 & & \operatorname{Var}(X_p) \end{pmatrix}^{\frac{1}{2}} \Sigma \begin{pmatrix} \operatorname{Var}(X_1) & & 0 \\ & \ddots & \\ 0 & & \operatorname{Var}(X_p) \end{pmatrix}^{\frac{1}{2}} = \operatorname{diag}\left(\sigma_1^{-1}, ..., \sigma_p^{-1}\right) \Sigma \operatorname{diag}\left(\sigma_1^{-1}, ..., \sigma_p^{-1}\right).$$

Thus, correlation matrices are non-negative definite, i.e. $R \succeq 0$. As a closely related notion, we may now define the cross-correlation matrix.

Definition 2.4.4. *Let X and Y be two centred random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively. The **cross-correlation matrix** of X and Y is the $n \times m$ matrix R_{XY} with entries $\frac{\mathrm{Cov}(X_i,Y_j)}{\sqrt{\mathrm{Var}(X_i)\mathrm{Var}(Y_j)}}$ for $i = 1,...,n$ and $j = 1,...,m$.*

Again, if we concatenate into an $(n+m)$-dimensional random vector $Z = (X^T \ Y^T)^T$ and use block notation, we may write,

$$\Sigma_Z = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} \quad \text{and,} \quad R_Z = \begin{pmatrix} R_X & R_{XY} \\ R_{XY}^T & R_Y \end{pmatrix}. \tag{2.37}$$

It is now easy to verify that:

$$R_{XY} = \big(\mathrm{diag}(\Sigma_X)\big)^{-\frac{1}{2}} \Sigma_{XY} \big(\mathrm{diag}(\Sigma_Y)\big)^{-\frac{1}{2}}. \tag{2.38}$$

We now end this section with the so-called moment generating function (MGF). Recall that the MGF $M_X(t) : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ of a scalar random variable X is defined as $M_X(t) := \mathbb{E}\big(e^{tX}\big)$ for $t \in \mathbb{R}$. It need not be finite for $t \neq 0$, however when $t = 0$ the following Theorem follows.

Theorem 2.4.5. *Let X and Y be scalar random variables, and assume that $M_X(t) < \infty$ and $M_Y(t) < \infty$ for all $t \in B = (-\epsilon, \epsilon)$, for some $\epsilon > 0$. Then the following points hold:*
(1) M_X is infinitely differentiable on B.
(2) $\mathbb{E}[|X|^k] < \infty$ and $\mathbb{E}(X^k) = \frac{d^k M_X}{dt^k}(0)$ for all $k \geq 1$.
(3) $F_X = F_Y$ on $\mathbb{R} \iff M_Y = M_X$ on B.
(4) If $X \perp Y$ then M_{X+Y} is finite and $M_{X+Y} = M_X M_Y$ on B.

Let us now present the MGF for a random vector.

Definition 2.4.6. *The **moment generating function** of a random vector W in $\mathbb{R}^d$ is defined as $M_W(\theta) := \mathbb{E}\big(e^{\theta^T W}\big)$ for $\theta \in \mathbb{R}^d$, and need not be finite for $\theta \neq 0$.*

When the MGF exists on an open ball at the origin, it characterizes the distribution of the corresponding random vector (as in the scalar case). Consequently, it factorizes into two marginal MGFs if and only if the corresponding random vectors are independent, i.e.:

$$X_{n\times 1} \perp Y_{m\times 1} \iff \mathbb{E}\left(e^{\beta^T X + \gamma^T Y}\right) = \mathbb{E}\left(e^{\beta^T X}\right)\mathbb{E}\left(e^{\gamma^T Y}\right), \quad \forall \beta \in \mathbb{R}^n, \forall \gamma \in \mathbb{R}^m.$$

Chapter 3

Gaussian Random Vectors, Quadratic Forms and Concentration of Measures

In this chapter, we investigate some basic and more evolved properties of Gaussian random vectors. We will then turn to the definition of the chi-squared distribution seen as a sum of squared Gaussians. We end this chapter by briefly introducing some notions on the concentration of measures.

3.1 Introduction - Gaussian Random Variables

If for some $\mu \in \mathbb{R}$ and some $\sigma \in (0,\infty)$ a random variable X has density,

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \tag{3.1}$$

for $x \in \mathbb{R}$, then X is a **Gaussian (or Normal) random variable** and write $X \sim \mathcal{N}(\mu,\sigma^2)$. This is indeed a valid probability density establishing existence. By a simple change of variable we have:

$$\begin{aligned}\left[\int_{\mathbb{R}} f_X(x)dx\right]^2 = \int_{\mathbb{R}}\int_{\mathbb{R}} f(x)f(y)dxdy &= \frac{1}{2\pi\sigma^2}\int_{\mathbb{R}}\int_{\mathbb{R}} \exp\left(-\frac{(x-\mu)^2+(y-\mu)^2}{2\sigma^2}\right)dxdy \\ &= \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\theta d\theta \int_0^{\infty} r\exp\left(-\frac{r^2}{2\sigma^2}\right)dr = 1.\end{aligned}$$

By convention, a constant $\mu \in \mathbb{R}$ is considered to be $\mathcal{N}(\mu,0)$ random variable. Hence, Gaussian random variables need not have density unless $\sigma > 0$. Now, a random variable $Z \sim \mathcal{N}(0,1)$ is called a **standard Gaussian random variable**. We write ϕ for its PDF and Φ for its CDF.

Lemma 3.1.1. *Given $\mu \in \mathbb{R}$ and $\sigma \in (0,\infty)$ one has $X \sim \mathcal{N}(\mu,\sigma^2)$ if and only if $X = \sigma Z + \mu$ for some Gaussian random variable $Z \sim \mathcal{N}(0,1)$.*

Proof. Changing variables to $z = \frac{x-\mu}{\sigma}$, one has:

$$F_X(y) = \int_{-\infty}^{y} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)dx = \int_{-\infty}^{\frac{y-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{z^2}{2}\right)dz = \Phi\left(\frac{y-\mu}{\sigma}\right). \tag{3.2}$$

Thus, if one defines $Z = \sigma^{-\frac{1}{2}}(X-\mu)$ then the ($\Longrightarrow$) part is proven. Starting with $Z \sim \mathcal{N}(0,1)$ and following the same steps in reverse gives the ($\Longleftarrow$) part, with $X = \sigma Z + \mu$. □

Lemma 3.1.2. *Given $\mu \in \mathbb{R}$ and $\sigma \in [0,\infty)$, the MGF of $X \sim \mathcal{N}(\mu,\sigma^2)$ satisfies,*

$$M_X(t) = \exp\left(t\mu + t^2\frac{\sigma^2}{2}\right). \tag{3.3}$$

This being finite for all $t \in \mathbb{R}$ implies that all moments of X exist, and its central moments are:

$$\mathbb{E}[(X-\mathbb{E}(X))^k] = \mathbb{E}[(X-\mu)^k] = \mathbb{E}[\sigma^k Z^k] = \begin{cases} 0, & \text{for } k \text{ odd} \\ \sigma^k(k-1)!! & \text{for } k \text{ even} \end{cases}. \tag{3.4}$$

Consequently, $\mu = \mathbb{E}(X)$ is the mean and $\sigma^2 = \mathrm{Var}(X)$ is the variance.

Proof. By definition and the change of variables $y = x - \mu - \sigma^2 t$ we find that:

$$\begin{aligned} M_X(t) = \mathbb{E}\left(e^{tX}\right) = \frac{1}{\sigma\sqrt{2\pi}}\int_{\mathbb{R}} \exp\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right)dx &= \frac{1}{\sigma\sqrt{2\pi}}\int_{\mathbb{R}} \exp\left(-\frac{(x-\mu-\sigma^2 t)^2 - 2\mu\sigma^2 t - \sigma^4 t^2}{2\sigma^2}\right)dx \\ &= \exp\left(t\mu + t^2\frac{\sigma^2}{2}\right)\left(\frac{1}{\sigma\sqrt{2\pi}}\int_{\mathbb{R}}\exp\left(-\frac{y^2}{2\sigma^2}\right)dy\right) \\ &= \exp\left(t\mu + t^2\frac{\sigma^2}{2}\right). \end{aligned}$$

With the MGF in hand, we may now calculate the moments of a $Z \sim \mathcal{N}(0,1)$:

$$\mathbb{E}[Z^k] = \frac{dM_X}{du}\Big|_{u=0} = \begin{cases} 0, & \text{for } k \text{ odd} \\ (k-1)!! & \text{for } k \text{ even} \end{cases}. \tag{3.5}$$

With a change of variables, this will yield the central moments of $X \sim \mathcal{N}(\mu,\sigma^2)$ as:

$$\mathbb{E}[(X-\mathbb{E}(X))^k] = \mathbb{E}[(X-\mu)^k] = \mathbb{E}[\sigma^k Z^k] = \begin{cases} 0, & \text{for } k \text{ odd} \\ \sigma^k(k-1)!! & \text{for } k \text{ even} \end{cases}. \tag{3.6}$$

□

3.2 Gaussian Random Vectors and Affine Transformations

Let us now formally define a Gaussian vector, where we will also list some key properties that will turn out to be very useful in the later sections.

Definition 3.2.1. *A random vector Y in $\mathbb{R}^d$ is **Gaussian** if and only if $\beta^T Y$ is Gaussian random variable for all deterministic vectors $\beta \in \mathbb{R}^d$.*

Note that from the definition, it follows that Y must have some well-defined mean vector μ and some well-defined covariance matrix Σ. To see this, note that since $\mathbb{E}\big((\beta^T Y)^2\big) < \infty$ for all β, then we can successively pick β to be equal to each canonical basis vector and conclude that each coordinate has finite variance, and thus $\mathbb{E}\|Y\|^2 < \infty$. So all the means, variances and covariances of its coordinates are well-defined. Thus the mean vector μ and covariance matrix Σ may be (uniquely) determined entrywise by equating $\mu_i = \mathbb{E}(e_i^T Y)$ and $\Sigma_{ij} = \mathrm{Cov}(e_i^T Y, e_j^T Y)$, where e_j is the j-th canonical basis vector (i.e. a vector of zeroes with a single 1 in the j-th entry).

Proposition 3.2.2 (MGF)**.** *The MGF of $Y \sim \mathcal{N}(\mu, \Sigma)$ is,*

$$M_Y(u) = \exp\left(u^T\mu + \frac{1}{2}u^T\Sigma u\right). \tag{3.7}$$

Proof. Let $v \in \mathbb{R}^d$ be arbitrary. Then v^TY is scalar Gaussian with mean $v^T\mu$ and variance $v^T\Sigma v$. Hence it has MGF,

$$M_{v^TY}(t) = \mathbb{E}\left(e^{tv^TY}\right) = \exp\left(t(v^T\mu) + \frac{t^2}{2}(v^T\Sigma v)\right). \tag{3.8}$$

Now, take $t = 1$ and observe that $M_{v^TY}(1) = M_Y(v)$. Combining the two yields the desired MGF for Y. □

We now show that Gaussian vectors are closed under affine transformations.

Proposition 3.2.3 (Closure under Affine Transformations)**.** *For $Y \sim \mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$ and given $B_{n\times p}$, $\theta_{n\times 1}$ we have:*

$$\theta + BY \sim \mathcal{N}(\theta + B\mu, B\Sigma B^T). \tag{3.9}$$

Proof. Let us compute the MGF of $\theta + BY$:

$$\begin{aligned} M_{\theta+BY}(u) = \mathbb{E}\left[\exp\Big(u^T(\theta+BY)\Big)\right] &= \exp\Big(u^T\theta\Big)\mathbb{E}\left[\exp\Big((B^Tu)^TY\Big)\right] \\ &= \exp\Big(u^T\theta\Big)M_Y(B^Tu) \\ &= \exp\Big(u^T\theta\Big)\exp\left\{(B^Tu)^T\mu + \frac{1}{2}u^TB\Sigma B^Tu\right\} \\ &= \exp\left\{u^T(\theta+B\mu) + \frac{1}{2}u^TB\Sigma B^Tu\right\}. \end{aligned}$$

This last expression is the MGF of a $\mathcal{N}(\theta + B\mu, B\Sigma B^T)$ distribution. □

Proposition 3.2.4. *If $Y \sim \mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$ and $A_{m\times p}$, $B_{d\times p}$ be real matrices. Then:*

$$AY \perp BY \iff A\Sigma B^T = 0. \tag{3.10}$$

Proof. It suffices to prove the result assuming $\mu = 0$.

($\Longleftarrow$) First assume that $A\Sigma B^T = 0$. Let $W_{(m+d)\times 1} = \begin{pmatrix} AY \\ BY \end{pmatrix}$ and $\theta_{(m+d)\times 1} = \begin{pmatrix} u_{m\times 1} \\ v_{d\times 1} \end{pmatrix}$. Then:

$$\begin{aligned} M_W(\theta) = \mathbb{E}\left[\exp\Big(W^T\theta\Big)\right] &= \mathbb{E}\big[\exp\big(Y^TA^Tu + Y^TB^Tv\big)\big] \\ &= \mathbb{E}\big[\exp\big(Y^T(A^Tu + B^Tv)\big)\big] \\ &= M_Y(A^Tu + B^Tv) \\ &= \exp\left(\frac{1}{2}(A^Tu + B^Tv)^T\Sigma(A^Tu + B^Tv)\right) \\ &= \exp\left(\frac{1}{2}\big(u^TA\Sigma A^Tu + v^TB\Sigma B^Tv + u^TA\Sigma B^Tv + v^TB\Sigma A^Tu\big)\right) \\ &= \exp\left(\frac{1}{2}\big(u^TA\Sigma A^Tu + v^TB\Sigma B^Tv\big)\right) = M_{AY}(u)M_{BY}(v). \end{aligned}$$

Thus the joint MGF being of product of the marginal MGFs, it proves independence.

($\Longrightarrow$) For the converse, assume that $AY \perp BY$. Then for all u, v: $M_W(\theta) = M_{AY}(u)M_{BY}(v)$. Thus:

$$\exp\left(\frac{1}{2}\left(u^T A\Sigma A^T u + v^T B\Sigma B^T v + u^T A\Sigma B^T v + v^T B\Sigma A^T u\right)\right) = \exp\left(\frac{1}{2}u^T A\Sigma A^T u\right)\exp\left(\frac{1}{2}v^T B\Sigma B^T v\right)$$

Therefore, from the LHS we require,

$$\exp\left(\frac{1}{2}u^T A\Sigma B^T v\right) = \exp\left(\frac{1}{2}v^T B\Sigma A^T u\right) = 1 \Longrightarrow u^T A\Sigma B^T v = 0,$$

for all $u \in \mathbb{R}^d, v \in \mathbb{R}^m$. Hence from the definition of the orthogonal complement (1.1.3), the column space of $A\Sigma B^T$ is the whole of $\mathbb{R}^m$. This implies that $A\Sigma B^T = 0$, as required. □

Let us now present the density function of a Gaussian vector.

Proposition 3.2.5 (Density function)**.** *Let $\Sigma_{p\times p}$ be non-singular. The density of $\mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$ is:*

$$f_Y(y) = \frac{1}{(2\pi)^{\frac{p}{2}}|\Sigma|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}(y-\mu)^T\Sigma^{-1}(y-\mu)\right). \tag{3.11}$$

Proof. Let $Z = (Z_1, ..., Z_p)^T$ be a vector of i.i.d. $\mathcal{N}(0,1)$ random variables. Since we have independence, we have that the density of Z is,

$$f_Z(z) = \prod_{i=1}^{p} f_{Z_i}(z_i) = \prod_{i=1}^{p}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}z_i^2\right) = \frac{1}{(2\pi)^{\frac{p}{2}}}\exp\left(-\frac{1}{2}z^T z\right). \tag{3.12}$$

Furthermore, we also have that the MGF of Z is given by:

$$M_Z(u) = \mathbb{E}\left(\exp\left(\sum_{i=1}^{p} u_i Z_i\right)\right) = \prod_{i=1}^{p}\mathbb{E}\left(\exp(u_i Z_i)\right) = \exp\left(\frac{1}{2}u^T u\right), \tag{3.13}$$

which corresponds to the MGF of a p-variate $\mathcal{N}(0, \mathbb{I})$ distribution. Thus, by combining the two previous facts we obtain that the density of $\mathcal{N}(0, \mathbb{I})$ is $f_Z(z) = \frac{1}{(2\pi)^{\frac{p}{2}}}\exp\left(-\frac{1}{2}z^T z\right)$. By the Spectral Theorem, Σ admits a square root $\Sigma^{\frac{1}{2}}$. Furthermore, since Σ is non-singular then $\Sigma^{\frac{1}{2}}$ is also non-singular. Now, from Proposition (3.2.3) we have that $Y \overset{d}{=} \Sigma^{\frac{1}{2}}Z + \mu \sim \mathcal{N}(\mu, \Sigma)$. By the change of variables formula we obtain:

$$f_Y(y) = f_{\Sigma^{\frac{1}{2}}Z+\mu}(y) = \left|\Sigma^{-\frac{1}{2}}\right| f_Z\left(\Sigma^{-\frac{1}{2}}(y-\mu)\right) = \frac{1}{(2\pi)^{\frac{p}{2}}|\Sigma|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}(y-\mu)^T\Sigma^{-1}(y-\mu)\right). \tag{3.14}$$

□

We now present an important Theorem that will be the basis for Principal Components Analysis. This theorem tells us what happens if we do a change of basis and express $Y \sim \mathcal{N}(\mu, \Sigma)$ in the basis of eigenvectors of Σ. As a result, we observe that the new coordinates become independent Gaussians with means $u_j^T\mu$ (the coordinates of μ in the U basis) and variances λ_j.

Theorem 3.2.6 (Karhunen-Loève Expansion)**.** *Let $\mu \in \mathbb{R}^p$ and $\Sigma \in \mathbb{R}^{p\times p}$ be a covariance with spectral decomposition,*

$$\Sigma = U\Lambda U^T = \begin{pmatrix} u_1 & \cdots & u_p \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{pmatrix} \begin{pmatrix} u_1^T \\ \vdots \\ u_p^T \end{pmatrix} = \sum_{j=1}^{p} \lambda_j u_j u_j^T, \tag{3.15}$$

where $\{u_j\}_{j=1}^p$ are the eigenvectors and $\{\lambda_j\}_{j=1}^p$ the eigenvalues. Then:

$$Y \sim \mathcal{N}(\mu,\Sigma) \iff Y = \mu + \sum_{j=1}^{p} \lambda_j^{\frac{1}{2}} Z_j u_j = \sum_{j=1}^{p} \left(\lambda_j^{\frac{1}{2}} Z_j + u_j^T \mu\right) u_j, \qquad Z_j \overset{iid}{\sim} \mathcal{N}(0,1).$$

Proof. Recall that $Y \sim \mathcal{N}(\mu,\Sigma) \iff Y = \Sigma^{\frac{1}{2}} W + \mu = U\Lambda^{\frac{1}{2}} U^T W + \mu$ for $W \sim \mathcal{N}(0, \mathbb{I}_{p\times p})$. Furthermore, defining $Z = U^T W$ we get $Z \sim \mathcal{N}(0, U^T U) \equiv \mathcal{N}(0, \mathbb{I}_{p\times p})$. Thus:

$$\Sigma^{\frac{1}{2}} W + \mu = U\Lambda^{\frac{1}{2}} Z + \mu = \begin{pmatrix} u_1 & \cdots & u_p \end{pmatrix} \begin{pmatrix} \lambda_1^{\frac{1}{2}} & & \\ & \ddots & \\ & & \lambda_p^{\frac{1}{2}} \end{pmatrix} \begin{pmatrix} Z_1 \\ \vdots \\ Z_p \end{pmatrix} + \mu = \mu + \sum_{j=1}^{p} \lambda_j^{\frac{1}{2}} Z_j u_j,$$

for $Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. We finally note that,

$$\mu = U^T U \mu = \begin{pmatrix} u_1 & \cdots & u_p \end{pmatrix} \begin{pmatrix} u_1^T \\ \vdots \\ u_p^T \end{pmatrix} \mu = \sum_{j=1}^{n} (u_j^T \mu) u_j.$$

□

Proposition 3.2.7. *If $Y \sim \mathcal{N}(\mu,\Sigma)$, then supp$(Y) = \mathcal{R}(\Sigma) + \mu$.*

Proof. Assume $\mu = 0$ wlog. When Σ is of full rank, $\mathcal{N}(\mu,\Sigma)$ admits a density whose form shows that any open ball is assigned positive probability. Now, for the possibility of the reduced rank case let $r = \text{rank}(\Sigma)$ where $y \in \mathcal{R}(\Sigma)$ and $\epsilon > 0$. Then observe that since $y \in \mathcal{R}(\Sigma)$ and given Lemma (2.3.3) giving us that $\text{supp}(Y) \subseteq \mathcal{R}(\Sigma)$ we have $H_Y Y = Y$ a.s. and $H_Y y = y$. Thus $\mathbb{P}(\|Y - y\| < \epsilon) = \mathbb{P}(\|H_Y Y - H_Y y\| < \epsilon)$. Now, using the Karhunen-Loève Expansion (3.2.6) we find:

$$\mathbb{P}(\|Y-y\| < \epsilon) = \mathbb{P}(\|H_Y Y - H_Y y\| < \epsilon) = \mathbb{P}\left(\left\| \begin{pmatrix} \lambda_1^{\frac{1}{2}} Z_1 \\ \vdots \\ \lambda_r^{\frac{1}{2}} Z_r \end{pmatrix} - \begin{pmatrix} y^T u_1 \\ \vdots \\ y^T u_r \end{pmatrix} \right\| < \epsilon \right) = \mathbb{P}(\|W - w\| < \epsilon) > 0,$$

where in the last equality we have used the fact that $W = \left(\lambda_1^{\frac{1}{2}} Z_1, ..., \lambda_r^{\frac{1}{2}} Z_r\right)^T \sim \mathcal{N}\big(0, \text{diag}(\lambda_1, ..., \lambda_r)\big)$ is a Gaussian on $\mathbb{R}^r$ with non-singular covariance. So we obtain $y \in \mathcal{R}(\Sigma)$ implying that $y \in \text{supp}(Y)$, and thus $\mathcal{R}(\Sigma) \subseteq \text{supp}(Y)$. □

We now generalize the probability density of a Gaussian in the case where $\Sigma_{p\times p}$ is singular. In this particular case, the $\mathcal{N}(\mu,\Sigma)$ distribution does not admit a density with respect to the Lebesgue measure on $\mathbb{R}^p$. Note that the support of $\mathcal{N}(\mu,\Sigma)$ is $\mathcal{R}(\Sigma) + \mu$. This is an affine set of dimension $r = \text{rank}(\Sigma)$ and thus it admits an

r-dimensional Lebesgue volume measure. Let us now define a density with respect to the Lebesgue measure on the support.

Proposition 3.2.8. *Let $X \sim \mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$. Then X admits a probability density with respect to the Lebesgue measure on $\mathcal{R}\left(\Sigma^{\frac{1}{2}}\right)+\mu$ given by:*

$$f_X(x) = \frac{1}{\prod_{j=1}^{r}(2\pi\lambda_j)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{\dagger}(x-\mu)\right), \tag{3.16}$$

for $x \in \mathcal{R}(\Sigma)+\mu$ and $r = \operatorname{rank}(\Sigma) \leq p$ and $\{\lambda_1, ..., \lambda_r\}$ are the non-zero eigenvalues of Σ.

Proof. The full rank case is already established and thus consider $r = \operatorname{rank}(\Sigma) < p$. We start with the case $\mu = 0$ and $\Sigma = \operatorname{diag}(\lambda_1, ..., \lambda_r, 0, ..., 0)$, for $\lambda_j > 0$, $j = 1, ..., r$. Given $y \in \mathcal{R}(\Sigma)$ we let $N_\epsilon(y) = \prod_{i=1}^{p}(y_i - \epsilon, y_i + \epsilon)$ be the open rectangle of sidelength $2\epsilon > 0$ centred at y. Then:

$$\mathbb{P}(X \in N_\epsilon(y)) = \mathbb{P}\left(\bigcap_{i=1}^{p}\{X_i \in (y_i-\epsilon, y_i+\epsilon)\}\right) = \underbrace{\left(\prod_{i=1}^{r}\int_{y_i-\epsilon}^{y_i+\epsilon}\frac{e^{-\frac{x_i^2}{2\lambda_i}}}{\sqrt{2\pi\lambda_i}}dx_i\right)}_{:=A(\epsilon)}\underbrace{\left(\prod_{j=r+1}^{p}\mathbb{P}(|Y_j - y_j| < \epsilon)\right)}_{:=B(\epsilon)}.$$

However, for $j > r$ we have that $y_j = 0$ and $Y_j = 0$ a.s. and so $B(\epsilon) = 1$. This establishes the form of the density in the mean zero and diagonal covariance case. For the general case $\mu \neq 0$, $\Sigma = U\Lambda U^T$ we note that $\mathbb{P}(Y \in A(\epsilon)) = \mathbb{P}(U^T(Y-\mu) \in U^T(A(\epsilon)-\mu))$ and $U^T(Y-\mu) \sim \mathcal{N}(0, U^T\Sigma U) \equiv \mathcal{N}(0, \operatorname{diag}(\lambda_1, ..., \lambda_r, 0, ..., 0))$. So the density is obtained by the change of variables $x \mapsto U^T(x-\mu)$ and observing that the term that will appear in the exponential's quadratic form $U\operatorname{diag}(\lambda_1^{-1}, ..., \lambda_r^{-1}, 0, ..., 0)U^T$ equals $\Sigma^{\dagger}$, by definition of the pseudoinverse. □

3.3 Regression Representation and Gaussian Regression

Since linear dependence is the only king of dependence between two jointly Gaussian random vectors X and Y, we may want to know what happens if we extract from X the part that is a perfect linear function of Y. In particular we would like to know if the corresponding remainder is independent of Y.

Theorem 3.3.1 (Regression Representation)**.** *Let $\mu \in \mathbb{R}^{n+m}$ and Σ be a covariance on $\mathbb{R}^{n+m}$ expressed in block form as:*

$$\mu = \begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix} \quad \text{and,} \quad \Sigma = \begin{pmatrix}\Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y\end{pmatrix}. \tag{3.17}$$

If X and Y are random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, then the following two statements are equivalent:

- $\begin{pmatrix}X \\ Y\end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$.
- $X = \mu_X + \Sigma_{XY}\Sigma_Y^{\dagger}(Y-\mu_Y)+\epsilon$ *with* $Y \sim \mathcal{N}(\mu_Y, \Sigma_Y)$, $\epsilon \sim \mathcal{N}(0, \Sigma_X - \Sigma_{XY}\Sigma_Y^{\dagger}\Sigma_{XY}^T)$ *and* $\epsilon \perp Y$.

Proof. We will prove this theorem in the case where $\mu = 0$ and $\Sigma \succ 0$. The general case may be established by using the Support and Covariance Lemma (2.3.3). First, since $\Sigma \succ 0$ we have $\Sigma_Y^{\dagger} = \Sigma_Y^{-1}$.

($\Longleftarrow$) Note that if X is defined as stipulated by the representation then,

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \Sigma_{XY}\Sigma_Y^{-1}Y + \epsilon \\ Y \end{pmatrix} = \begin{pmatrix} \mathbb{I}_{n\times n} & \Sigma_{XY}\Sigma_Y^{-1} \\ 0 & \mathbb{I}_{m\times m} \end{pmatrix} \begin{pmatrix} \epsilon \\ Y \end{pmatrix}, \tag{3.18}$$

where the conditions on ϵ and Y imply that:

$$\begin{pmatrix} \epsilon \\ Y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T & 0 \\ 0 & \Sigma_Y \end{pmatrix}\right). \tag{3.19}$$

This implies that $(X^T \;\; Y^T)^T$ is jointly normally distributed with mean zero and covariance,

$$\begin{pmatrix} \mathbb{I}_{n\times n} & \Sigma_{XY}\Sigma_Y^{-1} \\ 0 & \mathbb{I}_{m\times m} \end{pmatrix} \begin{pmatrix} \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T & 0 \\ 0 & \Sigma_Y \end{pmatrix} \begin{pmatrix} \mathbb{I}_{n\times n} & 0 \\ \Sigma_Y^{-1}\Sigma_{XY}^T & \mathbb{I}_{m\times m} \end{pmatrix}$$
$$= \begin{pmatrix} \mathbb{I}_{n\times n} & \Sigma_{XY}\Sigma_Y^{-1} \\ 0 & \mathbb{I}_{m\times m} \end{pmatrix} \begin{pmatrix} \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T & 0 \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} = \Sigma.$$

($\Longrightarrow$) We show that if $W = (X^T \;\; Y^T)^T \sim \mathcal{N}(0, \Sigma)$ then (i) $X - \Sigma_{XY}\Sigma_Y^{-1}Y \perp Y$ and (ii) $X - \Sigma_{XY}\Sigma_Y^{-1}Y \sim \mathcal{N}(0, \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T)$. To this aim we note that $X - \Sigma_{XY}\Sigma_Y^{-1}Y = \begin{pmatrix} \mathbb{I}_{n\times n} & -\Sigma_{XY}\Sigma_Y^{-1} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = QW$ and $Y = \begin{pmatrix} 0 & \mathbb{I}_{m\times m} \end{pmatrix} W = PW$. Therefore: (i) $X - \Sigma_{XY}\Sigma_Y^{-1}Y \perp Y$ if and only if $P\Sigma Q^T = 0$ which is verified since,

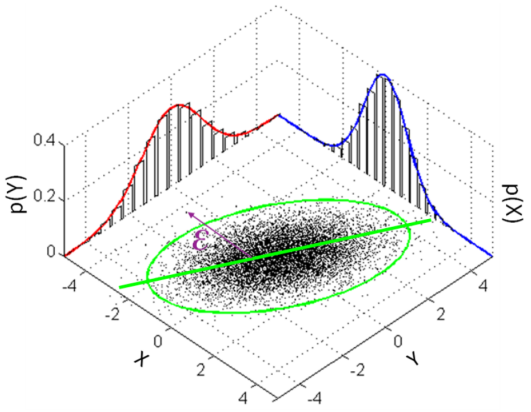
$$P\Sigma Q^T = \begin{pmatrix} 0 & \mathbb{I}_{m\times m} \end{pmatrix} \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} \begin{pmatrix} \mathbb{I}_{n\times n} \\ -\Sigma_Y^{-1}\Sigma_{XY}^T \end{pmatrix} = \begin{pmatrix} 0 & \mathbb{I}_{m\times m} \end{pmatrix} \begin{pmatrix} \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T \\ 0 \end{pmatrix} = 0. \tag{3.20}$$

Furthermore for (ii) we have that $QW \sim \mathcal{N}(0, Q\Sigma Q^T)$ and importing our previous calculation of ΣQ^T we find,

$$Q\Sigma Q^T = \begin{pmatrix} \mathbb{I}_{n\times n} & -\Sigma_{XY}\Sigma_Y^{-1} \end{pmatrix} \begin{pmatrix} \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T \\ 0 \end{pmatrix} = \Sigma_X - \Sigma_{XY}\Sigma_Y^{-1}\Sigma_{XY}^T. \tag{3.21}$$

□

The above theorem provides intuition as to how the pair $(X^T \;\; Y^T)^T$ arises. First, we affinely transform a realisation of $Y \sim \mathcal{N}(\mu_Y, \Sigma_Y)$ by a deterministic affine transformation. Then, we add an independent (of Y) zero mean Gaussian random variable ϵ. Considering the simplest case where X and Y are scalar, and $\mathrm{Var}(Y) > 0$. Then the representation in the theorem reduces to the familiar expression $X = \beta_0 + \beta Y + \epsilon$ where $Y \sim \mathcal{N}(\mu_Y, \mathrm{Var}(Y))$ independent of $\epsilon \sim \mathcal{N}(0, \mathrm{Var}(\epsilon))$. Furthermore, $\beta_0 = \mu_X - \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}\mu_Y$ is called the **intercept**, $\beta = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}$ is called the **regression coefficient** and ϵ is called the **error** or **innovation** that is *homoskedastic* in that $\mathrm{Var}(\epsilon) = \mathrm{Var}(X) - \frac{\mathrm{Cov}^2(X,Y)}{\mathrm{Var}(Y)}$ does not vary with Y. This explains why we call it the **regression representation**.



Theorem 3.3.2 (Gaussian Conditional Distributions)**.** *Let $(X^T \ Y^T)^T$ be jointly Gaussian, comprised of concatenated random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, with mean and covariance:*

$$\mu = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} \quad \text{and,} \quad \Sigma = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}. \tag{3.22}$$

Then:

$$X|\{Y=y\} \sim \mathcal{N}\left(\mu_X + \Sigma_{XY}\Sigma_Y^{\dagger}(y-\mu_Y), \Sigma_X - \Sigma_{XY}\Sigma_Y^{\dagger}\Sigma_{XY}^T\right). \tag{3.23}$$

Consequently, when Σ_Y is non-singular:

$$X|\{Y=y\} \sim \mathcal{N}\left(\mu_X + \Sigma_{XY}\Sigma_Y^{-1}(y-\mu_Y), \Theta_X^{-1}\right). \tag{3.24}$$

Proof. $(*)$ To ease the computations, we will prove this in terms of the precision matrix. Now, the density of $(X^T \ Y^T)^T$ may be written in terms of $\mu = (\mu_X, \mu_Y)$ and the precision matrix $\Sigma^{-1} = \Psi = \begin{pmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{pmatrix}$. We are interested in the density of $X|\{Y=y\}$ which is given by $f_{X|\{Y=y\}}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$. Note that the denominator carries no information regarding the functional dependence on the variable x. In what follows, we omit the terms which don't depend on x for the sake of simplicity (by the symbol $\overset{x}{\propto}$).

$$\begin{aligned} f_{X|\{Y=y\}}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)} &\overset{x}{\propto} \exp\left(-\frac{1}{2}(x-\mu_X)^T\Psi_{11}(x-\mu_X) - (x-\mu_X)^T\Psi_{12}(y-\mu_Y)\right) \\ &\overset{x}{\propto} \exp\left(-\frac{1}{2}x^T\Psi_{11}x + x^T(\Psi_{11}\mu_X - \Psi_{12}(y-\mu_Y))\right). \end{aligned}$$

We now recognize the form of a multivariate Gaussian density. We deduce that $X|\{Y=y\}$ is Gaussian and thus it is characterized by two parameters: the mean $\mu_{1|2}$ and its precision matrix $\Psi_{1|2}$ which we now need to identify. In general, the density of a multivariate Gaussian with mean $\mu_{1|2}$ and precision matrix $\Psi_{1|2}$ is proportional to,

$$\exp\left(-\frac{1}{2}(x-\mu_{1|2})^T\Psi_{1|2}(x-\mu_{1|2})\right) \overset{x}{\propto} \exp\left(-\frac{1}{2}x^T\Psi_{1|2}x + x^T\Psi_{1|2}\mu_{1|2}\right).$$

Thus, by identification we have that for the precision matrix $\Psi_{1|2} = \Psi_{11}$ implying that $\Sigma_X = \Theta_X^{-1}$ and the mean is $\mu_{1|2} = \mu_X - \Psi_{11}^{-1}\Psi_{12}(y-\mu_Y)$. Finally plugging in the coordinates of the precision matrix from Lemma (2.4.2) yields the result. □

We highlight that the covariance sub-block Σ_X of Σ is the covariance matrix of the marginal law of X, and the precision sub-block Θ_X of Θ is the precision matrix of the conditional law of $X|Y$.

Theorem 3.3.3 (Gaussian Conditional Independence)**.** *Let $(X^T \ Y^T \ Z^T)^T$ be an $(n+m+p)$-dimensional Gaussian vector with non-singular covariance matrix $\Sigma \succ 0$ and precision matrix $\Theta = \Sigma^{-1}$, expressed in block format as:*

$$\Sigma = \begin{pmatrix} \Sigma_X & \Sigma_{XY} & \Sigma_{XZ} \\ \Sigma_{XY}^T & \Sigma_Y & \Sigma_{YZ} \\ \Sigma_{XZ}^T & \Sigma_{YZ}^T & \Sigma_Z \end{pmatrix} \quad \text{and,} \quad \Sigma^{-1} = \Theta = \begin{pmatrix} \Theta_X & \Theta_{XY} & \Theta_{XZ} \\ \Theta_{XY}^T & \Theta_Y & \Theta_{YZ} \\ \Theta_{XZ}^T & \Theta_{YZ}^T & \Theta_Z \end{pmatrix}. \tag{3.25}$$

Then:

$$\{X \perp Y\}|Z \iff \Theta_{XY} = 0. \tag{3.26}$$

Proof. Set $W = (X^T \;\; Y^T)^T$. Since $(X^T \;\; Y^T \;\; Z^T)^T$ is Gaussian with non-singular covariance, we conclude that $F_{X,Y|Z} \equiv F_{W|Z}$ is Gaussian with non-singular covariance $\Gamma = \Theta_W^{-1}$ [by (3.24)]. In turns, this equals

$$\Gamma = \Theta_W^{-1} = \begin{pmatrix} \Theta_X & \Theta_{XY} \\ \Theta_{XY}^T & \Theta_Y \end{pmatrix}^{-1}, \tag{3.27}$$

i.e. the inverse of the top left $(n+m)\times(n+m)$ sub-matrix of Θ. We furthermore observe that:

- If $\Theta_{XY} = 0$ then:

$$\Gamma = \begin{pmatrix} \Theta_X & 0 \\ 0 & \Theta_Y \end{pmatrix}^{-1} = \begin{pmatrix} \Theta_X^{-1} & 0 \\ 0 & \Theta_Y^{-1} \end{pmatrix}, \tag{3.28}$$

which implies that $F_{X,Y|Z}$ factorizes as $F_{X|Z}F_{Y|Z}$ since $\Theta_{XY} = 0$ implies that we have independence in the Gaussian case.

- If $F_{X,Y|Z} = F_{X|Z}F_{Y|Z}$, then using the equivalence in the independence of Gaussians we know that Γ is block-diagonal. Therefore, its inverse is also block-diagonal, implying that $\Theta_{XY} = 0$. □

Let us give a regression interpretation of the Gaussian conditional independence. From our last theorem, we have that $X = \mu_X + \Sigma_{XZ}\Sigma_Z^\dagger(Z - \mu_Z) + \epsilon_X$ and $Y = \mu_Y + \Sigma_{YZ}\Sigma_Z^\dagger(Z - \mu_Z) + \epsilon_Y$ where $Z \sim \mathcal{N}(\mu_Z, \Sigma_Z)$, $\epsilon_X \sim \mathcal{N}(0, \Sigma_X - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{XZ}^T$ and $\epsilon_Y \sim \mathcal{N}(0, \Sigma_Y - \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{YZ}^T)$ such that $\epsilon_X \perp Z$, $\epsilon_Y \perp Z$. We notice that $\epsilon_X \perp \epsilon_Y$ in general, however we may consider the following proposition.

Proposition 3.3.4 (Regression and Gaussian Conditional Independence)**.** *In the context of the Gaussian regression representation, we have that:*

$$\begin{pmatrix} \epsilon_X \\ \epsilon_Y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \Theta_X & \Theta_{XY} \\ \Theta_{XY}^T & \Theta_Y \end{pmatrix}^{-1}\right). \tag{3.29}$$

Therefore:

$$\epsilon_X \perp \epsilon_Y \iff \Theta_{XY} = 0 \iff \{X \perp Y\}|Z. \tag{3.30}$$

Proof. Assume wlog that the mean is zero. By the regression representation:

$$\begin{pmatrix} \epsilon_X \\ \epsilon_Y \end{pmatrix} = \begin{pmatrix} X - \Sigma_{XZ}\Sigma_Z^{-1}Z \\ Y - \Sigma_{YZ}\Sigma_Z^{-1}Z \end{pmatrix} = \begin{pmatrix} \mathbb{I}_{n\times n} & 0 & -\Sigma_{XZ}\Sigma_Z^{-1} \\ 0 & \mathbb{I}_{m\times m} & -\Sigma_{YZ}\Sigma_Z^{-1} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

Thus $(\epsilon_X^T \;\; \epsilon_Y^T)^T$ is jointly Gaussian. The form of $\mathrm{Cov}(\epsilon_X)$ and $\mathrm{Cov}(\epsilon_Y)$ has already been established. The cross-covariance is given by:

$$\begin{aligned}
\mathrm{Cov}(\epsilon_X, \epsilon_Y) = \mathbb{E}\big(\epsilon_X\epsilon_Y^T\big) &= \mathbb{E}\big[(X - \Sigma_{XZ}\Sigma_Z^{-1}Z)(Y - \Sigma_{YZ}\Sigma_Z^{-1}Z)\big] \\
&= \mathbb{E}\big(XY^T\big) - \mathbb{E}\big[X(\Sigma_{YZ}\Sigma_Z^{-1}Z)^T\big] - \mathbb{E}\big[\Sigma_{XZ}\Sigma_Z^{-1}ZY^T\big] + \mathbb{E}\big[\Sigma_{XZ}\Sigma_Z^{-1}Z(\Sigma_{YZ}\Sigma_Z^{-1}Z)^T\big] \\
&= \mathbb{E}\big(XY^T\big) - \mathbb{E}\big(XZ^T\big)\Sigma_Z^{-1}\Sigma_{YZ}^T - \Sigma_{XZ}\Sigma_Z^{-1}\mathbb{E}\big(ZY^T\big) + \Sigma_{XZ}\Sigma_Z^{-1}\mathbb{E}\big(ZZ^T\big)\Sigma_Z^{-1}\Sigma_{YZ}^T \\
&= \Sigma_{XY} - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T + \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_Z\Sigma_Z^{-1}\Sigma_{YZ}^T \\
&= \Sigma_{XY} - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T.
\end{aligned}$$

As a result:

$$\mathrm{Cov}\left[\begin{pmatrix} \epsilon_X \\ \epsilon_Y \end{pmatrix}\right] = \begin{pmatrix} \Sigma_X - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_{XY} - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \\ \Sigma_{XY}^T - \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_Y - \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \end{pmatrix}. \tag{3.31}$$

We finally need to show that (3.31) $= \begin{pmatrix} \Theta_X & \Theta_{XY} \\ \Theta_{XY}^T & \Theta_Y \end{pmatrix}^{-1}$. To do this, we recall (3.24) to notice that this equals Θ_W^{-1}, i.e. the covariance of $W = (X^T \;\; Y^T)^T$ given Z. We can then use (3.24) to calculate it explicitly. Namely, we may write,

$$W|\{Z=z\} \sim \mathcal{N}\big(\Sigma_{WZ}\Sigma_Z z, \Theta_W^{-1}\big) \equiv \mathcal{N}\big(\Sigma_{WZ}\Sigma_Z z, \Sigma_W - \Sigma_{WZ}\Sigma_Z^{-1}\Sigma_{WZ}^T\big).$$

We can now write out:

$$\begin{aligned}
\Sigma_W - \Sigma_{WZ}\Sigma_Z^{-1}\Sigma_{WZ}^T &= \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} - \mathbb{E}\left[\begin{pmatrix} XZ^T \\ YZ^T \end{pmatrix}\right]\Sigma_Z^{-1}\mathbb{E}\left[\begin{pmatrix} ZX^T & ZY^T \end{pmatrix}\right] \\
&= \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} - \begin{pmatrix} \Sigma_{XZ} \\ \Sigma_{YZ} \end{pmatrix}\Sigma_Z^{-1}\begin{pmatrix} \Sigma_{XZ}^T & \Sigma_{YZ}^T \end{pmatrix} \\
&= \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix} - \begin{pmatrix} \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \\ \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \end{pmatrix} \\
&= \begin{pmatrix} \Sigma_X - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_{XY} - \Sigma_{XZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \\ \Sigma_{XY}^T - \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{XZ}^T & \Sigma_Y - \Sigma_{YZ}\Sigma_Z^{-1}\Sigma_{YZ}^T \end{pmatrix},
\end{aligned}$$

and we are done. The second part of the proposition follows from the fact that $\epsilon_X \perp \epsilon_Y$ if and only if the covariance of $\begin{pmatrix} \epsilon_X^T & \epsilon_Y^T \end{pmatrix}^T$ is block-diagonal. □

We now end this section by going beyond Gaussian regression, and consider the so-called *best linear prediction*. Let X and Y be random vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively and write,

$$\mathrm{Cov}(X,Y) = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}.$$

The problem at hand is to find a way to best predict Y using an affine function of X. Formally, we seek an affine map $x \mapsto B_* x + \beta_* \in \mathbb{R}^m$ where $x \in \mathbb{R}^n$ such that $\mathbb{E}\|Y - \beta_* - B_* X\|^2 \leq \mathbb{E}\|Y - \beta - BX\|^2$ is minimal over all choices of $\beta \in \mathbb{R}^m$ and $B \in \mathbb{R}^{m\times n}$. Such a map is called a *best linear predictor*.
Recall that when X and Y are Gaussian, we saw that the conditional expectation of $Y|X$ is an affine transformation of X. Thus it is the *de facto* best linear predictor, seeing as it is the best predictor: $\beta_* = \mu_Y - \Sigma_{YX}\Sigma_X^\dagger \mu_X$ and $B_* = \Sigma_{YX}\Sigma_X^\dagger$. Let us now see what happens beyond the jointly Gaussian vectors.

Theorem 3.3.5 (Best Linear Prediction)**.** *Regardless of the joint law of X and Y, the best linear predictor of $Y|X$ is the same as in the jointly Gaussian case.*

Proof. Writing $\mathbb{E}\|Y - \beta - BX\|^2 = \sum_{i=1}^m \mathbb{E}\big[(Y_i - \beta_i - B_i X)^2\big]$ for $B_i \in \mathbb{R}^{1\times n}$ is the i-th row of B, we see that it is sufficient to consider the scalar Y and $B \in \mathbb{R}^{1\times n}$. Now, the key step is to show that X and $\epsilon := Y - \beta_* - B_* X$ are uncorrelated. However, we have already done this in the proof of the Gaussian regression representation.
Now, $\mathbb{E}\big[(Y - \beta - BX)^2\big]$ may be written as:

$$\mathrm{Var}(Y - \beta - BX) + \big[\mathbb{E}(Y - \beta - BX)\big]^2 = \mathrm{Var}(Y - BX) + (\mu_Y - \beta - B\mu_X)^2.$$

We can immediately check that $(\beta, B) = (\beta_*, B_*)$ minimizes the second term (yielding zero). Let us now check that this choice minimizes also the first term:

$$\begin{aligned}\mathrm{Var}(Y-BX) = \mathrm{Var}(Y-\beta_*-B_*X+\beta_*+B_*X-B_X) &= \mathrm{Var}(\epsilon+\beta_*+(B_*-B)X)\\ &= \mathrm{Var}(\epsilon+(B_*-B)X)\\ &= \mathrm{Var}(\epsilon)+\mathrm{Var}((B_*-B)X)\\ &= \mathrm{Var}(\epsilon)+(B_*-B)\Sigma_X(B_*-B)^T.\end{aligned}$$

The first term does not depend on (β, B) and the second is clearly minimized at $B = B_*$ since we have $(B_*-B)\Sigma_X(B_*-B)^T \succeq 0$ for all B. □

Definition 3.3.6. *Let $X = (X_1, ..., X_p)^T$ be a random p-vector with covariance $\Sigma \succ 0$ and precision matrix $\Theta = \{\theta_{ij}\} = \Sigma^{-1}$. The **partial correlation** of X_i and X_j given $\{X_k\}_{k\neq i,j}$ is defined as,*

$$\rho_{ij|k\neq i,j} := -\frac{\theta_{ij}}{\sqrt{\theta_{ii}\theta_{jj}}}. \tag{3.32}$$

It expresses the correlation between X_i and X_j when controlling for the linear effects of the remaining variables $\{X_k\}_{k\neq i,j}$ on X_i and X_j. Now, when the partial correlation vanishes, the corresponding variables are called *partially uncorrelated*. When $X = (X_1^T, X_2^T, X_3^T)^T$ is a partitioned Gaussian, X_1 and X_2 are uncorrelated given X_3 if and only if $\{X_1 \perp X_2\}|X_3$. However, when $X = (X_1^T, X_2^T, X_3^T)^T$ is possibly non-Gaussian, X_1 and X_2 are partially uncorrelated given X_3 if and only if $X_1 - X_{1|3}^*$ and $X_2 - X_{2|3}^*$ are uncorrelated, where $X_{k|3}^*$ is the best linear predictor of X_k given X_3 $(k = 1, 2)$. This last case could be given the title of *regression and Gaussian conditional independence theorem* .

3.4 Quadratic Forms - The χ^2 Distribution

Let us first recall the construction of the χ_1^2-distribution. Starting with $Z \sim \mathcal{N}(0,1)$, we note that $F_{Z^2}(y) = 0$ if $y < 0$ and for $y \geq 0$ the CDF is,

$$F_{Z^2}(y) = \mathbb{P}\big(Z^2 \leq y\big) = \mathbb{P}\big(|Z| \leq \sqrt{y}\big) = \mathbb{P}\big(-\sqrt{y} \leq Z \leq \sqrt{y}\big) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = \Phi(\sqrt{y}) - (1-\Phi(\sqrt{y})) = 2\Phi(\sqrt{y}) - 1.$$

Differentiating yields the PDF,

$$f_{Z^2}(y) = 2\frac{d}{dy}\Phi(\sqrt{y}) = 2\frac{d}{d\sqrt{y}}\Phi(\sqrt{y})\frac{d}{dy}\sqrt{y} = \frac{1}{\sqrt{2\pi}}e^{-\frac{y}{2}}y^{-\frac{1}{2}}.$$

The MGF is,

$$M_{Z^2}(t) = \frac{1}{\sqrt{2\pi}}\int_0^\infty e^{ty}y^{-\frac{1}{2}}e^{-\frac{y}{2}}dy = \frac{1}{\sqrt{2\pi}}\int_0^\infty y^{-\frac{1}{2}}e^{-\frac{(1-2t)y}{2}}dy,$$

and provided that $1-2t > 0$ we have that the integral is finite and we can make the substitution $u = (1-2t)y$ to get:

$$M_{Z^2}(t) = (1-2t)^{-\frac{1}{2}}\int_0^\infty \frac{1}{\sqrt{2\pi}}u^{-\frac{1}{2}}e^{-\frac{u}{2}}du = (1-2t)^{-\frac{1}{2}}, \quad t < \frac{1}{2}.$$

The corresponding distribution is called the $\boldsymbol{\chi_1^2}$**-distribution**.

Proposition 3.4.1. *If $Z \sim \mathcal{N}(0, \mathbb{I}_{k\times k})$, then the MGF of the random variable Z^TZ is given by:*

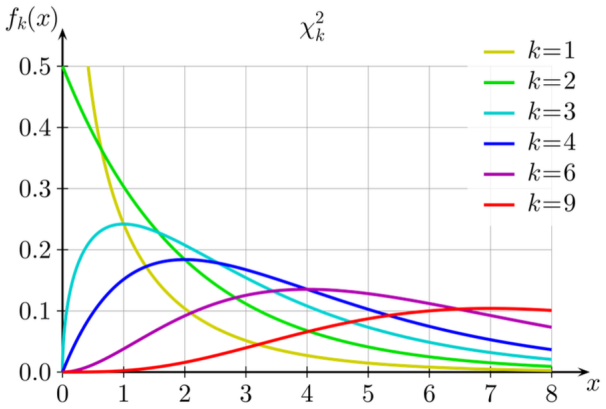
$$M_{Z^TZ}(t) = (1-2t)^{-\frac{k}{2}}, \quad t < \frac{1}{2}. \tag{3.33}$$

Proof. $(*)$ Note that $Z^TZ = \sum_{i=1}^k Z_i^2$ where Z_i are i.i.d. standard Gaussians. Thus, using independence and the MGF of the univariate χ_1^2, we find:

$$M_{Z^TZ}(t) = M_{\sum_{i=1}^k Z_i^2}(t) = \prod_{i=1}^k M_{Z_i^2}(t) = \left((1-2t)^{-\frac{1}{2}}\right)^k = (1-2t)^{-\frac{k}{2}}. \tag{3.34}$$

□

The corresponding distribution is called the **χ_k^2-distribution** with k degrees of freedom. As some useful facts it holds that $\mathbb{E}(Z^TZ) = k$ and $\mathrm{Var}(Z^TZ) = 2k$.



We finally present some useful facts about the Chi-squared distribution.

Proposition 3.4.2. *If $X \sim \mathcal{N}(0, \Sigma_{p\times p})$ and Σ invertible, then $X^T\Sigma^{-1}X \sim \chi_p^2$.*

Proof. $(*)$ Let $Y = \Sigma^{-\frac{1}{2}}X$. Since Σ is invertible then $\Sigma^{-\frac{1}{2}}$ exists, and Y follows a multivariate normal distribution with mean 0 and covariance matrix $\mathbb{I}_p$. Therefore $Y_i^2 \sim \chi_1^2$ for $i = 1,...,p$. Furthermore, the sum of independent χ^2 variables follows a χ^2-distribution with degrees of freedom equal to the sum of the individual degrees of freedom. Mathematically this gives:

$$X^T\Sigma^{-1}X = Y^TY = \sum_{i=1}^p Y_i^2 \sim \sum_{i=1}^p \chi_1^2 \equiv \chi_p^2. \tag{3.35}$$

□

Proposition 3.4.3. *If $X \sim \mathcal{N}(0, \mathbb{I}_p)$ and H is a projection, then $X^THX \sim \chi^2_{\mathrm{rank}(H)}$.*

Proof. $(*)$ We employ the spectral decomposition theorem and we may write $H = UJU^T$ where J is a diagonal matrix with only zeroes and ones, with r non-zero entries. Then, $X^THX = X^T(UJU^T)X = (U^TX)^TJ(U^TX)$. Now note that $Y = U^TX$ follows the same law as X since U is orthogonal. Therefore:

$$X^THX = Y^TJY = \sum_{i:J_{ii}\neq 0} Y_i^2 \sim \sum_{i:J_{ii}\neq 0} \chi_1^2 \equiv \chi^2_{\mathrm{rank}(H)}, \tag{3.36}$$

where we have used the fact that $|i : J_{ii} \neq 0| = \mathrm{rank}(H)$. □

Proposition 3.4.4. *If $X \sim \mathcal{N}(0, \Sigma_{p\times p})$, then $X^T\Sigma^\dagger X \sim \chi^2_{\text{rank}(\Sigma)}$.*

Proof. $(*)$ Denote the eigenvalue decomposition of Σ as $\Sigma = UDU^T$ where D is a diagonal matrix with the non-zero eigenvalues of Σ on the diagonal. Now consider the transformation $Y = U^T X$. Then, $Y \sim \mathcal{N}(0, U^T\Sigma U) \equiv \mathcal{N}(0, D)$. That is, Y_i are independent centred Gaussians with variance D_{ii}. Thus:

$$X^T\Sigma^\dagger X = Y^T D^\dagger Y = \sum_{i:D_{ii}\neq 0} D_{ii}^{-1} Y_i^2 \sim \sum_{i:D_{ii}\neq 0} \chi_1^2 \equiv \chi^2_{\text{rank}(\Sigma)}, \tag{3.37}$$

where we have used the fact $|i : D_{ii} \neq 0| = \text{rank}(\Sigma)$. □

3.5 Concentration of Measure

In this last section, we present a surprising yet very important phenomenon for a Gaussian measure. We first present a Lemma consisting of a Chernoff Bound for a chi-squared distribution.

Lemma 3.5.1 (Chernoff Bound for χ^2_p)**.** *Given $Q \sim \chi^2_p$ one has:*

$$\mathbb{P}\left(\left|\frac{Q-p}{p}\right| > u\right) \leq 2\exp\left(-\frac{pu^2}{8}\right), \quad \forall u \in (0,1). \tag{3.38}$$

Proof. A Chernoff bound combines the MGF with Markov's Inequality.

$$\begin{aligned}\mathbb{P}\big(Q-p > pu\big) = \mathbb{P}\left(e^{\lambda(Q-p)} > e^{\lambda pu}\right) &\leq e^{-\lambda pu}\mathbb{E}\left(e^{\lambda(Q-p)}\right) = e^{-\lambda pu}e^{-\lambda p}M_Q(\lambda) \\ &= e^{-\lambda pu}\left(\frac{e^{-\lambda}}{\sqrt{1-2\lambda}}\right)^p \\ &\leq \exp\big(-\lambda pu + 2\lambda^2 p\big),\end{aligned}$$

for all $|\lambda| < 1/4$ (in the last step we have used the inequality $e^{-\lambda}/\sqrt{1-2\lambda} \leq e^{2\lambda^2}$ which is valid for $|\lambda| < 1/4$). Now, we optimise the upper bound with respect to λ. It can be checked that $\exp\big(-\lambda pu + 2\lambda^2 p\big)$ has a global minimum at the value $\lambda^* = u/4$. Since $u \in (0,1)$, we have $|\lambda^*| < 1/4$ and we can plug λ^* into our inequality to obtain $\mathbb{P}\big(Q-p > pu\big) \leq \exp\big(-pu^2/8\big)$. Now we go in the other direction and consider,

$$\mathbb{P}\big(Q-p < -pu\big) = \mathbb{P}\big(-\theta(Q-p) > \theta pu\big) = \mathbb{P}\left(e^{-\theta(Q-p)} > e^{\theta pu}\right) \leq e^{-\theta pu}\mathbb{E}\left(e^{-\theta(Q-p)}\right) = e^{-\theta pu}e^{-\theta p}M_Q(-\theta).$$

Provided $|\theta| < 1/4$ we may define $\lambda = -\theta$ and repeat the exact same steps as before to obtain $\mathbb{P}\big(Q-p < -pu\big) \leq \exp\big(-pu^2/8\big)$. To complete the proof, we put the two inequalities together to observe that,

$$\mathbb{P}\left(\left|\frac{Q-p}{p}\right| > u\right) = \mathbb{P}\left(\frac{Q-p}{p} > u\right) + \mathbb{P}\left(\frac{Q-p}{p} < -u\right) \leq 2\exp\left(-\frac{pu^2}{8}\right).$$

□

Theorem 3.5.2 (Gaussian Concentration of Measure)**.** *Let $Z \sim \mathcal{N}(0, \mathbb{I}_{p\times p})$ and $X = \Sigma^{\frac{1}{2}}Z + \mu$ with Σ non-singular. Then, for all $\epsilon \in (0,1)$:*

$$\mathbb{P}\big(\|\Sigma^{-\frac{1}{2}}(X-\mu)\|^2 \notin [(1-\epsilon)p, (1+\epsilon)p]\big) = \mathbb{P}\big(\|Z\|^2 \notin [(1-\epsilon)p, (1+\epsilon)p]\big) \leq 2\exp\left(-\frac{\epsilon^2 p}{8}\right). \tag{3.39}$$

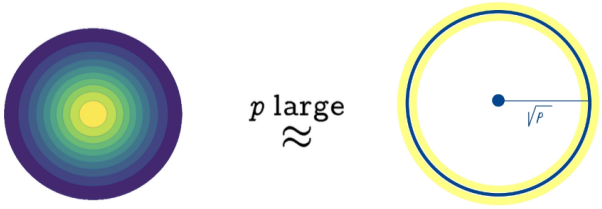
Proof. Using the Chernoff bound for the χ^2_k-distribution, we get:

$$\begin{aligned} 2\exp\left(-\frac{pu^2}{8}\right) \geq \mathbb{P}\left(\left|\frac{\|Z\|^2-p}{p}\right| > u\right) &= \mathbb{P}\left(\{\|Z\|^2 - p > pu\} \cup \{\|Z\|^2 - p < -pu\}\right) \\ &= 1 - \mathbb{P}\left(\{\|Z\|^2 - p \leq pu\} \cap \{\|Z\|^2 - p \geq -pu\}\right) \\ &= 1 - \mathbb{P}\left(\{\|Z\|^2 \leq (1+u)p\} \cap \{\|Z\|^2 \geq (1-u)p\}\right) \\ &= 1 - \mathbb{P}\left((1-u)p \leq \|Z\|^2 \leq (1+u)p\right), \end{aligned}$$

and the result follows taking $\epsilon = u \in (0,1)$. □

Upon closer inspection, the result is striking. Indeed, given $\epsilon \in (0,1)$ the bound $2\exp\left(-\epsilon^2 p/8\right)$ rapidly approaches 0 as p grows. Morally this says that **in high dimensions (p large), the realisation of $Z \sim \mathcal{N}(0, \mathbb{I}_{p\times p})$ highly concentrates near the surface of the sphere of radius $\sqrt{p}$**. In other words, the standard normal distribution in high dimension of p is close to the uniform distribution on the sphere of radius $\sqrt{p}$:

$$\mathcal{N}(0,\mathbb{I}_p) \overset{p \text{ large}}{\approx} \text{Uniform}\left(\sqrt{p}\mathbb{S}^{p-1}\right) \tag{3.40}$$



This may seem surprising given that the mode of the PDF is always at zero. Furthermore, in the case $X \sim \mathcal{N}(\mu, \Sigma_{p\times p})$ with $\Sigma = U\text{diag}(\lambda_1, ..., \lambda_p)U^T$ invertible, we have a similar concentration but this time around an ellipsoid. This is an ellipsoid centred at μ, with eigenvectors u_i of Σ as principle axes, and with principle axis lengths $2\lambda_i^{-\frac{1}{2}}\sqrt{p}$.

Chapter 4

Spherical and Elliptical Distributions

This relatively short chapter is dedicated to defining spherical random vectors and also the notion of elliptical distributions. Recall that a key feature of a Gaussian vector $X \sim \mathcal{N}(\mu, \Sigma)$ is that of the representation $X \overset{d}{=} \mu + \Sigma^{\frac{1}{2}} Z$ for $Z \sim \mathcal{N}(0, \mathbb{I})$. Let us see what happens if we replace $Z \sim \mathcal{N}(0, \mathbb{I})$ by some *spherical* random vector W.

Definition 4.0.1. *A **spherical random vector** W is one that satisfies $UW \overset{d}{=} W$ for all orthogonal U.*

We can alternatively give two other characterisations of random vectors.

Proposition 4.0.2. *All the following statements are equivalent:*
(i) W is an orthogonal random vector.
(ii) $W = \xi U$ where $U \sim \text{Unif}\{x : \|x\| = 1\}$, $\xi > 0$ is a random scalar (called the radial part) and $U \perp \xi$.
(iii) $v^T W \overset{d}{=} \|v\| W_1$, for all v where $W = (W_1, ..., W_p)^T$.

Proof. $(*)$ We prove this by showing $(i) \implies (ii) \implies (iii) \implies (i)$.

• $(i) \implies (ii)$ Let us write $W = \xi U$ where $U = W/\|W\|$ is a random unit vector and $\xi = \|W\|$. First note that the distribution of U need be rotationally invariant, and is thus necessarily the uniform distribution over the unit sphere. Now we need to show that $\xi \perp U$, i.e. for any measurable sets $A \subseteq \mathbb{R}^+$ and $B \subseteq \mathbb{S}^{p-1}$: $\mathbb{P}(\xi \in \mathcal{E}, U \in \mathcal{U}) = \mathbb{P}(\xi \in \mathcal{E})\mathbb{P}(U \in \mathcal{U})$, for suitable measurable sets $\mathcal{E} \subset \mathbb{R}^+$ and $\mathcal{U} \subset [0, 2\pi)$. Now:

$$\mathbb{P}(\xi \in \mathcal{E} | U \in \mathcal{U}) = \frac{\mathbb{P}(\|W\| \in \mathcal{E}, W/\|W\| \in \mathcal{U})}{\mathbb{P}(U \in \mathcal{U})} = \frac{\mathbb{P}(\|OW\| \in \mathcal{E}, OW/\|OW\| \in \mathcal{U})}{\mathbb{P}(OU \in \mathcal{U})} = \frac{\mathbb{P}(\|W\| \in \mathcal{E}, W/\|W\| \in O\mathcal{U})}{\mathbb{P}(U \in O\mathcal{U})}$$
$$= \mathbb{P}(\xi \in \mathcal{E} | U \in O\mathcal{U}),$$

where we have used the invariance under rotations in the second equality. Overall, this proves that the law of ξ is constant under conditioning on U, and hence establishes their independence.

• $(ii) \implies (iii)$ We represent v in polar form $v = \|v\| u$, where u is a unit vector in the direction of v. Then, $v^T W = (\|v\| u)^T W = \xi \|v\| u^T U$ and it is not difficult to see that $u^T U$ has distribution of the marginals U_i (by invariance of the uniform distribution on the sphere with respect to the coordinate system). Thus, $v^T W \overset{d}{=} \xi \|v\| U_1 = \|v\| W_1$.

• $(iii) \implies (i)$ First, recall that the distribution of all possible one-dimensional marginals uniquely characterizes the joint distribution of a random vector. That is, the distribution of X is uniquely determined by the family of distributions $v^T X$ for all vectors v. Now, for any orthogonal U and vector v, note that

$\|(v^T U)^T\| = \|v\|$ and then observe that:

$$v^T(UX) = (v^T U)X \stackrel{d}{=} \|v\| X_1 \stackrel{d}{=} v^T X,$$

for all $v \in \mathbb{R}^p$. This directly implies that $X \stackrel{d}{=} UX$. □

If a spherical law has density f, then necessarily $f(x) = f(Ux)$ for all orthogonal U. Hence $f(x) = g(\|x\|^2)$ for some $g : [0,\infty) \to [0,\infty)$.

Definition 4.0.3. *A random vector $X = (X_1, ..., X_p)^T$ is called **elliptical** with location μ and dispersion $\Sigma = AA^T$ if and only if $X \stackrel{d}{=} \mu + AW$ for W a spherical random vector, and $A_{p\times d}$ with $p \le d$.*

Elliptical distributions are affine transformations of spherical distributions, as Gaussians that are affine transformations of standard Gaussians. Now, since any spherical random vector is represented as ξU where $\xi \perp U$ then we conclude that X is elliptical if and only if $X \stackrel{d}{=} \mu + \xi AU$ where $U \sim \text{Unif}\{x : \|x\| = 1\}$. Notice that in the elliptical case (contrary to the spherical case) the radial part ξ is unique up to rescaling since $\xi A = (\xi/c)(cA)$ for any $c \neq 0$.

Let us now turn to elliptical densities, and consider $X = \mu + AW$ for a spherical random vector W in $\mathbb{R}^d$. If A is non-singular then $W = A^{-1}(X - \mu)$, which implies that if W has a density $\psi(u) = g_\psi(\|u\|^2)$ then X has density:

$$f_X(x) = \frac{1}{|\Sigma|^{\frac{1}{2}}} \psi\left(\Sigma^{-\frac{1}{2}}(x-\mu)\right) = \frac{1}{|\Sigma|^{\frac{1}{2}}} g_\psi\left((x-\mu)^T\Sigma^{-1}(x-\mu)\right), \tag{4.1}$$

where $\Sigma = AA^T$. We will name ψ the **generating density** of f_X.

Remark 4.0.4. *The density depends on x only via $(x-\mu)^T\Sigma^{-1}(x-\mu)$ hence it is constant on ellipsoids (i.e. it has elliptical contours). Since $X = \mu + cAc^{-1}W$ for any $c \neq 0$, the dispersion matrix Σ is not unique (it is unique only up to rescaling). Note also that we have not assumed existence of second (or even first) moments. If a first moment exists then μ is the expectation, and if second moments exist then some rescaled version of Σ is the covariance.*

We now provide a couple of examples of elliptical distributions.

Example 4.0.5. *All spherical distributions and Gaussian distributions are also elliptical.*

Example 4.0.6. *Gaussian variance mixtures $Y = \mu + \sqrt{\xi}AZ$, where $0 < \xi \perp Z \sim \mathcal{N}(0, \mathbb{I})$ are elliptical [one can check that $\text{Cov}(Y) = \mathbb{E}(\xi)\text{Cov}(AZ)$ and $\text{corr}(Y) = \text{corr}(AZ)$].*

Example 4.0.7. *A special case of Gaussian variance mixture (with $\nu/\xi \sim \chi^2_\nu$) is given by the multivariate t-distribution $t(\nu, \mu, \Sigma)$ with density,*

$$f(x) = \frac{\Gamma\left(\frac{(\nu+p)}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)(\nu\pi)^{\frac{p}{2}}|\Sigma|^{\frac{1}{2}}} \left(1 + \frac{(x-\mu)^T\Sigma^{-1}(x-\mu)}{\nu}\right)^{-\frac{\nu+p}{2}}, \tag{4.2}$$

for $x \in \mathbb{R}^p$, where $\mu \in \mathbb{R}^p$, $\Sigma_{p\times p} \succ 0$ and $\nu \in \mathbb{N}$ are the degrees of freedom. Note that $\nu > 1$ is required for first moments to exist, and $\nu > 3$ for second moments.

The general advantage of elliptical distributions is that they retain some nice properties of Gaussians while they allow for greater generality, some heavier tails and extremal dependence. The Gaussian however does not: if

$(X_1, X_2)^T \sim \mathcal{N}(0, \Sigma)$ with $\text{corr}(X_1, X_2) = \rho$ and $\text{Var}(X_1) = \text{Var}(X_2) = 1$, then $X_2|X_1 \sim \mathcal{N}(\rho x, 1-\rho^2)$ and so:

$$\mathbb{P}(X_2 > x | X_1 = x) = 1 - \Phi\left(x\sqrt{\frac{1-\rho}{1+\rho}}\right) \xrightarrow{x\to\infty} 0.$$

Proposition 4.0.8. *Elliptical distribution vectors are closed under marginalisation and affine transformations.*

Proof. $(*)$ Let X be an elliptical random vector. That is, we may write $X = \mu + AU$ for some spherical U. Then:

- $X_j = e_j^T X = \mu_j + (e_j^T A)U$, thus concluding that we have closure under marginalisation.
- $BX + b = (B\mu + b) + BAU$, thus concluding that we have closure under affine transformations. □

Remark 4.0.9. *Elliptical distributions are also closed under conditioning, possibly with a different generating density.*

Proposition 4.0.10 (Closure under Marginalization)**.** *If $W = (X^T, Y^T)^T$ is jointly elliptical with location and dispersion,*

$$\mu_W = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} \quad \text{and,} \quad \Sigma_W = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix},$$

then:

(i) X (respectively Y) is also elliptical with location μ_X (respectively μ_Y) and dispersion Σ_X (respectively Σ_Y). Furthermore, if W has generating density ψ, so do X and Y.

(ii) $Q = BW + \theta$ is also elliptical, with location $B\mu + \theta$ and dispersion $B\Sigma B^T$. When they exist, the generating densities of W and Q coincide.

(iii) $X|\{Y = y\}$ is also elliptical, with location $\mu_X - \Sigma_{XY}\Sigma_Y^\dagger(y - \mu_Y)$ and dispersion $\Sigma_X - \Sigma_{XY}\Sigma_Y^\dagger\Sigma_{XY}^T$. Should they exist, the generating densities of W and $X|Y$ need <u>not</u> coincide.

Chapter 5

Sampling Theory: Gaussians and Asymptotic Approximations

We now turn to some probabilistic tools that will be used in a statistical context in the next chapter. We will first present the data matrix formalism before engaging into Gaussian sampling. We will then introduce Wishart matrices and the Hotelling's T^2-distribution. We finally end the chapter by introducing (/recalling) the notions of weak convergence, the Strong Law of Large Numbers and the Central Limit Theorem (in different cases).

5.1 Gaussian Data Matrices and Gaussian Sampling

In this section, we consider an i.i.d. Gaussian sample which corresponds to a collection $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$ in $\mathbb{R}^p$. We may stack then row-wise to build a very useful matrix.

Definition 5.1.1. *A **Gaussian data matrix** has the form:*

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1p} \\ X_{21} & \cdots & X_{2p} \\ \vdots & & \vdots \\ X_{n1} & \cdots & X_{np} \end{pmatrix} = \begin{pmatrix} X_1^T \\ X_2^T \\ \vdots \\ X_n^T \end{pmatrix} = \begin{pmatrix} X_1 & X_2 & \cdots & X_n \end{pmatrix}^T. \tag{5.1}$$

In this matrix, the n rows represent observations/cases, and the p columns represent variables/features.

When sampling a Gaussian data matrix X, we wish to know sampling laws of (i) linear transformations of X, i.e. matrices of the form AXB for $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$ or also (ii) quadratic forms of X, i.e. matrices of the form X^TCX for $C = C^T \in \mathbb{R}^{n\times n}$. The reason is that we have multiple statistical quantities that can be written in this form. For instance, the sample mean $\overline{X}$ is a linear form:

$$\overline{X}^T = \frac{1}{n}\mathbb{1}_n^T X \equiv AXB, \tag{5.2}$$

with $A = \frac{1}{n}\mathbb{1}_n^T$ and $B = \mathbb{I}_{p\times p}$. Or we have that the sample covariance is a quadratic form:

$$\widehat{\Sigma} = \frac{1}{n}X^T H X, \tag{5.3}$$

where the centring matrix H_n of dimension n is defined as $H_n = \mathbb{I}_{n\times n} - \frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T$. We note that $\frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T = \mathbb{1}_n(\mathbb{1}_n^T\mathbb{1}_n)^{-1}\mathbb{1}_n^T$ is the projection matrix onto the span$\{\mathbb{1}_n\}$ and thus H_n is the projection onto the span$^\perp\{\mathbb{1}_n\}$. Here, under some conditions, the sample mean/covariance are sufficient for their population counterparts.

Theorem 5.1.2 (Gaussian Sampling). *Let $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu, \Sigma)$ be a random sample of size n of Gaussian d-vectors. Then:*
(i) The sample mean is a Gaussian vector $\overline{X} = \frac{1}{n} X^T \mathbb{1}_n \sim \mathcal{N}(\mu, n^{-1}\Sigma)$.
(ii) The sample covariance is a Wishart matrix $n\widehat{\Sigma} = X^T H X \sim W_p(\Sigma, n-1)$.
(iii) The sample mean and sample covariance are independent $\overline{X} \perp \widehat{\Sigma}$.

Before stating the proof [which can be found below (5.2)], we require several tools. This theorem could follow from the behaviour of Gaussians under linear/quadratic transformations. However, it might be easier to restate such theorems in terms of data matrices where properties of linear forms and quadratic forms both involve Gaussian data matrices. Let us then first see when a linear form AXB of a Gaussian data matrix X is also a Gaussian data matrix. Clearly, AXB is always a Gaussian matrix since entries are jointly Gaussian but to call it Gaussian data matrix we require i.i.d. rows.

Theorem 5.1.3 (Linear forms of Gaussian data matrices). *If $X_{n\times p}$ is a Gaussian data matrix from $\mathcal{N}(\mu, \Sigma)$, then $A_{m\times n} X B_{p\times q}$ is a $m \times q$ Gaussian data matrix if and only if the following two conditions hold true:*
*(i) $A\mathbb{1}_n = \alpha \mathbb{1}_n$ for some $\alpha \in \mathbb{R}$ **OR** $B^T \mu = 0$.*
*(ii) $AA^T = \beta \mathbb{I}_{m\times m}$ for some $\beta \in \mathbb{R}$ **OR** $B^T \Sigma B = 0$.*
Clearly, when AXB is a Gaussian data matrix, it is from a $\mathcal{N}(\alpha B^T \mu, \beta B^T \Sigma B)$.

Proof. $(*)$ We first note the following:

$$(AXB)_{ij} = [A(XB)]_{ij} = \sum_{i=1}^{n} A_{il}(XB)_{lj} = \sum_{l=1}^{n} A_{il} \sum_{k=1}^{p} X_{lk} B_{kj} = \sum_{l=1}^{n} \sum_{k=1}^{p} A_{il} X_{lk} B_{kj}. \tag{5.4}$$

Thus, it is immediate to see that each element is univariate normal. However, as mentioned, we also require the rows of Y to be independent and to have the same distribution. The heuristic idea is the following: post-multiplication of X involves adding weighted variables. Hence the rows of XB^T are independent. The transformed objects are also independent unless the pre-multiplication by A introduces some inter-dependence, so each row can be transformed only by scalar multiplication. To have that the distribution is the same across rows, the scalar factor must be the same. More concretely, we need each transformed row to have the same mean and covariance, and the cross-covariance between different rows to be zero. To this end, let Y be the transformed data matrix and denote by Y_i the i-th row of Y.

- We compute the mean:

$$\mathbb{E}(Y_i) = A_i \begin{pmatrix} \mu^T \\ \vdots \\ \mu^T \end{pmatrix} B = \begin{pmatrix} A_i \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_1 \end{pmatrix} & \cdots & A_i \begin{pmatrix} \mu_p \\ \vdots \\ \mu_p \end{pmatrix} \end{pmatrix} B = \begin{pmatrix} \mu_1(A_i \mathbb{1}_n) & \cdots & \mu_p(A_i \mathbb{1}_n) \end{pmatrix} B = (A_i \mathbb{1}_n)\mu^T B,$$

which shows the first part.

- Without loss of generality assume that X is centred. First we have:

$$\mathbb{E}(Y_i^T Y_i) = \mathbb{E}\big[(A_i X B)^T (A_i X B)\big] = B^T \mathbb{E}\big[X^T A_i^T A_i X\big] B.$$

Now remark that,

$$\mathbb{E}\big[(X^T A_i^T A_i X)_{kl}\big] = \mathbb{E}\left[\left(\sum_j X_{jk}A_{ij}\right)\left(\sum_{j'} A_{ij'}X_{j'l}\right)\right] = \mathbb{E}\left[\sum_{j,j'} X_{jk}A_{ij}A_{ij'}X_{j'l}\right] = \Sigma_{lk}\sum_j A_{ij}^2.$$

As a result we obtain $\mathbb{E}(Y_i^T Y_i) = (A_i A_i^T)B^T\Sigma B$ which does not depend on i if and only if $A_i A_i^T$ does not depend on i, or $B^T\Sigma B = 0$. Similarly, $\mathbb{E}(Y_i^T Y_i) = (A_i A_{i'}^T)B^T\Sigma B$ which is 0 if and only if distinct rows of A are orthogonal, or $B^T\Sigma B = 0$. □

Now, a natural question would be to know in what instance are two different linear forms AXB and CXD of a Gaussian data matrix independent.

Theorem 5.1.4 (Independence of linear forms of Gaussian data matrices)**.** *Let X be a Gaussian data matrix from $\mathcal{N}(\mu,\Sigma)$. Then:*

$$AXB \perp CXD \iff AC^T = 0 \quad \textbf{\textit{OR}} \quad B^T\Sigma D = 0.$$

Proof. $(*)$ Given that X is an $n\times p$ data matrix from a multivariate normal distribution $\mathcal{N}(\mu,\Sigma)$, and we define $Y = AXB$ and $Z = CXD$ where A,B,C,D are matrices of appropriate dimensions. Using vectorization, $\text{vec}(Y)$ and $\text{vec}(Z)$ may be written as $(B^T\otimes A)\text{vec}(X)$ and $(D^T\otimes C)\text{vec}(X)$ respectively. Hence, the covariance between $\text{vec}(Z)$ and $\text{vec}(Y)$ is given by:

$$\text{Cov}(\text{vec}(Y),\text{vec}(Z)) = (B^T\otimes A)\text{Cov}(\text{vec}(X))(D^T\otimes C)^T.$$

Given that $\text{Cov}(\text{vec}(X)) = \Sigma\otimes\mathbb{I}$, the above expansion becomes:

$$\text{Cov}(\text{vec}(Y),\text{vec}(Z)) = (B^T\otimes A)(\Sigma\otimes\mathbb{I})(D^T\otimes C)^T = ((B^T\Sigma)\otimes A)(D\otimes C^T) = B^T\Sigma D\otimes AC^T.$$

Thus, the elements $\text{vec}(Y)$ and $\text{vec}(Z)$ are uncorrelated if and only if the above matrix is the zero matrix; i.e. if and only if $B^T\Sigma D = 0$ or $AC^T = 0$. □

5.2 Wishart Random Matrices and the Hotelling's T^2

In this section, we first present the Wishart random matrix which roughly corresponds to the square of a Gaussian data matrix X. We will then do a small parenthesis on the Hotelling's T^2-distribution, that will be a useful tool for Inference.

Definition 5.2.1. *Let $X_{n\times p}$ be a Gaussian data matrix from a $\mathcal{N}(0,\Sigma)$ distribution. The $p\times p$ random matrix X^TX is said to follow a p-dimensional **Wishart distribution** with scale Σ and n degrees of freedom, such that $X^TX \sim W_p(\Sigma_{p\times p}, n)$.*
*When $\Sigma = \mathbb{I}_{p\times p}$ we speak of the standard p-Wishart distribution with n degrees of freedom. We call the matrix X^TX the **Wishart matrix**.*

Some immediate properties are the following.

Proposition 5.2.2. *For $W_i \overset{iid}{\sim} W_p(\Sigma_{p\times p}, 1)$ we have:*

$$W \sim W_p(\Sigma_{p\times p}, n) \iff W \overset{d}{=} \sum_{i=1}^n W_i. \tag{5.5}$$

Proof. $(*)$ We first set:

$$X^TX = \begin{pmatrix} X_1^T & \cdots & X_n^T \end{pmatrix} \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} = \sum_{i=1}^{n} X_i^T X_i.$$

Defining $W_i = X_i^T X_i \sim W_p(\Sigma, 1)$ and since the rows are independent, the matrices W_i are independent which completes the claim. □

Proposition 5.2.3.

$$W \sim W_p(\Sigma_{p\times p}, n) \Longrightarrow \mathbb{E}(W) = n\Sigma. \tag{5.6}$$

Proof. $(*)$ This is a simple calculation:

$$\mathbb{E}(W) = \mathbb{E}(X^TX) = \sum_{i=1}^{n} \mathbb{E}(X_i^T X_i) = n\Sigma.$$

□

Proposition 5.2.4. *The lower triangular part of $W \sim W_p(\Sigma_{p\times p}, n)$ has density if and only if $n \geq p$.*

Proof. $(*)$ When $n < p$, the Wishart matrix $W = \sum_{i=1}^{n} X_i X_i^T$ is the sum of at most n rank-one matrices. Since the rank of each $X_i X_i^T$ is at most 1, the total rank of W is at most n. Therefore, W is a.s. singular because its rank never reaches the full dimension p. If it is a.s. singular, then it is constrained to a lower-dimensional subset of the space of symmetric $p \times p$ matrices. This means that it cannot have a density since it is supported on a Lebesgue-zero measure set. □

Proposition 5.2.5.

$$W \sim W_p(\Sigma_{p\times p}, n) \Longrightarrow \frac{\theta^T W \theta}{\theta^T \Sigma \theta} \sim \chi^2_n, \qquad \forall \theta \notin \ker(\Sigma). \tag{5.7}$$

Proof. $(*)$ We first write $W = X^TX$ where $X_{n\times p}$ is a matrix with i.i.d. rows $X_i \sim \mathcal{N}(0, \Sigma)$. Furthermore, for any $\theta \notin \ker(\Sigma)$, define $Y = X\theta$ satisfying $Y_i \sim \mathcal{N}(0, \theta^T\Sigma\theta)$. Then:

$$\theta^T W \theta = \theta^T X^T X \theta = Y^T Y = \sum_{i=1}^{n} Y_i^2.$$

Now, since $Y_i \sim \mathcal{N}(0, \theta^T\Sigma\theta)$, normalizing by $\theta^T\Sigma\theta$ yields:

$$\frac{\theta^T W \theta}{\theta^T \Sigma \theta} = \sum_{i=1}^{n} \left(\frac{Y_i}{\sqrt{\theta^T\Sigma\theta}} \right)^2.$$

Since each term follows $\mathcal{N}(0,1)$, we conclude from the definition of the χ^2-distribution that $\frac{\theta^T W\theta}{\theta^T\Sigma\theta} \sim \chi^2_n$. □

Proposition 5.2.6.

$$W \sim W_p(\Sigma_{p\times p}, n) \Longrightarrow B^T W B \sim W_p(B^T\Sigma B, n). \tag{5.8}$$

Proof. Writing $W = X^T X$ for an $n \times p$ data matrix from $\mathcal{N}(0,\Sigma)$, we have:

$$B^T W = B^T X^T X B = Y^T Y,$$

where $Y = \mathbb{I}_{p\times p} X B$ is also Gaussian data matrix, from a $\mathcal{N}(0, B^T \Sigma B)$, by Theorem (5.1.3) [evidently $\mathbb{I}_{p\times p}\mathbb{1}_p = \mathbb{1}_p$ and $\mathbb{I}_{p\times p}^T \mathbb{I}_{p\times p} = \mathbb{I}_{p\times p}$]. □

Corollary 5.2.7. *The random matrix W has a $W_p(\Sigma, n)$ distribution if and only if $\left(\Sigma^{\frac{1}{2}}\right)^\dagger W \left(\Sigma^{\frac{1}{2}}\right)^\dagger$ has a $W_q(H, n)$ distribution, where $H = \Sigma^\dagger \Sigma$ is the projection onto the range of Σ.*

Now, recall from Proposition (3.4.3) that when $Z \sim \mathcal{N}(0, \mathbb{I}_p)$ and H is a projection, then $Z^T H Z \sim \chi^2_{\text{rank}(H)}$. Let us now find an analogue of this proposition for Wishart matrices. Note that $Z^T H Z = \|HZ\|^2$ so the corresponding Wishart quantity would be $Z^T H Z$ where $Z_{n\times p} = (Z_1^T, ..., Z_n^T)^T$.

Theorem 5.2.8 (Cochran's Theorem)**.** *Let $X_{n\times p}$ be a Gaussian data matrix from $\mathcal{N}(0,\Sigma)$ and H be symmetric. Then:*

$$X^T H X \text{ is a Wishart matrix of mean } \Sigma \iff H \text{ is a projection.}$$

When H is indeed a projection, $X^T H X \sim W_p(\Sigma, \text{rank}(H))$.

Proof. Let $H = U \Lambda U^T$ be the spectral decomposition of H. Then $X^T H X = X^T U \Lambda U^T X$ and $Y = U^T X \mathbb{I}$ is a $\mathcal{N}(0,\Sigma)$ Gaussian data matrix, since $U^T U = \mathbb{I}$ and $\mathbb{I}0 = 0$. Thus, $X^T H X = Y^T \Lambda Y = \sum_{i=1}^n \lambda_i Y_i Y_i^T$ for $Y_i \overset{iid}{\sim} \mathcal{N}(0,\Sigma)$. Now:

- if H is a projection, then λ_i is either 0 or 1. Hence,

$$\sum_{i=1}^n \lambda_i Y_i Y_i^T = \sum_{j=1}^{\text{rank}(H)} Y_i Y_i^T \sim W_p(\Sigma, \text{rank}(H)).$$

- Now let $\text{rank}(H) = q \leq n$. Then only its first q eigenvalues λ_i are non-zero. Assume now, as in the statement, that for some (yet unspecified) d, $\sum_{i=1}^q \lambda_i Y_i Y_i^T \sim W_p(\Sigma, d)$. Multiplying both sides from the left by e_1^T (i.e. the first canonical vector) and from the right by e_1, then dividing both sides by $e_1^T \Sigma e_1$ and finally using Proposition (5.2.5) we finally arrive at an equality in distribution of (i) on the LHS a weighted sum of q independent χ_1^2 with λ_i as weights, and (ii) on the RHS a single χ_d^2. Thus, equating the corresponding MGFs yields, for all t sufficiently small:

$$\prod_{i=1}^q (1-2\lambda_i t)^{-\frac{1}{2}} = (1-2t)^{-\frac{d}{2}} \implies \prod_{i=1}^q (1-2\lambda_i t) = (1-2t)^d.$$

Now, this last equality can only happen over an interval of t's if both polynomials have the same degree and same roots. Hence $d = q$ and $\lambda_i = 1$. □

We are finally ready to prove the Gaussian Sampling Theorem (5.1.2).

Proof of (5.1.2). The distribution of the sample mean vector follows directly from Theorem (5.1.3) since $\overline{X}^T = n^{-1}\mathbb{1}_n^T X$ where $A = n^{-1}\mathbb{1}_n^T$ is such that $A\mathbb{1}_n = n^{-1}\mathbb{1}_n^T \mathbb{1}_n = 1$ and $AA^T = n^{-1}\mathbb{1}_n^T n^{-1}\mathbb{1}_n = n^{-1}$. The distribution of the sample covariance follows directly from Cochran's Theorem (5.2.8). Independence of the sample mean and sample covariance follow from Theorem (5.1.4) by considering $CX := HX$ and $AX := n^{-1}\mathbb{1}_n^T X$, noting that $AC^T = H\mathbb{1}_n = 0$ since H projects onto $\text{span}^\perp\{\mathbb{1}_n\}$. □

We now present the Hotelling's T^2-distribution along with some of its properties. When doing testing, we will be interested in constructing statistics of the form $(\overline{X}-\mu)^T\Sigma^{-1}(\overline{X}-\mu)$. But, more often than not, Σ will be unknown (nuisance parameter). Thus we will replace it by the sample version $(\overline{X}-\mu)^T\widehat{\Sigma}^{-1}(\overline{X}-\mu)$. Obviously, this will yield a different sampling distribution for the statistic than when using Σ itself.

Definition 5.2.9. *For $n \geq p$, let $\tau^2 = nY^TW^{-1}Y$ where $Y_{p\times 1} \sim \mathcal{N}(0,\mathbb{I})$ and $W_{p\times p} \sim W_p(\mathbb{I},n)$ are independently distributed. Then τ^2 is said to follow the **Hotelling** T^2 **distribution** with parameter p and n, written $\tau^2 \sim T^2(p,n)$.*

Lemma 5.2.10. *If $X \sim \mathcal{N}(\mu,\Sigma_{p\times p})$ independently of $W \sim W_p(\Sigma,n)$ with Σ non-singular and $n \geq p$ then:*

$$n(X-\mu)^TW^{-1}(X-\mu) \sim T^2(p,n). \tag{5.9}$$

Proof. $(*)$ Note that if $W \sim W_p(\Sigma,n)$ then $\left(\Sigma^{-\frac{1}{2}}\right)^T W\left(\Sigma^{-\frac{1}{2}}\right) \sim W_p(\mathbb{I},n)$ and $\left(\Sigma^{-\frac{1}{2}}\right)(X-\mu) \sim \mathcal{N}(0,\mathbb{I})$. Now, by the definition of the Hotelling distribution, we have that:

$$n\left(\left(\Sigma^{-\frac{1}{2}}\right)(X-\mu)\right)^T\left(\left(\Sigma^{-\frac{1}{2}}\right)^T W\left(\Sigma^{-\frac{1}{2}}\right)\right)^{-1}\left(\left(\Sigma^{-\frac{1}{2}}\right)(X-\mu)\right) = n(X-\mu)^TW^{-1}(X-\mu) \sim T^2(p,n).$$

□

Corollary 5.2.11. *If $\overline{X}$ and $\widehat{\Sigma}$ be the sample mean and covariance of a $\mathcal{N}(\mu,\Sigma)$ i.i.d. sample. If $n \geq p$ and Σ non-singular then:*

$$(n-1)(\overline{X}-\mu)^T\widehat{\Sigma}^{-1}(\overline{X}-\mu) \sim T^2(p,n-1). \tag{5.10}$$

Proof. $(*)$ We use Theorem (5.1.2) on Gaussian sampling along with Lemma (2.3.4) for the covariance. We have that $\left(\left(n^{-1}\Sigma\right)^{-\frac{1}{2}}\right)(\overline{X}-\mu) = \sqrt{n}\left(\Sigma^{-\frac{1}{2}}\right)(\overline{X}-\mu) \sim \mathcal{N}(0,\mathbb{I})$ and $n\left(\Sigma^{-\frac{1}{2}}\right)^T\widehat{\Sigma}\left(\Sigma^{-\frac{1}{2}}\right) \sim W_p(\mathbb{I},n-1)$. Now, by the definition of the Hotelling distribution, we have that:

$$\begin{aligned}(n-1)\left(\sqrt{n}\left(\Sigma^{-\frac{1}{2}}\right)(\overline{X}-\mu)\right)^T\left(n\left(\Sigma^{-\frac{1}{2}}\right)^T\widehat{\Sigma}\left(\Sigma^{-\frac{1}{2}}\right)\right)^{-1}\left(\sqrt{n}\left(\Sigma^{-\frac{1}{2}}\right)(\overline{X}-\mu)\right) &= (n-1)(\overline{X}-\mu)^T\widehat{\Sigma}^{-1}(\overline{X}-\mu)\\ &\sim T^2(p,n-1).\end{aligned}$$

□

Recall that the square of a Student t_n distribution yields a $F_{1,n}$ distribution. As a result $T^2(1,n) \equiv F_{1,n} \equiv (t_n)^2$, and more generally:

$$T^2(p,n) \equiv \frac{np}{n-p+1}F_{p,n-p+1}. \tag{5.11}$$

5.3 Asymptotic Approximations

It may happen that X is a data matrix though non-Gaussian. In this case, we might want to know what we can say about $\overline{X}$ and $\widehat{\Sigma}$. In general, the answer depends on the row distribution, however we are looking for universality. For this we need to consider an approximate/asymptotic law, where the two parameters we may play with are n and p. To this end, we first recall some useful tools to state asymptotic results.

Definition 5.3.1 (Weak convergence). *Let $\{X_n\}$ be a sequence of random vectors of $\mathbb{R}^d$, and X a random vector of $\mathbb{R}^d$. Let $F_n, F : \mathbb{R}^d \to [0,1]$ be the corresponding joint CDFs. We say that X_n **converges in distribution** to X as $n \to \infty$ (and write $X_n \xrightarrow{d} X$) if $F_{X_n}(x) \xrightarrow{n\to\infty} F_X(x)$ for every continuity point $x \in \mathbb{R}^d$ of F_X.*

There is a link between univariate and multivariate weak convergence.

Theorem 5.3.2 (Cramér-Wold Device). *Let $\{X_n\}$ be a sequence of random vectors of $\mathbb{R}^d$, and X a random vector of $\mathbb{R}^d$. Then:*

$$X_n \xrightarrow{d} X \iff \theta^T X_n \xrightarrow{d} \theta^T X, \qquad \forall \theta \in \mathbb{R}^d. \tag{5.12}$$

Example 5.3.3 (Separate weak convergence of each coordinate $\not\Longrightarrow$ weak convergence of the random vector). *We construct a counterexample and consider the Gaussian random vectors X, Y with same marginals but different covariance structure. Then, the sequence $Z_{2n} := X$ and $Z_{2n+1} := Y$ establishes this counterexample.*

When it comes to matrices, the definition is essentially the same.

Theorem 5.3.4 (Cramér-Wold Device for $n \times p$ matrices). *Let $\{M_n\}$ be a sequence of random matrices of $\mathbb{R}^{n\times p}$, and M a random matrix of $\mathbb{R}^{n\times p}$. Then:*

$$M_n \xrightarrow{d} M \iff \operatorname{trace}\big(A^T M_n\big) \xrightarrow{d} \operatorname{trace}\big(A^T M\big), \qquad \forall A \in \mathbb{R}^{n\times p}. \tag{5.13}$$

We now review two important results from Probability Theory.

Theorem 5.3.5 (Strong Law of Large Numbers). *Let $\{X_n\}$ be pairwise i.i.d. random vectors with $\mathbb{E}(X_k) = \mu$ and $\mathbb{E}\|X_k\| < \infty$ for all $k \geq 1$. Then:*

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow{a.s.} \mu. \tag{5.14}$$

Theorem 5.3.6 (Weak Law of Large Numbers). *Let $\{X_n\}$ be pairwise i.i.d. random vectors with $\mathbb{E}(X_k) = \mu$ and $\mathbb{E}(X_k^2) < \infty$ for all $k \geq 1$. Then:*

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow{\mathbb{P}} \mu. \tag{5.15}$$

Theorem 5.3.7 (Central Limit Theorem). *Let $\{X_n\}$ be an i.i.d. sequence of random vectors in $\mathbb{R}^d$ with mean μ and covariance Σ with $\operatorname{trace}(\Sigma) < \infty$. Let $\overline{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$ be their sample mean. Then:*

$$\sqrt{n}\big(\overline{X} - \mu\big) \xrightarrow{d} \mathcal{N}(0, \Sigma). \tag{5.16}$$

Let us now briefly investigate the relationship between convergence rates and high dimensions. For the Law of Large Numbers, assuming finite variance the rate is $n^{-\frac{1}{2}}$. Let us see the Central Limit Theorem.

Theorem 5.3.8 (Berry-Essen-Benktus). *Let $\{X_i\}_{i=1}^n$ be an i.i.d. sequence of p-random vectors with mean 0 and covariance $\mathbb{I}_{p\times p}$. Define $S_n := \frac{1}{\sqrt{n}}\sum_{i=1}^n X_i$. If $\mathcal{A}_p$ denotes the class of convex subsets of $\mathbb{R}^p$, then for $Z\sim\mathcal{N}_p(0,\mathbb{I}_p)$ and some universal constant $c\leq 400$:*

$$\sup_{A\in\mathcal{A}_p}\left|\mathbb{P}(S_n\in A)-\mathbb{P}(Z\in A)\right|\leq c\cdot\frac{p^{\frac{1}{4}}\mathbb{E}\|X_i\|^3}{\sqrt{n}}. \tag{5.17}$$

Note that by Jensen's Inequality, when $\mathbb{E}(XX^T)=\mathbb{I}_{p\times p}$ we have:

$$\mathbb{E}\big(\|X\|^3\big)\geq\big(\mathbb{E}\big(\|X\|^2\big)^{\frac{3}{2}}=p^{\frac{3}{2}}.$$

Thus to make the upper-bound shrink to 0 it is necessary that $p=o\left(n^{\frac{2}{7}}\right)$, i.e.

$$\frac{p_n}{n^{\frac{2}{7}}}\xrightarrow{n\to\infty}0.$$

Now, let us see what happens when p is of same order or larger than n.

Theorem 5.3.9 (Chernozhukov, Chetverikov and Kato, 2017). *In the context of the last Theorem, assume that (i) $\mathbb{E}(X_{ij}^2)\geq b$ for all $j=1,...,p$, (ii) $\mathbb{E}(|X_{ij}|^{2+k})\leq B$ for $k=1,2$ and all $j=1,...p$, and (iii) $\mathbb{E}(\exp(|X_{ij}|/B))\leq 2$ for all $i\leq n$ and $j\leq p$. Then, if we focus only on rectangles $\mathcal{R}_p$ of $\mathbb{R}^p$ we obtain:*

$$\sup_{R\in\mathcal{R}_p}\left|\mathbb{P}(S_n\in R)-\mathbb{P}(Z\in R)\right|\leq C\cdot\left(\frac{\log^7(pn)}{n}\right)^{\frac{1}{6}}, \tag{5.18}$$

where $Z\sim\mathcal{N}(0,\mathbb{I}_{p\times p})$ and the constant C depends only on b and B.

Observe that CCK, compared to BEB, simply requires $\frac{\log(p_n)}{n^{1/7}}\xrightarrow{n\to\infty}0$, i.e. $\log(p_n)=o(n^{1/7})$ allowing for $p\gg n$. However, the improvement comes at a cost of smaller class of sets $\mathcal{R}_d\subset\mathcal{A}_d$. However, this is not a CLT in a traditional sense of a convergence in distribution.

Let us now investigate the empirical covariance, defined by the sample average of the random matrices $X_1X_1^T,...,X_nX_n^T$, i.e. $\frac{1}{n}\sum_{i=1}^n X_iX_i^T$. These are i.i.d. with mean Σ and some covariance. In the Gaussian (and elliptical) case, this depends only on second moments but in general it will depend on fourth moments (as we will see shortly). They are also elements of a real vector space of dimension $p(p+1)/2$ and therefore, the usual Law of Large Numbers holds and the usual CLT holds. Also, provided we can standardise, the high dimensional CLT [Theorem (5.3.9)] holds unchanged (i.e. with $\log(p(p+1))\approx\log\big(p^2\big)=2\log(p)=o(n^{1/7})$). For the record, let us now specify the covariance structure of $W=XX^T$ for $X\sim\mathcal{N}(0,\Sigma)$.

Proposition 5.3.10 (Isserlis' formula - Covariance of Wishart matrix). *If $W\sim W(\Sigma,1)$, its element-wise covariance is:*

$$\mathrm{Cov}(w_{ij},w_{kl})=\sigma_{ik}\sigma_{jl}+\sigma_{il}\sigma_{jk}. \tag{5.19}$$

Sketch of Proof. We have:

$$\mathrm{Cov}(w_{ij},w_{kl})=\mathrm{Cov}(X_iX_j,X_kX_l)=\mathbb{E}(X_iX_jX_kX_l)-\mathbb{E}(X_iX_j)\mathbb{E}(X_kX_l).$$

Now, by considering the mixed partial derivatives of order 4 of the MGF of a 4-dimensional Gaussian, one

can show that $\mathbb{E}(X_iX_jX_kX_l) = \sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk} + \sigma_{ij}\sigma_{kl}$. Thus:

$$\mathrm{Cov}(w_{ij}, w_{kl}) = \mathbb{E}(X_iX_jX_kX_l) - \sigma_{ij}\sigma_{kl} = (\sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk} + \sigma_{ij}\sigma_{kl}) - \sigma_{ij}\sigma_{kl} = \sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk}.$$

□

The above formula may be arranged in vectorized form as:

$$\mathrm{Cov}(\mathrm{vec}(W)) = (\Sigma \otimes \Sigma)\big(\mathbb{I}_{p^2 \times p^2} + K_{pp}\big), \tag{5.20}$$

where $K_{pp} := \sum_{i=1}^{p}\sum_{j=1}^{p} H_{ij} \otimes H_{ij}^T$ is the **commutation matrix**, i.e. $K_{pp}\mathrm{vec}(W) = \mathrm{vec}(W^T)$.

Proposition 5.3.11. *Σ diagonal implies that the Wishart entries are uncorrelated where the ij element has variance $\big(1 + \mathbb{1}_{\{i=j\}}\big)\sigma_{ii}\sigma_{jj}$ [recall $\sigma_{ii} \equiv \Sigma_{ii} = \sigma_i^2$].*

Proof. $(*)$ Since Σ is diagonal, $\sigma_{ik} = 0$ when $i \neq k$. Thus $\mathrm{Cov}(w_{ij}, w_{kl}) = \sigma_{ii}\sigma_{jj}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$. Thus, $\mathrm{Cov}(w_{ij}, w_{ij}) = \mathrm{Cov}(w_{ij}, w_{ji}) = \sigma_{ii}\sigma_{jj}$ and otherwise 0 (recall that W is symmetric). □

Proposition 5.3.12.

$$\Sigma \succ 0 \Longrightarrow \mathrm{Cov}\big[\{w_{ij}\}_{i \leq j}\big] \succ 0 \quad \textit{(lower-triangular part of } W\textit{)}. \tag{5.21}$$

Proof. $(*)$ Let us first consider the diagonal case. Then the covariance of the vectorization of the lower-triangular part of W has diagonal covariance with strictly positive elements on the diagonal, which is clearly positive definite. For the general case, it suffices to observe that the reduction may be obtained by the SVD $\Sigma = U^T\Lambda U$ and observing that UWU^T is Wishart with diagonal scale Λ. □

Now note that in the non-Gaussian case, second moments do **not** determine fourth moments:

$$\mathbb{E}(X_iX_j, X_kX_l) = m_4(i,j,k,l) \neq \sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk} + \sigma_{ij}\sigma_{kl}.$$

Thus we can't use the covariance structure of a Wishart matrix in the CLT limit. Instead, we need essentially all mixed fourth moments (which are also very difficult to estimate). A notable exception is for centred X with elliptical law of dispersion $\Psi \propto \Sigma$. The proportionality constant can be deduced to be $\frac{1}{p}\mathbb{E}\big(X^T\Psi^{-1}X\big)$. Here, the fourth moments enter only through a simple scalar parameter,

$$\kappa := \frac{p}{p+2}\frac{\mathbb{E}\big([X^T\Psi^{-1}X]^2\big)}{\mathbb{E}^2\big(X^T\Psi^{-1}X\big)} - 1. \tag{5.22}$$

Indeed, one may calculate $\mathrm{Cov}(\mathrm{vec}(XX^T)) = (\kappa+1)(\Sigma\otimes\Sigma)\big(\mathbb{I}_{p^2\times p^2} + K_{pp}\big) + \kappa\mathrm{vec}(\Sigma)\mathrm{vec}(\Sigma)^T$, or element-wise $\mathbb{E}(X_iX_jX_kX_l) = (\kappa+1)\sigma_{ik}\sigma_{jl} + (\kappa+1)\sigma_{il}\sigma_{jk} + \kappa\sigma_{ij}\sigma_{lk}$. So if Σ is diagonal, the entries of XX^T are uncorrelated as in the Gaussian case. In the Gaussian case, one can check that $\kappa = 0$.

Occasionally, we need to studentise by an unknown but consistently estimable quantity. Other times, we have a CLT for a quantity but we are interested in some functional thereof. For such cases, statisticians rely on two essential tools.

Lemma 5.3.13 (Slutsky)**.** *Let X be a random vector in $\mathbb{R}^p$, $\theta \in \mathbb{R}^q$, and $g : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}$ be continuous on $\mathrm{supp}(X) \times \{\theta\}$. If $X_n \xrightarrow{d} X$ in $\mathbb{R}^p$ and $Y_n \xrightarrow{d} \theta$ then $g(X_n, Y_n) \xrightarrow{d} g(X, \theta)$.*

Theorem 5.3.14 (Delta Method)**.** *Let* $Z_n := a_n(X_n - \theta) \xrightarrow{d} Z$ *where* $0 < a_n \uparrow \infty$ *and* $\theta \in \mathbb{R}^p$*. Let* $g(\cdot) : \mathbb{R}^p \to \mathbb{R}^q$ *be differentiable at* θ*. Then,* $a_n(g(X_n) - g(\theta)) \xrightarrow{d} \big[(\nabla g)(\theta)\big] Z$*.*

We finally end this sections with some comments on low versus high dimensions. There is a fundamental difference between the low/high dimensional cases. To see this, consider the n i.i.d. p-vectors $\{X_i\}$ with mean 0 and covariance $\mathbb{I}_p$. Whether in low/high dimension, when it holds, the CLT heuristically says,

$$\sqrt{n}\,\overline{X} \overset{d}{\approx} \mathcal{N}(0, \mathbb{I}_p), \qquad \text{for large } n. \tag{5.23}$$

In the low dimensional case, $\overline{X}$ collapses to the true mean as $n \to \infty$. In the high dimensional case, $\overline{X}$ concentrates on the sphere of radius $\sqrt{\frac{p}{n}}$ (recall the concentration of measure phenomenon). Thus, depending on the ration of p/n we might very well not have a law of large numbers. This fundamental difference has immediate consequences when doing statistics at low versus high dimensions - they are two distinct and very different regimes.

Chapter 6

Inference

We now turn to inference, especially in the Gaussian case. We first start by presenting the Gaussian likelihood, along with its mean and covariance maximum likelihood estimators. We then focus on the likelihood ratio framework, before presenting an important tool for hypothesis testing being Wilk's Theorem. We then end the chapter by presenting an exhaustive list of hypothesis tests with different assumptions.

6.1 Gaussian Mean and Covariance Estimation

In this first section, we present the Gaussian likelihood, along with the maximum likelihood estimators (MLE) of the mean and covariance. We finally end the section by showing that the MLE is parametrisation equivariant.

Proposition 6.1.1 (Gaussian Likelihood)**.** *Let $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu, \Sigma)$ be a sample of Gaussian p-vectors. The likelihood $L(\mu, \Sigma)$ of (μ, Σ) is given by:*

$$\frac{\mathbb{1}_{\left\{\mathcal{R}\left(\widehat{\Sigma}+(\overline{x}-\mu)(\overline{x}-\mu)^T\right)\subseteq\mathcal{R}(\Sigma)\right\}}}{\left(\prod_{j=1}^{r}\sqrt{2\pi\lambda_j(\Sigma)}\right)^n}\exp\Big(-\frac{n}{2}(\overline{x}-\mu)^T\Sigma^\dagger(\overline{x}-\mu)-\frac{n}{2}\mathrm{trace}\big(\Sigma^\dagger\widehat{\Sigma}\big)\Big), \tag{6.1}$$

where $r = \mathrm{rank}(\Sigma) \leq p$ and $\{\lambda_1(\Sigma), ..., \lambda_r(\Sigma)\}$ are the positive eigenvalues of Σ. Consequently, the likelihood depends on the data only through $\big(\overline{X}, \widehat{\Sigma}\big)$.

Proof. We first observe that:

$$\begin{aligned}
\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T &= \sum_{i=1}^{n}(x_i-\overline{x}+\overline{x}-\mu)(x_i-\overline{x}+\overline{x}-\mu)^T \\
&= \sum_{i=1}^{n}\Big[(x_i-\overline{x})(x_i-\overline{x})^T+(x_i-\overline{x})(\overline{x}-\mu)^T+(\overline{x}-\mu)(x_i-\overline{x})^T+(\overline{x}-\mu)(\overline{x}-\mu)^T\Big] \\
&= \sum_{i=1}^{n}(x_i-\overline{x})(x_i-\overline{x})^T+\underbrace{\left(\sum_{i=1}^{n}(x_i-\overline{x})\right)(\overline{x}-\mu)^T}_{(*)}+\underbrace{(\overline{x}-\mu)\left(\sum_{i=1}^{n}(x_i-\overline{x})^T\right)}_{(**)}+n(\overline{x}-\mu)(\overline{x}-\mu)^T.
\end{aligned}$$

Now $(*) = \big(\sum_{i=1}^n (x_i - \overline{x})\big)(\overline{x}-\mu)^T = \sum_{i=1}^n x_i(\overline{x}-\mu)^T - n\overline{x}(\overline{x}-\mu)^T = n\overline{x}(\overline{x}-\mu)^T - n\overline{x}(\overline{x}-\mu)^T = 0$ and also $(**) = (\overline{x}-\mu)\big(\sum_{i=1}^n (x_i-\overline{x})^T\big) = (\overline{x}-\mu)\big(\sum_{i=1}^n x_i\big)^T - n(\overline{x}-\mu)\overline{x}^T = n(\overline{x}-\mu)\overline{x}^T - n(\overline{x}-\mu)\overline{x}^T = 0$. As a result:

$$\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T = \sum_{i=1}^{n}(x_i-\overline{x})(x_i-\overline{x})^T + n(\overline{x}-\mu)(\overline{x}-\mu)^T. \tag{6.2}$$

Recall that when $X \sim \mathcal{N}(\mu_{p\times 1}, \Sigma_{p\times p})$, X admits a density on $\mu + \mathcal{R}(\Sigma)$ given by:

$$f_X(x) = \frac{1}{\prod_{j=1}^{r}\sqrt{2\pi\lambda_j(\Sigma)}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^\dagger(x-\mu)\right)\mathbb{1}_{\{x\in\mu+\mathcal{R}(\Sigma)\}}, \tag{6.3}$$

where $r = \text{rank}(\Sigma) \leq p$ and $\{\lambda_1(\Sigma), ..., \lambda_r(\Sigma)\}$ are the positive eigenvalues of Σ. Therefore, we obtain the joint density with respect to the Lebesgue measure on $\mu + \mathcal{R}(\Sigma)$ as:

$$f_{X_1,...,X_n}(x_1,...,x_n) = \frac{\prod_{i=1}^{n}\mathbb{1}_{\{x\in\mu+\mathcal{R}(\Sigma)\}}}{\left(\prod_{j=1}^{r}\sqrt{2\pi\lambda_j(\Sigma)}\right)^n} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^\dagger(x_i-\mu)\right). \tag{6.4}$$

Let us first focus on the exponential term. Re-writing the quadratic form as a trace and using (6.2) we get:

$$\begin{aligned}
\exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^\dagger(x_i-\mu)\right) &= \exp\left(-\frac{1}{2}\sum_{i=1}^{n}\text{trace}(\Sigma^\dagger(x_i-\mu)(x_i-\mu)^T)\right) \\
&= \exp\left(-\frac{1}{2}\text{trace}\left(\Sigma^\dagger\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T\right)\right) \\
&= \exp\Big(-\frac{n}{2}\text{trace}(\Sigma^\dagger(\overline{x}-\mu)(\overline{x}-\mu)^T)\Big)\exp\left(-\frac{1}{2}\text{trace}\left(\Sigma^\dagger\sum_{i=1}^{n}(x_i-\overline{x})(x_i-\overline{x})^T\right)\right) \\
&= \exp\Big(-\frac{n}{2}(\overline{x}-\mu)^T\Sigma^\dagger(\overline{x}-\mu)\Big)\exp\Big(-\frac{n}{2}\text{trace}\big(\Sigma^\dagger\widehat{\Sigma}\big)\Big),
\end{aligned}$$

which is exactly the sought form. As for the product in the numerator, we need to show that $\prod_{i=1}^{n}\mathbb{1}_{\{x\in\mu+\mathcal{R}(\Sigma)\}}$, or equivalently $\prod_{i=1}^{n}\mathbb{1}_{\{x-\mu\in\mathcal{R}(\Sigma)\}}$, is as stipulated. To do this, we will use (6.2) again combined with Proposition (1.2.7), to immediately have that: since $\Sigma \succeq 0$,

$$x_1-\mu,...,x_n-\mu \in \mathcal{R}(\Sigma) \iff \mathcal{R}\left(\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T\right) \subseteq \mathcal{R}(\Sigma) \iff \mathcal{R}\left(n\widehat{\Sigma}+n(\overline{x}-\mu)(\overline{x}-\mu)^T\right) \subseteq \mathcal{R}(\Sigma).$$

Since the factor n has no bearing on the inclusion, the proof is complete. □

Some comments: If/when they exist, the MLEs of (μ, Σ) will be functions of $\big(\overline{X}, \widehat{\Sigma}\big)$. And when $\Sigma \succ 0$, we immediately conclude that $\big(\overline{X}, \widehat{\Sigma}\big)$ is sufficient for (μ, Σ) by the Fisher-Neyman factorisation Theorem. Without restrictions on the support of the Gaussian law, the model is non-regular (i.e. no common dominating measure). More commonly, we will use the **log-likelihood** instead of the standard likelihood to make the computations easier. Thus, the log-likelihood (up to constants) for μ and Σ on a Gaussian data matrix $X_{n\times p}$ for $n > 1$, equals $-\infty$ when $\mathcal{R}\left(n\widehat{\Sigma}+n(\overline{x}-\mu)(\overline{x}-\mu)^T\right) \not\subseteq \mathcal{R}(\Sigma)$, and otherwise equals:

$$l(\mu,\Sigma) = -\frac{n}{2}\sum_{j=1}^{\text{rank}(\Sigma)}\log(\lambda_j(\Sigma)) - \frac{n}{2}(\overline{x}-\mu)^T\Sigma^\dagger(\overline{x}-\mu) - \frac{n}{2}\text{trace}\big(\Sigma^\dagger\widehat{\Sigma}\big), \tag{6.5}$$

which is finite for all $\overline{x}-\mu \in \mathcal{R}(\Sigma)$. When $\Sigma \succ 0$, the log-likelihood is positive for all $\overline{x}-\mu$ and equals:

$$l(\mu,\Sigma) = -\frac{n}{2}\log|\Sigma| - \frac{n}{2}(\overline{x}-\mu)^T\Sigma^{-1}(\overline{x}-\mu) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big). \tag{6.6}$$

Imposing an assumption on the range of Σ (equivalently on the support of the random vector) represents imposing a restriction on the model $\mathcal{N}_p = \{\mathcal{N}(\mu,\Sigma) : \mu \in \mathbb{R}^p, \Sigma_{p\times p} \succeq 0\}$. Because the model $\mathcal{N}_p$ is non-regular, the maximum likelihood estimation depends heavily on such restrictions.

Theorem 6.1.2 (Gaussian mean/covariance MLE). *Let $X_1,...,X_n \overset{iid}{\sim} \mathcal{N}(\mu,\Sigma)$ be a sample of Gaussian p-vectors. Then:*
(i) The unique MLE of μ is always $\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i$.
(ii) Under the restriction $\mathcal{R}(\Sigma) = \mathcal{R}\big(\widehat{\Sigma}\big)$, the unique MLE of (μ,Σ) is $\big(\overline{X},\widehat{\Sigma}\big)$.
(iii) Without any range restriction: (a) if $\widehat{\Sigma}$ is non-singular, the unique MLE of (μ,Σ) is $\big(\overline{X},\widehat{\Sigma}\big)$; otherwise (b) if $\widehat{\Sigma}$ is singular, the MLE of μ is $\overline{X}$ but the MLE of Σ does not exist.

Proof. We prove the statements in order.
Notice that, unless $\Sigma \succ 0$, we need to be careful about the support of the joint density. The joint density vanishes unless $\mathcal{R}\left(n\widehat{\Sigma} + n(\overline{x}-\mu)(\overline{x}-\mu)^T\right) \subseteq \mathcal{R}(\Sigma)$, or equivalently $x_i - \mu \in \mathcal{R}(\Sigma)$ for all $i \leq n$. When considering the joint density, this is seen as a condition of the observations. However when considering the likelihood, this is seen as a condition on Σ and μ. When (μ,Σ) fails to satisfy it, the likelihood becomes zero (and the log-likelihood is negatively infinite). We call this the support condition.

(i) Now consider the estimation of μ first. The support condition compels us to only consider μ that satisfies $\overline{x} - \mu \in \mathcal{R}(\Sigma)$. Now, regardless of the choice of Σ, the middle term in the log-likelihood (the only term depending on μ),

$$-\frac{n}{2}(\overline{x}-\mu)^T\Sigma^\dagger(\overline{x}-\mu),$$

attains its maximum of 0 when $\mu = \overline{x}$. This choice of μ trivially satisfies the support condition $\overline{x}-\mu \in \mathcal{R}(\Sigma)$, regardless of the choice of Σ. Thus $\overline{X}$ is an MLE of μ. For uniqueness, let y be a candidate estimator satisfying the support condition, i.e. $\overline{x} - y \in \mathcal{R}(\Sigma)$. Since $\Sigma^\dagger \succ 0$ on $\mathcal{R}(\Sigma)$, we have $(\overline{x}-\mu)^T\Sigma^\dagger(\overline{x}-\mu) = 0 \Longrightarrow y = \overline{x}$.

(ii) Note that wlog we can take $\mathcal{R}\big(\widehat{\Sigma}\big) = \mathbb{R}^p$ (otherwise we rotate the space). Equivalently, we can take Σ non-singular. Therefore, any candidate Σ is also non-singular, so our search space is that of strictly positive definite matrices. When we plug in the MLE of μ, the likelihood reduces to,

$$-\frac{n}{2}\log|\Sigma| - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big) = \frac{n}{2}\log|\Theta| - \frac{n}{2}\text{trace}\big(\Theta\widehat{\Sigma}\big),$$

where $\Theta = \Sigma^{-1}$. Now let $\widehat{V} = \widehat{\Sigma}^{\frac{1}{2}} \succ 0$. Define $\Psi = \widehat{V}\Theta\widehat{V}$ and note that,

$$\log|\Psi| = \log\big(|\widehat{V}||\Theta||\widehat{V}|\big) = \log\big(|\widehat{V}|\widehat{V}|\big) + \log|\Theta| = \text{constant} + \log|\Theta|,$$

while,

$$\text{trace}\big(\Theta\widehat{\Sigma}\big) = \text{trace}\big(\widehat{V}^{-1}\Psi\widehat{V}^{-1}\widehat{V}\widehat{V}\big) = \text{trace}(\Psi).$$

Hence, up to constants, the log-likelihood may be expressed as,

$$\log|\Psi| - \text{trace}(\Psi) = \sum_{i=1}^p \log(\lambda_i(\Psi)) - \sum_{i=1}^p \lambda_i(\Psi) = \sum_{i=1}^p \big(\log(\lambda_i(\Psi)) - \lambda_i(\Psi)\big),$$

which can be optimized for each λ_i separately. Noting that 1 is the unique maximum of $\log(x) - x$ over $x > 0$, we get the unique MLE for $\lambda_i(\Psi) = 1$ for all i, i.e. $\Psi = \mathbb{I}_{p\times p}$, i.e. at $\Theta = \widehat{\Sigma}^{-1}$, i.e. at $\Sigma = \widehat{\Sigma}$.

(iii) We plug in the MLE for the mean in the general form of the log-likelihood to get (up to constants),

$$l(\overline{x},\Sigma) = \begin{cases} -\sum_{j=1}^{\text{rank}(\Sigma)}\log(\lambda_j(\Sigma)) - \text{trace}\big(\Sigma^\dagger\widehat{\Sigma}\big), & \text{when } \mathcal{R}\big(\widehat{\Sigma}\big) \subseteq \mathcal{R}(\Sigma) \\ -\infty, & \text{otherwise} \end{cases}.$$

We now consider the sets $C_1 = \{\Sigma : \mathcal{R}(\widehat{\Sigma}) \subseteq \mathcal{R}(\Sigma)\}^c$, $C_2 = \{\mathcal{R}(\widehat{\Sigma}) = \mathcal{R}(\Sigma)\}$ and $C_3 = \{\Sigma : \mathcal{R}(\widehat{\Sigma}) \subset \mathcal{R}(\Sigma)\}$. Plugging in the MLE for the mean, we see that: over C_1, the likelihood is zero. Over C_2, the maximal log-likelihood is finite and is attained uniquely at $\widehat{\Sigma}$. Finally, over C_3, we may obtain a sequence with log-likelihood diverging to ∞ as follows. Take $\Sigma_m := \widehat{\Sigma} + \alpha_m v v^T$ where $v \in \mathcal{R}^{\perp}(\widehat{\Sigma})$ is a unit vector and $0 < \alpha_m < \lambda_{\widehat{r}}(\widehat{\Sigma})$ with $\alpha_m \downarrow 0$. For all α_m, the trace term yields the same value $-\widehat{r}$. The log-det term on the other hand equals $-\left(\sum_{j=1}^{\widehat{r}} \log\left(\lambda_j(\widehat{\Sigma})\right)\right) - \log(\alpha_m)$ which diverges. □

Corollary 6.1.3 (The non-singular and low-dimensional case)**.** *If $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu, \Sigma)$ with $n > p$ and $\Sigma \succ 0$, then the unique MLE of μ and Σ are $\overline{X}$ and $\widehat{\Sigma}$ respectively. Furthermore, $\overline{X}$ and $\frac{n\widehat{\Sigma}}{(n-1)}$ are minimum variance unbiased estimators of μ and Σ.*

Proof. $(*)$ We only prove this Corollary for the mean, where the proof for the covariance follows similar ideas. Let $\overline{X}$ be the MLE of μ. Note that it is clearly unbiased, since $\mathbb{E}(\overline{X}) = \mu$. Now, let T be any other unbiased estimator of μ. We need to show that $v^T \mathrm{Cov}(\overline{X}) v \leq v^T \mathrm{Cov}(T) v$ for all $v \in \mathbb{R}^p$. The key is to observe that $v^T \overline{X}$ is the unique minimum variance unbiased estimator for $v^T \mu$, by known 1-dimensional results. Therefore, observing that $v^T T$ is unbiased for $v^T \mu$ by linearity, we have that $\mathrm{Cov}(v^T \overline{X}) \leq \mathrm{Cov}(v^T T)$, or equivalently $v^T \mathbb{E}\left[(\overline{X} - \mu)(\overline{X} - \mu)^T\right] v \leq v^T \mathbb{E}\left[(T - \mu)(T - \mu)^T\right] v$ for all $v \in \mathbb{R}^p$, which is precisely the claim. □

Proposition 6.1.4 (Parametrisation equivariance of the MLE)**.** *For any transformation g, if $\widehat{\theta}$ is MLE of θ then $g(\widehat{\theta})$ is MLE of $g(\theta)$. Furthermore, if g is 1-1, then this MLE is unique.*

Proof. $(*)$ Consider a statistical model parametrized by θ, and let $\widehat{\theta}$ be the MLE of θ:

$$\widehat{\theta} = \arg\max_{\theta} L(\theta | \boldsymbol{x}). \tag{6.7}$$

Define for $g(\theta)$ the induced likelihood function L^* given by,

$$L^*(\eta | \boldsymbol{x}) = \sup_{\{\theta : g(\theta) = \eta\}} L(\theta | \boldsymbol{x}). \tag{6.8}$$

Now, let $\widehat{\eta}$ denote the value that maximizes $L^*(\eta, \boldsymbol{x})$, which is the MLE of $\eta = g(\theta)$. We must show that $L^*(\widehat{\eta} | \boldsymbol{x}) = L^*\left[g(\widehat{\theta}) | \boldsymbol{x}\right]$. Note that:

$$L^*(\widehat{\eta} | \boldsymbol{x}) = \sup_{\eta} \sup_{\{\theta : g(\theta) = \eta\}} L(\theta | \boldsymbol{x}) = \sup_{\theta} L(\theta | \boldsymbol{x}) = L(\widehat{\theta} | \boldsymbol{x}), \tag{6.9}$$

where the second equality follows since the iterated maximization over θ, which is attained at $\widehat{\theta}$. Furthermore,

$$L(\widehat{\theta} | \boldsymbol{x}) = \sup_{\{\theta : g(\theta) = g(\widehat{\theta})\}} L(\theta | \boldsymbol{x}) = L^*\left[g(\widehat{\theta}) | \boldsymbol{x}\right]. \tag{6.10}$$

Hence the string of equalities shows that $L(\widehat{\eta} | \boldsymbol{x}) = L^*\left[g(\widehat{\theta}) | \boldsymbol{x}\right]$ and that $g(\widehat{\theta})$ is the MLE of $g(\theta)$. Concerning the uniqueness, note that if $\widehat{\eta}, \widehat{\eta}'$ both maximize the transformed likelihood, then necessarily $g^{-1}(\widehat{\eta}) = g^{-1}(\widehat{\eta}') = \widehat{\theta}$ so that, by injectivity of the inverse, $\widehat{\eta} = \widehat{\eta}'$. □

Thus, whenever $\Sigma \succ 0$ and $n > p$, we obtain the immediate corollaries:

- The **MLE of the precision matrix** $\Theta = \Sigma^{-1}$ is given by $\widehat{\Theta} := \widehat{\Sigma}^{-1}$.
- The **MLE of the correlation matrix** R is given by $\widehat{R} := \mathrm{diag}(\widehat{\sigma}_1^{-1}, ..., \widehat{\sigma}_p^{-1}) \widehat{\Sigma} \, \mathrm{diag}(\widehat{\sigma}_1^{-1}, ..., \widehat{\sigma}_p^{-1})$, where $\widehat{\sigma}_j$ is the j-th diagonal element of $\widehat{\Sigma}$.

6.2 The Likelihood Ratio Test and Gaussian Hypothesis Testing

In this section, unless we have a very special hypothesis structure (e.g. simple versus simple, or one-sided concerning a single coordinate of the mean), there will generally be no unequivocal choice of test (i.e. no optimal test). In special cases related to the mean (e.g. when Σ is known to be diagonal), one can do separate univariate tests and combine them with a careful correction. A general and usually sensible general method is based on the likelihood ratio. This is not the only approach, and other methods may have some advantages. However it will suffice for the purpose of this section.

We consider the following general framework: $X = (X_1^T, ..., X_n^T)^T$ random vectors with likelihood $L(\mathcal{V}) = f_{\mathcal{V}}(X_1, ..., X_n)$ for $\mathcal{V} \in \Theta \subseteq \mathbb{R}^d$ where $\Theta = \Theta_0 \cup \Theta_1$ and $\Theta_0 \cap \Theta_1 = \varnothing$.

Definition 6.2.1 (Likelihood ratio test statistic)**.** *The **likelihood ratio statistic** for $H_0 : \mathcal{V} \in \Theta_0$ versus $H_1 : \mathcal{V} \in \Theta_1$ is:*

$$\Lambda := \frac{\sup_{\mathcal{V} \in \Theta} L(\mathcal{V})}{\sup_{\mathcal{V} \in \Theta_0} L(\mathcal{V})}. \tag{6.11}$$

The intuition is to know how much better do we do if we do not restrict the maximization of the likelihood to be over the subset Θ_0. In this framework, we reject for large Λ or of some monotone increasing continuous function of Λ. Often, the following increasing function is simple to calibrate:

$$2\log(\Lambda) = 2\left(\sup_{\mathcal{V} \in \Theta} \log(L(\mathcal{V})) - \sup_{\mathcal{V} \in \Theta_0} \log(L(\mathcal{V}))\right) =: 2(l^* - l_0^*). \tag{6.12}$$

Now, implementing a likelihood ratio test requires two steps: (1) determining the test statistic, and (2) calibrating the test statistic (or a monotone transformation $\tau(\Lambda)$ thereof). In this case, **calibration** refers to finding the distribution $\tau(\Lambda)$ under H_0 so that we can choose an appropriate quantile q to define the critical region for rejection:

$$\tau(\Lambda) \overset{H_0}{\sim} F \Longrightarrow \text{reject } H_0 \text{ at level } \alpha \text{ whenever } \tau(\Lambda) > q_{1-\alpha}(F). \tag{6.13}$$

Often, especially in exponential families (e.g. Gaussians), we can find the exact sampling law under H_0. However, more often, it is not tractable and we need an asymptotic approximation. A general such result for $\tau(\Lambda) = 2\log(\Lambda)$ is given by Wilks' Theorem. Before presenting this Theorem (without proof), we provide some regularity conditions that we will name **usual regularity conditions**:

(A1) The parameter space $\Theta \in \mathbb{R}^p$ is open.

(A2) The support of $f_{\mathcal{V}}$ is invariant with respect to $\mathcal{V}$.

(A3) All mixed partial derivatives of l with respect to $\mathcal{V}$ up to degree 3 exist and are continuous.

(A4) $\mathbb{E}[\nabla_{\mathcal{V}} l(X_i; \mathcal{V})] = 0$ for all $\mathcal{V}$ and $\text{Cov}[\nabla_{\mathcal{V}} l(X_i; \mathcal{V})] \succ 0$ for all $\mathcal{V}$.

(A5) $-\mathbb{E}[\nabla_{\mathcal{V}}^2 l(X_i; \mathcal{V})] \succ 0$ for all $\mathcal{V}$.

(A6) For $\|\mathcal{V} - u\| \leq \delta$ with M_{jkl} where $\mathbb{E}(M_{jkl}(X_i)) < \infty$ is such that there exists $\delta > 0$ such that for all $\mathcal{V} \in \Theta$ and all $1 \leq j, k, l \leq p$:

$$\left| \frac{\partial^3}{\partial \mathcal{V}_j \partial \mathcal{V}_k \partial \mathcal{V}_l} l(x; u) \right| \leq M_{jkl}(x).$$

Theorem 6.2.2 (Wilks' Theorem)**.** *Let $X_1, ..., X_n$ be i.i.d. random vectors with density (frequency) depending on $\mathcal{V} \in \mathbb{R}^p$ and satisfying the usual regularity conditions. If the MLE sequence $\widehat{\mathcal{V}}_n$ is consistent for $\mathcal{V}$, then the likelihood ratio statistic Λ_n for $H_0 : \{\mathcal{V}_j = \mathcal{V}_{j,0}\}_{j=1}^s$, $s \leq p$, satisfies $2\log(\Lambda_n) \xrightarrow{d} V \sim \chi_s^2$ when H_0 is true.*

Note that Wilks' Theorem applies **only** for a simple null (not composite), though this does not need to fix all the parameters ($s < p$ is allowed). Now, hypotheses of the form $H_0 : \{g_j(\mathcal{V}) = a_j\}_{j=1}^s$ for g_j differentiable real functions, may also be handled by Wilks' Theorem: (1) Define $(\phi_1, ..., \phi_p) = g(\mathcal{V}) = (g_1(\mathcal{V}), ..., g_p(\mathcal{V}))$, then (2) $g_{s+1}, ..., g_p$ are defined so that $\mathcal{V} \mapsto g(\mathcal{V})$ is 1-1, and finally (3) apply Wilks' Theorem with parameter ϕ.

6.2.1 One Sample - $\{H_0 : \mu = \mu_0\}$ versus $\{H_1 : \mu \neq \mu_0\}$ (Σ is known)

Let $X_1, ..., X_n$ be a random sample from $\mathcal{N}(\mu, \Sigma_{p\times p})$ with $\Sigma \succ 0$ known. Now, as Σ is known and H_0 is simple, the maximal log-likelihood under H_0 is given by (6.6):

$$l_0^* = l(\mu_0, \Sigma) = -\frac{n}{2}\log\det(2\pi\Sigma) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big) - \frac{n}{2}\big(\overline{X} - \mu_0\big)^T\Sigma^{-1}\big(\overline{X} - \mu_0\big). \tag{6.14}$$

On the other hand, the unrestricted maximal log-likelihood occurs at the unrestricted MLE,

$$l^* = l\big(\overline{X}, \Sigma\big) = -\frac{n}{2}\log\det(2\pi\Sigma) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big). \tag{6.15}$$

Hence $2\log(\Lambda) = 2(l^* - l_0^*) = n\big(\overline{X} - \mu_0\big)^T\Sigma^{-1}\big(\overline{X} - \mu_0\big)$ which under H_0 follows the χ_p^2-distribution. As a result, we reject H_0 at the level α if and only if,

$$2\log(\Lambda) = n\big(\overline{X} - \mu_0\big)^T\Sigma^{-1}\big(\overline{X} - \mu_0\big) > q_{1-\alpha}(\chi_p^2). \tag{6.16}$$

Remark 6.2.3. *Note that we can invert this test to get a 95% confidence region for μ in the form,*

$$\big\{\mu_0 \in \mathbb{R}^p : n\big(\overline{X} - \mu_0\big)^T\Sigma^{-1}\big(\overline{X} - \mu_0\big) \leq q_{1-\alpha}(\chi_p^2)\big\}, \tag{6.17}$$

i.e. all the possible μ_0 for which H_0 is not rejected.

6.2.2 One Sample - $\{H_0 : \mu = \mu_0\}$ versus $\{H_1 : \mu \neq \mu_0\}$ (Σ is unknown)

Before diving into the study of this test, we first state and prove two preliminary results.

Proposition 6.2.4 (Matrix determinant)**.** *If $\Sigma \succ 0$, then $\big|\Sigma + uu^T\big| = |\Sigma|\big(1 + u^T\Sigma^{-1}u\big)$.*

Proof. $(*)$ Let us first consider the case $\Sigma = \mathbb{I}$ and recall that the determinant corresponds to the product of the eigenvalues. Then note that the matrix $\mathbb{I} + vv^T$ has v as eigenvector, with corresponding eigenvalue $1 + v^Tv$. Being a positive matrix, we may complete to find a basis of orthogonal eigenvectors, all of which will be orthogonal to v, and thus with eigenvalue 1. Therefore, $|\mathbb{I} + vv^T| = 1 + v^Tv$. The general case then follows by using the multiplicativity of the determinant, and taking $u = \Sigma^{\frac{1}{2}}v$:

$$|\Sigma + uu^T| = \left|\Sigma^{\frac{1}{2}}(\mathbb{I} + vv^T)\Sigma^{\frac{1}{2}}\right| = |\Sigma||\mathbb{I} + vv^T| = |\Sigma||1 + v^Tv| = |\Sigma|\left|1 + u^T\Sigma^{-1}u\right|.$$

□

Proposition 6.2.5 (Sherman-Morrison formula)**.** *If $\Sigma \succ 0$, then:*

$$\big(\Sigma + uu^T\big)^{-1} = \Sigma^{-1} - \frac{1}{1 + u^T\Sigma^{-1}u}\Sigma^{-1}uu^T\Sigma^{-1}. \tag{6.18}$$

Proof. $(*)$ First note that when $\Sigma = \mathbb{I}$ we may expand by power series:

$$(\mathbb{I}+vv^T)^{-1} = \sum_{j\geq 0}(-vv^T)^j = \mathbb{I}+\sum_{j\geq 1}(-vv^T)^j = \mathbb{I}+\sum_{k\geq 0}(-vv^T)^{k+1} = \mathbb{I}-\sum_{k\geq 0}(-vv^T)^k(vv^T) = \mathbb{I}-\frac{1}{1+v^Tv}vv^T.$$

Thus in the general case it follows by taking $u = \Sigma^{\frac{1}{2}}v$:

$$(\Sigma+uu^T)^{-1} = \Sigma^{-\frac{1}{2}}(\mathbb{I}+vv^T)^{-1}\Sigma^{-\frac{1}{2}} = \Sigma^{-\frac{1}{2}}\left(\mathbb{I}-\frac{1}{1+v^Tv}vv^T\right)\Sigma^{-\frac{1}{2}} = \Sigma^{-1}-\frac{1}{1+u^T\Sigma^{-1}u}\Sigma^{-1}uu^T\Sigma^{-1}.$$

□

Let $X_1,...,X_n$ be a random sample from $\mathcal{N}(\mu,\Sigma_{p\times p})$ with $\Sigma \succ 0$ unknown, $n > p$. Σ must be estimated under H_0 and also under H_1, therefore both hypotheses are composite. Based on our results,

- Under H_0 the MLE for μ and Σ are μ_0 and $\widehat{\Sigma}_0 = \frac{1}{n}\sum_{i=1}^n(X_i-\mu_0)(X_i-\mu_0)^T = \widehat{\Sigma}+\delta\delta^T$, where $\delta = \overline{X}-\mu_0$.
- Under H_1, the MLE for μ and Σ are $\overline{X}$ and $\widehat{\Sigma}_1 = \widehat{\Sigma}$.

Plugging into the log-likelihood (6.6) we obtain,

$$l_0^* = l(\mu_0,\widehat{\Sigma}+\delta\delta^T) = -\frac{n}{2}\left[p\log(2\pi)+\log|\widehat{\Sigma}+\delta\delta^T|+\delta^T\widehat{\Sigma}^{-1}\delta+\text{trace}\left((\widehat{\Sigma}+\delta\delta^T)^{-1}\widehat{\Sigma}\right)\right]. \tag{6.19}$$

As $\widehat{\Sigma} \succ 0$ a.s. when $n > p$, its rank 1 perturbed determinant is given by $\left|\widehat{\Sigma}+\delta\delta^T\right| = |\widehat{\Sigma}|\left(1+\delta^T\widehat{\Sigma}^{-1}\delta\right)$ [Proposition (6.2.4)], which yields by Proposition (6.2.5):

$$l_0^* = -\frac{n}{2}\left[p\log(2\pi)+\log|\widehat{\Sigma}|+\log\left(1+\delta^T\widehat{\Sigma}^{-1}\delta\right)+p\right]. \tag{6.20}$$

In the unrestricted case, we calculate:

$$l^* = l\big(\overline{X},\widehat{\Sigma}\big) = -\frac{n}{2}\left[p\log(2\pi)+\log|\widehat{\Sigma}|+p\right]. \tag{6.21}$$

Thus,

$$2\log(\Lambda) = 2(l^*-l_0^*) = n\log\left(1+\delta^T\widehat{\Sigma}\delta\right) = n\log\left(1+\frac{1}{n-1}(n-1)\delta^T\widehat{\Sigma}^{-1}\delta\right). \tag{6.22}$$

So this statistic depends upon $(n-1)\delta^T\widehat{\Sigma}^{-1}\delta$ which follows the $T^2(p,n-1)$ distribution (hence often referred to as the Hotelling one-sample T^2 statistic). As $2\log(\Lambda)$ is a strictly increasing function of the Hotelling statistic, we reject H_0 at the level α if and only if $(n-1)\delta^T\widehat{\Sigma}^{-1}\delta > q_{1-\alpha}\big(T^2(p,n-1)\big)$.

6.2.3 Two Samples - $\{H_0 : \mu_X = \mu_Y\}$ versus $\{H_1 : \mu_X \neq \mu_Y\}$ with $\Sigma_X = \Sigma_Y = \Sigma$ known

Suppose that for $\Sigma_{p\times p} \succ 0$ known we have independent samples $X_1,...,X_n \overset{iid}{\sim} \mathcal{N}(\mu_X,\Sigma)$ and $Y_1,...,Y_m \overset{iid}{\sim} \mathcal{N}(\mu_Y,\Sigma)$ and we wish to discern whether they share the same mean or not. The two samples are not identically distributed under both hypothesis. Still a global likelihood with respect to $(\mu_X^T,\mu_Y^T)^T \in \mathbb{R}^{2p}$ is obtained by multiplication:

$$l(\mu_X,\mu_Y) = -\frac{n+m}{2}\log|\Sigma| - \frac{n}{2}(\overline{x}-\mu_X)^T\Sigma^{-1}(\overline{x}-\mu_X) - \frac{m}{2}(\overline{y}-\mu_Y)^T\Sigma^{-1}(\overline{y}-\mu_Y) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_X\big) - \frac{m}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_Y\big).$$

The null $\{H_0 : \mu_X = \mu_Y\}$ corresponds to a restriction on the parameter space, and the corresponding restricted likelihood is,

$$l(\mu_X) = -\frac{n+m}{2}\log|\Sigma| - \frac{n}{2}(\overline{x}-\mu_X)^T\Sigma^{-1}(\overline{x}-\mu_X) - \frac{m}{2}(\overline{y}-\mu_X)^T\Sigma^{-1}(\overline{y}-\mu_X) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_X\big) - \frac{m}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_Y\big).$$

Observe that the restricted likelihood corresponds to that of the parameter μ_X in a $\mathcal{N}(\mu_X, \Sigma)$ model, based on an i.i.d. sample of size $n+m$. This is maximized with respect to μ_X at the *pooled sample mean*,

$$M = \frac{1}{n+m}\left(\sum_{i=1}^{n} X_i + \sum_{j=1}^{m} Y_j\right) = \frac{n}{n+m}\overline{X} + \frac{m}{n+m}\overline{Y}. \tag{6.23}$$

It is not difficult to see that the unrestricted likelihood is maximized at $\big(\widehat{\mu}_X, \widehat{\mu}_Y\big) = \big(\overline{X}, \overline{Y}\big)$, and thus the difference $2(l^* - l_0^*)$ may be seen to equal to:

$$\begin{aligned}
2(l^* - l_0^*) &= n\big(\overline{X}-M\big)^T\Sigma^{-1}\big(\overline{X}-M\big) + m\big(\overline{Y}-M\big)^T\Sigma^{-1}\big(\overline{Y}-M\big) \\
&= n\frac{m}{n+m}\big(\overline{X}-\overline{Y}\big)^T\Sigma^{-1}\frac{m}{n+m}\big(\overline{X}-\overline{Y}\big) + m\frac{n}{n+m}\big(\overline{Y}-\overline{X}\big)^T\Sigma^{-1}\frac{n}{n+m}\big(\overline{Y}-\overline{X}\big) \\
&= \frac{nm}{n+m}\big(\overline{X}-\overline{Y}\big)^T\left(\frac{n}{n+m}\Sigma^{-1} + \frac{m}{n+m}\Sigma^{-1}\right)\big(\overline{X}-\overline{Y}\big) \\
&= \frac{nm}{n+m}\big(\overline{X}-\overline{Y}\big)^T\Sigma^{-1}\big(\overline{X}-\overline{Y}\big).
\end{aligned}$$

Under H_0, $\sqrt{\frac{nm}{n+m}}\big(\overline{X}-\overline{Y}\big) \sim \mathcal{N}(0,\Sigma)$, thus we reject H_0 at the level α if and only if,

$$2\log(\Lambda) = \frac{nm}{n+m}\big(\overline{X}-\overline{Y}\big)^T\Sigma^{-1}\big(\overline{X}-\overline{Y}\big) > q_{1-\alpha}(\chi^2_p). \tag{6.24}$$

6.2.4 Two Samples - $\{H_0 : \mu_X = \mu_Y\}$ versus $\{H_1 : \mu_X \neq \mu_Y\}$ with $\Sigma_X = \Sigma_Y = \Sigma$ unknown

Suppose that for $\Sigma_{p\times p} \succ 0$ unknown, $n+m > p$, and we have independent samples $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu_X, \Sigma)$ and $Y_1, ..., Y_m \overset{iid}{\sim} \mathcal{N}(\mu_Y, \Sigma)$ where we wish to discern whether they share the same mean or not. In this setting, one has an a.s. invertible *pooled empirical covariance* such that,

$$\widehat{\Sigma} = \frac{1}{n+m}\left(\sum_{i=1}^{n} X_i X_i^T + \sum_{j=1}^{m} Y_j Y_j^T\right) - MM^T = \frac{n}{n+m}\widehat{\Sigma}_X + \frac{m}{n+m}\widehat{\Sigma}_Y, \tag{6.25}$$

where we have used $M = \frac{n}{n+m}\overline{X} + \frac{m}{n+m}\overline{Y}$. So, the likelihood of the pooled sample is:

$$\begin{aligned}
l(\mu_X, \mu_Y, \Sigma) &= -\frac{n+m}{2}\log|\Sigma| - \frac{n}{2}(\overline{x}-\mu_X)^T\Sigma^{-1}(\overline{x}-\mu_X) - \frac{m}{2}(\overline{y}-\mu_Y)^T\Sigma^{-1}(\overline{y}-\mu_Y) - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_X\big) \\
&\qquad - \frac{m}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_Y\big) \\
&= -\frac{n+m}{2}\log|\Sigma| - \frac{n}{2}(\overline{x}-\mu_X)^T\Sigma^{-1}(\overline{x}-\mu_X) - \frac{m}{2}(\overline{y}-\mu_Y)^T\Sigma^{-1}(\overline{y}-\mu_Y) \\
&\qquad - \frac{n+m}{2}\underbrace{\text{trace}\left(\Sigma^{-1}\left(\frac{n}{n+m}\widehat{\Sigma}_X + \frac{m}{n+m}\widehat{\Sigma}_Y\right)\right)}_{\equiv\text{trace}\left(\Sigma^{-1}\widehat{\Sigma}\right)}.
\end{aligned}$$

With this form in mind, we can now easily verify that: (i) in the unconstrained case, the MLEs are the pooled mean and pooled covariance, (ii) in the constrained case, the MLEs are the two separate sample means and the

pooled covariance, and (iii) repeating similar calculations as when Σ was known we arrive at:

$$2(l^* - l_0^*) = \frac{nm}{n+m}\big(\overline{X}-\overline{Y}\big)^T\widehat{\Sigma}^{-1}\big(\overline{X}-\overline{Y}\big) = nm\big(\overline{X}-\overline{Y}\big)^T\big(n\widehat{\Sigma}_X + m\widehat{\Sigma}_Y\big)^{-1}\big(\overline{X}-\overline{Y}\big) =: Q. \tag{6.26}$$

When suitably rescaled, this is known as the Hotelling's two-sample T^2. We provide the detailed result in the following proposition.

Proposition 6.2.6. *For $n+m > p > 1$, let $X_{n\times p}$ and $Y_{m\times p}$ be independent data matrices from $\mathcal{N}(\mu_X, \Sigma_X)$ and $\mathcal{N}(\mu_Y, \Sigma_Y)$ respectively. If $\mu_X = \mu_Y$ and $\Sigma_X = \Sigma_Y$ then,*

$$\left(1 - \frac{2}{n+m}\right) Q \sim T^2(p, n+m-2). \tag{6.27}$$

Proof. From the Gaussian Sampling Theorem (5.1.2) and independence of X and Y we have:

$$\delta := \sqrt{\frac{nm}{n+m}}\big(\overline{X}-\overline{Y}\big) \sim \mathcal{N}_p\left(\mu_X - \mu_Y, \frac{nm\Sigma_X}{n(n+m)} + \frac{nm\Sigma_Y}{m(n+m)}\right) \overset{H_0}{\equiv} \mathcal{N}(0, \Sigma)$$

$$n\widehat{\Sigma}_X \sim W_p(\Sigma_X, n-1) \overset{H_0}{\equiv} W_p(\Sigma, n-1)$$

$$m\widehat{\Sigma}_Y \sim W_p(\Sigma_Y, m-1) \overset{H_0}{\equiv} W_p(\Sigma, m-1),$$

where $\{H_0 : \mu_X = \mu_Y \text{ and } \Sigma_X = \Sigma_Y = \Sigma\}$. By independence, and the additivity property of the Wishart, we thus have under H_0 that:

$$n\widehat{\Sigma}_X + m\widehat{\Sigma}_Y \sim W_p(\Sigma, n-1+m-1) \equiv W_p(\Sigma, n+m-2).$$

Moreover, the Gaussian Sampling Theorem states that $\widehat{\Sigma}_X \perp \overline{X}$, $\widehat{\Sigma}_Y \perp \overline{Y}$ and as $X \perp Y$ then $\big(\overline{X}-\overline{Y}\big) \perp \big(n\widehat{\Sigma}_X + m\widehat{\Sigma}_Y\big)$. Thus, by the Hotelling Lemma (5.2.10):

$$\frac{n+m-2}{n+m} nm\big(\overline{X}-\overline{Y}\big)^T\big(n\widehat{\Sigma}_X + m\widehat{\Sigma}_Y\big)^{-1}\big(\overline{X}-\overline{Y}\big) \equiv \frac{n+m-2}{n+m} Q \sim T^2(p, n+m-2).$$

□

Thus, under H_0, we reject H_0 at level α if and only if,

$$\left(1 - \frac{2}{n+m}\right) Q > q_{1-\alpha}\big(T^2(p, n+m-2)\big). \tag{6.28}$$

6.2.5 Correlation Test - $\{H_0 : \rho_{ij} = 0\}$ versus $\{H_1 : \rho_{ij} \neq 0\}$

For $n > 2$ and $\Sigma \succ 0$, let $\mathrm{Var}(X_1) = \sigma_X^2$, $\mathrm{Var}(Y_1) = \sigma_Y^2$ and $|\rho| < 1$ such that it suffices to consider the following setting:

$$\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, ..., \begin{pmatrix} X_n \\ Y_n \end{pmatrix} \overset{iid}{\sim} \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \sigma_X\sigma_Y\rho \\ \sigma_Y\sigma_X\rho & \sigma_Y^2 \end{pmatrix}\right), \tag{6.29}$$

where we test $\{H_0 : \rho = 0\}$ versus $\{H_1 : \rho \neq 0\}$. The unrestricted log-likelihood occurs at the unrestricted MLE,

$$l^* = l\big(\overline{X}, \widehat{\Sigma}\big) = -\frac{n}{2}\log\det\big(2\pi\widehat{\Sigma}\big) - \frac{n}{2}\mathrm{trace}\big(\widehat{\Sigma}^{-1}\widehat{\Sigma}\big) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\big(\widehat{\sigma}_X^2\widehat{\sigma}_Y^2 - \widehat{\rho}^2\widehat{\sigma}_X^2\widehat{\sigma}_Y^2\big) - n. \tag{6.30}$$

The log-likelihood is always maximized with respect to the mean at the sample mean. Thus for the restricted log-likelihood (under H_0) it suffices to consider the function,

$$l\big(\overline{X},\sigma_X^2,\sigma_Y^2\big)=-\frac{n}{2}\log\big(2\pi\sigma_X^2\sigma_Y^2\big)-\frac{n}{2}\text{trace}\big(\text{diag}\big(\sigma_X^{-2},\sigma_Y^{-2}\big)\widehat{\Sigma}\big)=-\frac{n}{2}\log(2\pi)-\frac{n}{2}\log\big(\sigma_X^2\big)-\frac{n}{2}\log\big(\sigma_Y^2\big)-\frac{n}{2}\frac{\widehat{\sigma}_X^2}{\sigma_X^2}-\frac{n}{2}\frac{\widehat{\sigma}_Y^2}{\sigma_Y^2},$$

with unique maximum at $\sigma_X^2=\widehat{\sigma}_X^2$ and $\sigma_Y^2=\widehat{\sigma}_Y^2$ (recalling that 1 is the unique maximum of $\log(x)-x$ over $x>0$). This maximum equals,

$$l_0^*=-\frac{n}{2}\log(2\pi)-\frac{n}{2}\log\big(\widehat{\sigma}_X^2\big)-\frac{n}{2}\log\big(\widehat{\sigma}_Y^2\big)-n. \tag{6.31}$$

It follows that the likelihood ratio $2\log(\Lambda)=2(l^*-l_0^*)=-\frac{n}{2}\log\big(1-\widehat{\rho}^2\big)$ is a monotone function of the squared sample correlation $\widehat{\rho}^2$. This is, in turn, a monotone function of $\kappa=\widehat{\rho}^2/(1-\widehat{\rho}^2)$ and thus we reject H_0 for κ large. Furthermore, we can be even more precise by considering the following proposition.

Proposition 6.2.7. *In the context of a correlation test, we have:*

$$(n-2)\frac{\widehat{\rho}^2}{1-\widehat{\rho}^2}\overset{H_0}{\sim}T^2(1,n-2). \tag{6.32}$$

Proof. Let $X=(X_1,...,X_n)^T\sim\mathcal{N}(\mu_X\mathbb{1}_n,\sigma_X^2\mathbb{I}_n)$, $Y=(Y_1,...,Y_n)^T\sim\mathcal{N}(\mu_Y\mathbb{1}_n,\sigma_Y^2\mathbb{I}_n)$ and $H_n=\mathbb{I}_{n\times n}-\frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T$ is the centring matrix (projection onto $\text{span}^{\perp}(\mathbb{1}_n)$). In particular, $H_n=U\Omega U^T$ with the $n\times n$ diagonal matrix $\Omega=\text{diag}(1,...,1,0)$ and $U^TU=\mathbb{I}_{n\times n}$. Then, $\Omega U^TX\sim\mathcal{N}(0,\sigma_X^2\Omega)$, $\Omega U^TY\sim\mathcal{N}(0,\sigma_Y^2\Omega)$ and obviously $\Omega^2=\Omega$ thus,

$$\widehat{\rho}=\frac{X^TH_nY}{\sqrt{X^TH_nXY^TH_nY}}=\frac{X^TU\Omega\Omega U^TY}{\sqrt{X^TU\Omega\Omega U^TXY^TU\Omega\Omega U^TY}}\overset{d}{=}\left(\frac{W}{\|W\|}\right)^T\frac{V}{\|V\|}, \tag{6.33}$$

where $W,V\overset{iid}{\sim}\mathcal{N}\big(0,\mathbb{I}_{(n-1)\times(n-1)}\big)$ (independence comes from H_0). Consequently we have the following collection of facts:

- $\frac{W}{\|W\|},\frac{V}{\|V\|}\overset{iid}{\sim}$ Uniform(on the surface of the unit sphere in $\mathbb{R}^{n-1}$).
- $\frac{V}{\|V\|}=Ve_1$ where $V=\left(\frac{V}{\|V\|},V_2,...,V_{n-1}\right)$ is a random orthogonal matrix obtained by randomly extending $\frac{V}{\|V\|}$ to an orthonormal basis.
- $\left(\frac{W}{\|W\|}\right)^T\frac{V}{\|V\|}=\frac{(V^TW)^T}{\|W\|}e_1=\frac{(V^TW)^T}{\|V^TW\|}e_1\overset{d}{=}\left(\frac{W}{\|W\|}\right)^Te_1$ by independence.

Therefore, $\widehat{\rho}^2\overset{d}{=}\left(\frac{W^Te_1}{\|W\|}\right)^2\overset{d}{=}\frac{Z_1^2}{Z_1^2+...+Z_{n-1}^2}$ for Z_i i.i.d. standard normal random variables. Thus,

$$\frac{\widehat{\rho}^2}{1-\widehat{\rho}^2}=\frac{Z_1^2}{Z_2^2+...+Z_{n-1}^2}, \tag{6.34}$$

which is the ratio of two independent χ^2 random variables. When each is renormalized by their respective degrees of freedom, we obtain:

$$(n-2)\frac{\widehat{\rho}^2}{1-\widehat{\rho}^2}\sim F_{1,n-2}\equiv T^2(1,n-2). \tag{6.35}$$

□

Thus, since $(n-2)\kappa\overset{H_0}{\sim}T^2(1,n-2)$, we reject H_0 at level α if and only if,

$$(n-2)\frac{\widehat{\rho}^2}{1-\widehat{\rho}^2}>q_{1-\alpha}\big(T^2(1,n-2)\big). \tag{6.36}$$

Remark that $\frac{\widehat{\rho}^2}{1-\widehat{\rho}^2}$ (and the test) can be arrived at using the regression representation. Furthermore, assuming that $\Sigma \succ 0$ it implies that $|\rho_{ij}| < 1$. Thus the derived test does not apply in the boundary case. However this is not an issue since if $\rho_{ij} = \pm 1$, we will see it immediately in the data (the two coordinates will realise perfectly along a line).

6.2.6 Partial Correlation Test - $\{H_0 : \rho_{ij|k} = 0\}$ versus $\{H_1 : \rho_{ij|k} \neq 0\}$

Say we have a sample of size $n > 3$, $(X_i, Y_i, Z_i)^T \overset{iid}{\sim} \mathcal{N}(\mu, \Sigma_{3\times 3})$, with $\Sigma \succ 0$. We wish to test whether X is partially correlated with Y given Z, $\{H_0 : \rho_{XY|Z} = 0\}$ versus $\{H_1 : \rho_{XY|Z} \neq 0\}$ where we recall that $\rho_{XY|Z} = -\theta_{XY}/\sqrt{\theta_{XX}\theta_{YY}}$. Note that we do not have to perform likelihood again, since we can simply use the regression representation [Proposition (3.3.4)]:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \sim \mathcal{N}\left(\mu, \begin{pmatrix} \Sigma_X & \Sigma_{XY} & \Sigma_{XZ} \\ \Sigma_{XY}^T & \Sigma_Y & \Sigma_{YZ} \\ \Sigma_{XZ}^T & \Sigma_{YZ}^T & \Sigma_Z \end{pmatrix}\right) \iff \begin{pmatrix} \epsilon_X \\ \epsilon_Y \\ Z \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ \mu_Z \end{pmatrix}, \begin{pmatrix} \begin{pmatrix} \Theta_X & \Theta_{XY} \\ \Theta_{XY}^T & \Theta_Y \end{pmatrix}^{-1} & 0 \\ 0 & \Sigma_Z \end{pmatrix}\right), \quad (6.37)$$

with $\epsilon_X = X - \Sigma_{XZ}\Sigma_Z^\dagger(Z - \mu_Z)$ and $\epsilon_Y = Y - \Sigma_{YZ}\Sigma_Z^\dagger(Z - \mu_Z)$. In the 2×2 case, the inverse has the explicit formula,

$$\begin{pmatrix} \theta_X & \theta_{XY} \\ \theta_{XY}^T & \theta_Y \end{pmatrix}^{-1} = |\Theta|^{-1} \begin{pmatrix} \theta_Y & -\theta_{XY} \\ -\theta_{XY} & \theta_X \end{pmatrix}. \quad (6.38)$$

As a result, we have reduced the problem of testing whether the correlation is zero in a 2×2 Gaussian setting. Therefore, the likelihood ratio test will reject for large values of $\frac{\widehat{\rho}^2_{\epsilon_X,\epsilon_Y}}{1-\widehat{\rho}^2_{\epsilon_X,\epsilon_Y}}$ for $\widehat{\rho}_{\epsilon_X,\epsilon_Y}$ the MLE of the correlation of ϵ_X and ϵ_Y. We don't actually observe the sample of $(\epsilon_X, \epsilon_Y, Z)^T$ but the induced likelihood is equivalent to that induced by the observable sample (i.e. it gives the same values at the same parameter choices). Thus, using equivariance, since

$$\rho_{\epsilon_X,\epsilon_Y} = \frac{-|\Theta^{-1}|\theta_{XY}}{\sqrt{|\Theta^{-1}|\theta_{XX}|\Theta^{-1}|\theta_{YY}}} = -\frac{\theta_{XY}}{\sqrt{\theta_{XX}\theta_{YY}}} \quad (6.39)$$

we get that the MLE $\widehat{\rho}_{\epsilon_X,\epsilon_Y}$ equals $-\frac{\widehat{\theta}_{XY}}{\sqrt{\widehat{\theta}_{XX}\widehat{\theta}_{YY}}}$. It turns out that the distribution under the null is now $T^2(1, n-3)$ after re-scaling by $(n-3)$, thus we reject H_0 at level α if and only if,

$$(n-3)\frac{\widehat{\theta}^2_{XY}/(\widehat{\theta}_X\widehat{\theta}_Y)}{1-\widehat{\theta}^2_{XY}/(\widehat{\theta}_X\widehat{\theta}_Y)} > q_{1-\alpha}\big(T^2(1, n-3)\big). \quad (6.40)$$

6.2.7 One Sample - $\{H_0 : \Sigma = \Sigma_0\}$ versus $\{H_1 : \Sigma \neq \Sigma_0\}$ (with $\Sigma_0 \succ 0$)

Let $X_1, ..., X_n$ be a random sample from $\mathcal{N}(\mu, \Sigma_{p\times p})$ with $\Sigma \succ 0$. The restricted MLEs (under H_0) are $\overline{X}$ and Σ_0, and the unrestricted MLEs are $\overline{X}$ and $\widehat{\Sigma}$. Thus:

$$l_0^* = l\big(\overline{X}, \Sigma_0\big) = -\frac{n}{2}\log|2\pi\Sigma_0| - \frac{n}{2}\mathrm{trace}\big(\Sigma_0^{-1}\widehat{\Sigma}\big) \quad (6.41)$$

$$l^* = l\big(\overline{X}, \widehat{\Sigma}\big) = -\frac{n}{2}\log|2\pi\widehat{\Sigma}| - \frac{n}{2}\mathrm{trace}\big(\widehat{\Sigma}^{-1}\widehat{\Sigma}\big) = -\frac{n}{2}\log|2\pi\widehat{\Sigma}| - \frac{np}{2}, \quad (6.42)$$

which yields,

$$2\log(\Lambda) = 2(l^* - l_0^*) = -n\log\left(\frac{|\Sigma_0|}{|\widehat{\Sigma}|}\right) + n\mathrm{trace}\big(\Sigma_0^{-1}\widehat{\Sigma}\big) = n\log|\Sigma_0^{-1}\widehat{\Sigma}| + n\mathrm{trace}\big(\Sigma_0^{-1}\widehat{\Sigma}\big) - np. \quad (6.43)$$

If α and γ are the arithmetic and geometric means of the eigenvalues of $\Sigma_0^{-1}\widehat{\Sigma}$ respectively, then $\text{trace}\big(\Sigma_0^{-1}\widehat{\Sigma}\big) = p\alpha$ and $|\Sigma_0^{-1}\widehat{\Sigma}| = \gamma^p$. Thus,

$$2\log(\Lambda) = np(\alpha - \log(\gamma) - 1). \tag{6.44}$$

The exact distribution of this statistic is non-trivial to obtain, however Wilks' Theorem applies and thus we may use the asymptotic approximation χ^2_m, where $m = p(p+1)/2$. Thus, if n is large enough, we reject H_0 at level α if and only if,

$$2\log(\Lambda) = np(\alpha - \log(\gamma) - 1) > q_{1-\alpha}\big(\chi^2_{p(p+1)/2}\big). \tag{6.45}$$

6.2.8 Two Samples - $\{H_0 : \Sigma_X = \Sigma_Y\}$ versus $\{H_1 : \Sigma_X \neq \Sigma_Y\}$ (with $\Sigma_X, \Sigma_Y \succ 0$)

Suppose that for $\Sigma_X, \Sigma_Y \succ 0$ unknown, $n, m > p$, and we have independent samples $X_1, ..., X_n \overset{iid}{\sim} \mathcal{N}(\mu_X, \Sigma_X)$ and $Y_1, ..., Y_m \overset{iid}{\sim} \mathcal{N}(\mu_Y, \Sigma_Y)$ where we wish to discern whether they share the same covariance or not. The log-likelihood of the pooled sample is,

$$l(\mu_X, \mu_Y, \Sigma_X, \Sigma_Y) = -\frac{n}{2}\log|\Sigma_X| - \frac{m}{2}\log|\Sigma_Y| - \frac{n}{2}(\overline{x} - \mu_X)^T\Sigma_X^{-1}(\overline{x} - \mu_X) - \frac{m}{2}(\overline{y} - \mu_Y)^T\Sigma_Y^{-1}(\overline{y} - \mu_Y)$$
$$- \frac{n}{2}\text{trace}\big(\Sigma_X^{-1}\widehat{\Sigma}_X\big) - \frac{m}{2}\text{trace}\big(\Sigma_Y^{-1}\widehat{\Sigma}_Y\big).$$

In the unconstrained case, this separates into the sum of the two separate likelihoods corresponding to each sample, $l(\mu_X, \mu_Y, \Sigma_X, \Sigma_Y) = l(\mu_X, \Sigma_X) + l(\mu_Y, \Sigma_Y)$, which is maximized at the separate MLEs $\big(\overline{X}, \overline{Y}, \widehat{\Sigma}_X, \widehat{\Sigma}_Y\big)$ with maximum,

$$l^* = -\frac{n}{2}\log|\widehat{\Sigma}_X| - \frac{m}{2}\log|\widehat{\Sigma}_Y| - \frac{(n+m)p}{2}. \tag{6.46}$$

In the constrained case, the MLEs for the means are $\big(\overline{X}, \overline{Y}\big)$ regardless of the choice of covariance. Plugging these in, the log-likelihood for the common covariance Σ becomes,

$$l\big(\overline{X}, \overline{Y}, \Sigma\big) = -\frac{n+m}{2}\log|\Sigma| - \frac{n}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_X\big) - \frac{m}{2}\text{trace}\big(\Sigma^{-1}\widehat{\Sigma}_Y\big) = -\frac{n+m}{2}\Big(\log|\Sigma| + \text{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big)\Big), \tag{6.47}$$

where $\widehat{\Sigma} = \frac{n}{n+m}\widehat{\Sigma}_X + \frac{m}{n+m}\widehat{\Sigma}_Y$ is the weighted average of the two sample covariances (notice that this is no longer equal to the pooled covariance since the two means are possibly different). Therefore $\widehat{\Sigma}$ is the restricted MLE yielding $l_0^* = -\frac{n+m}{2}\log|\widehat{\Sigma}| - \frac{(n+m)p}{2}$. We thus arrive at,

$$2(l^* - l_0^*) = -n\log|\widehat{\Sigma}_X| - m\log|\widehat{\Sigma}_Y| + (n+m)\log|\widehat{\Sigma}|. \tag{6.48}$$

The exact distribution of this statistic is non-trivial to obtain, however Wilks's Theorem applies and thus we may use the asymptotic approximation χ^2_m. Here, $m = \{\#\text{ parameters} - \#\text{ free parameters under } H_0\} = 2p + p(p+1) - 2p - \frac{p(p+1)}{2} = \frac{p(p+1)}{2}$. Thus, if n is large enough, we reject H_0 at level α if and only if,

$$2\log(\Lambda) = -n\log|\widehat{\Sigma}_X| - m\log|\widehat{\Sigma}_Y| + (n+m)\log|\widehat{\Sigma}| > q_{1-\alpha}\big(\chi^2_{p(p+1)/2}\big). \tag{6.49}$$

6.2.9 One Sample - Sphericity Test $(*)$

Let $X_1, ..., X_n$ be a random sample from $\mathcal{N}(\mu, \Sigma_{p\times p})$ with $\Sigma \succ 0$ and $n > p$. We consider the hypothesis pair $\{H_0 : \Sigma = \lambda\mathbb{I} \text{ for some } \lambda > 0\}$ versus $\{H_1 : \Sigma \neq \lambda\mathbb{I} \text{ for all } \lambda > 0\}$. Note that both hypotheses are composite. The restricted MLEs are $\big(\overline{X}, \widehat{\lambda}\mathbb{I}\big)$ and the unrestricted MLEs are $\big(\overline{X}, \widehat{\Sigma}\big)$ where $\widehat{\lambda} = \frac{1}{p}\text{trace}\big(\widehat{\Sigma}\big)$ (by the equivariance of the MLE). Thus:

$$l_0^* = l\big(\overline{X}, \widehat{\lambda}\mathbb{I}\big) = -\frac{n}{2}\log|2\pi\widehat{\lambda}\mathbb{I}| - \frac{n}{2}\text{trace}\big(\widehat{\lambda}^{-1}\widehat{\Sigma}\big) \tag{6.50}$$

$$l^* = l\big(\overline{X}, \widehat{\Sigma}\big) = -\frac{n}{2}\log|2\pi\widehat{\Sigma}| - \frac{np}{2}. \tag{6.51}$$

As a result,

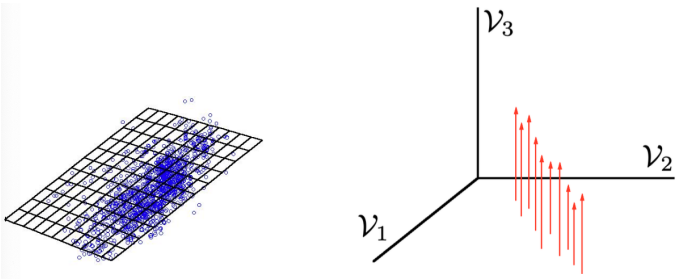
$$2\log(\Lambda) = n\left(\log|2\pi\widehat{\lambda}\mathbb{I}| + \text{trace}\big(\widehat{\lambda}^{-1}\widehat{\Sigma}\big) - \log|2\pi\widehat{\Sigma}| - p\right) = n\left(-\log\big|\widehat{\lambda}^{-1}\widehat{\Sigma}\big| + \text{trace}\big(\widehat{\lambda}^{-1}\widehat{\Sigma}\big) - p\right) = \log\left[\left(\frac{\left(\frac{\text{trace}\big(\widehat{\Sigma}\big)}{p}\right)}{\big|\widehat{\Sigma}\big|^{1/p}}\right)^{pn}\right].$$

As a result, for n large enough, the likelihood ratio test rejects H_0 for large values of $\Big(\gamma\big(\widehat{\Sigma}\big)/\alpha\big(\widehat{\Sigma}\big)\Big)^n$, where $\alpha\big(\widehat{\Sigma}\big)$ and $\gamma\big(\widehat{\Sigma}\big)$ are the arithmetic and geometric means of the eigenvalues of $\widehat{\Sigma}$. Note that the exact null sampling distribution of the ratio test is not available in closed form, and Wilks' Theorem does not apply since the null is composite. However, the asymptotic distribution may be obtained by different means, but it is too convoluted to state.

Chapter 7

Dimension Reduction

Dimension reduction is a means to introduce parsimony by way of projections or low-rank techniques. The key principle of dimension reduction is, roughly speaking that, *most of the statistical action is happening in some latent hyperplane of dimension far lower than the dimension p of the ambient space* $\mathbb{R}^p$. The name of the game is looking for good linear functionals (projections) which (i) capture most of the action, when all variables are treated equally, and (ii) distil most of the dependence, when variables are treated as input/output. To this effect, we will see two types of analysis: Principal Component Analysis (PCA) and Canonical Correlation Analysis (CCA). The two caricatures to have in mind are a *thin scatter plot* for PCA and a *picket fence* for CCA.



7.1 Principal Component Analysis (PCA)

PCA

- Let X be a random vector in $\mathbb{R}^p$ with covariance matrix Σ.
- We seek $v_1 \in \mathbb{S}^{d-1}$ such that X's projection onto v_1 has maximal variance.
- For $j > 1$, we also seek a direction $v_j \in \text{span}^{\perp}\{v_1, ..., v_{j-1}\}$ such that Y's projection onto v_j has maximal variance.

Solution to PCA. We want to maximize $\text{Var}(v_1^T X) = v_1^T \Sigma v_1$ over $\|v_1\| = 1$. We have:

$$v_1^T \Sigma v_1 = v_1^T U \Lambda U^T v_1 = \|\Lambda^{1/2} U^T v_1\|^2 = \sum_{i=1}^{d} \lambda_i (u_i^T v_1)^2. \tag{7.1}$$

Now since $\sum_{i=1}^{d} (u_i^T v_1)^2 = \|v_1\|^2 = 1$, then we have a convex combination of the $\{\lambda_j\}_{j=1}^{d}$. Thus we consider the form $\sum_{i=1}^{d} p_i \lambda_i$ such that $\sum_i p_i = 1$, $p_i \geq 0$ for $i = 1, ..., d$. Now if $\lambda_1 \geq \lambda_i \geq 0$, then the sum is clearly maximized when $p_1 = 1$ and $p_j = 0$ for all $j \neq 1$, i.e. $v_1 = \pm u_1$. Iteratively, we find $v_j = \pm u_j$, i.e. the eigenvectors of Σ. □

The eigenvectors of Σ are called the **principal components** (of variation). The ratio $\lambda_j/\text{trace}(\Sigma)$ gives the percentage of variance explained by the j-th component. The actual projection $\langle u_1, X\rangle = u_1^T X$ is called the **score** of X. Furthermore, the scores along different components are uncorrelated $\text{Cov}\big(u_i^T X, u_j^T X\big) = u_i^T \Sigma u_j = \lambda_j \mathbb{1}_{\{i=j\}}$.

Theorem 7.1.1 (Optimal Linear Dimension Reduction Theorem)**.** *Let X be a mean-zero random variable in $\mathbb{R}^p$ with $p \times p$ covariance Σ. Let H_k be the projection matrix onto the span of the first k eigenvectors of Σ. Then,*

$$\mathbb{E}\|X - H_k X\|^2 \leq \mathbb{E}\|X - QX\|^2, \tag{7.2}$$

for any $p \times p$ projection matrix Q with $\text{rank}(Q) \leq k$.

Intuitively, if we want to approximate a mean-zero random variable taking values in $\mathbb{R}^p$ by a random variable that ranges over a subspace of dimension at most $k < p$, the optimal choice is the projection of the random variable onto the space spanned by its first k principle components. Here, *optimal* is with respect to the mean squared error.

Proof. Recall that Q is a rank k projection if and only if $Q = \sum_{j=1}^k v_j v_j^T$ for orthonormal $\{v_j\}_{j=1}^k$. Then:

$$\begin{aligned}
\mathbb{E}\|X - QX\|^2 = \mathbb{E}\big[X^T(\mathbb{I}-Q)^T(\mathbb{I}-Q)X\big] &= \mathbb{E}\big[\text{trace}\big((\mathbb{I}-Q)XX^T(\mathbb{I}-Q)^T\big)\big] \\
&= \text{trace}\big[(\mathbb{I}-Q)\mathbb{E}\big(XX^T\big)(\mathbb{I}-Q)^T\big] \\
&= \text{trace}\big[(\mathbb{I}-Q)^T(\mathbb{I}-Q)\Sigma\big] \\
&= \text{trace}\big[(\mathbb{I}-Q)\Sigma\big] \\
&= \text{trace}(\Sigma) - \text{trace}(Q\Sigma) \\
&= \sum_{i=1}^n \lambda_i - \text{trace}\left(\sum_{j=1}^k v_j v_j^T \Sigma\right) \\
&= \sum_{i=1}^n \lambda_i - \sum_{j=1}^k \text{trace}(v_j v_j^T \Sigma) \\
&= \sum_{i=1}^n \lambda_i - \sum_{j=1}^k v_j^T \Sigma v_j = \sum_{i=1}^n \lambda_i - \sum_{j=1}^k \text{Var}\big(v_j^T X\big).
\end{aligned}$$

If we can minimize this expression over all $\{v_j\}_{j=1}^k$ with $v_j^T v_{j'} = \mathbb{1}_{\{j=j'\}}$, then we are done. By PCA, this is done by choosing the top k eigenvectors of Σ. □

More generally, we can show the following result:

Theorem 7.1.2 (Eckhard-Young-Schmidt-Mirsky, Hilbert-Schmidt Case)**.** *Let $\Sigma_{p\times p} = \sum_{i=1}^p \lambda_i u_i u_i^T \succeq 0$ and $\Sigma_k = \sum_{i=1}^k \lambda_i u_i u_i^T$ be its rank k spectral truncation. Then,*

$$\|\Sigma - \Sigma_k\|_{\mathbb{R}^{p\times p}} \leq \|\Sigma - \Gamma\|_{\mathbb{R}^{p\times p}}, \tag{7.3}$$

for any Γ of rank at most k (not necessarily non-negative definite). Here,

$$\|A\|^2_{\mathbb{R}^{p\times p}} = \text{trace}(A^T A) = \|\text{vec}(A)\|^2_{\mathbb{R}^{p^2}}. \tag{7.4}$$

Note that $\Sigma_k = H_k \Sigma H_k$ where $H_k = \sum_{i=1}^k u_i u_i^T$ projects onto $\text{span}\{u_1, ..., u_k\}$.

This shows that PCA can also be interpreted via the Optimal Low Rank Approximation of the covariance matrix. Before stating the proof of the above Theorem, we state and prove the Von Neumann's Trace Inequality.

Proposition 7.1.3 (Von Neumann's Trace Inequality)**.** *For $\sigma_i(A)$ and $\sigma_i(B)$ the non-negative singular values of the matrices A and B, we have:*

$$|\text{trace}(AB)| \leq \sum_i \sigma_i(A)\sigma_i(B). \tag{7.5}$$

Proof. By the SVD, the statement is equivalent to showing that,

$$|\text{trace}(\Lambda U \Omega V^T)| \leq \text{trace}(\Lambda\Omega),$$

for orthonormal matrices $\{U, V\}$ and (say) $p \times p$ diagonal $\{\Lambda, \Omega\}$. We express Λ and Ω as weighted averages of the projectors $P_k = \sum_{i=1}^k e_i e_i^T$ with $\{e_i\}$ the canonical basis of $\mathbb{R}^p$:

$$\Lambda = (\lambda_1 - \lambda_2)P_1 + (\lambda_2 - \lambda_3)P_2 + ... + (\lambda_{p-1} - \lambda_p)P_{p-1} + \lambda_p P_p = \sum_{i=1}^p \alpha_i P_i$$

$$\Omega = (\omega_1 - \omega_2)P_1 + (\omega_2 - \omega_3)P_2 + ... + (\omega_{p-1} - \omega_p)P_{p-1} + \omega_p P_p = \sum_{i=1}^p \beta_i P_i.$$

With this representation, our sought inequality becomes,

$$\left| \sum_{i,j=1}^p \alpha_i \beta_j \text{trace}\big(P_i U P_j V^T\big) \right| \leq \sum_{i,j=1}^p \alpha_i \beta_j \text{trace}(P_i P_j).$$

This will follow by the triangle inequality if we can bound each term as,

$$\left| \alpha_i \beta_j \text{trace}\big(P_i U P_j V^T\big) \right| \leq \alpha_i \beta_j \text{trace}(P_i P_j).$$

For $i \geq j$, $P_i U P_j = (P_i u_1, ..., P_i u_j, 0, ..., 0)$ and thus we must show that $\sum_{k=1}^j \langle P_i u_k, v_k \rangle \leq j$. This follows directly from the Cauchy-Schwarz Inequality since $\|P_i u_k\| \leq \|u_k\| = 1$. □

Proof of Theorem (7.1.2). Writing $\Gamma = W\Omega V^T$ in SVD form, we open the square and use the trace inequality (7.1.3):

$$\|\Sigma - \Gamma\|^2_{\mathbb{R}^{p\times p}} = \|\Sigma\|^2_{\mathbb{R}^{p\times p}} + \|\Gamma\|^2_{\mathbb{R}^{p\times p}} - 2\text{trace}(\Sigma\Gamma) \geq \sum_{j=1}^p \lambda_j^2 + \sum_{j=1}^p \omega_j^2 - 2\sum_{j=1}^p \lambda_j \omega_j = \sum_{j=1}^p (\lambda_j - \omega_j)^2,$$

where we have used the fact that $\lambda_j \omega_j \geq 0$ for all j. As a result,

$$\inf_{\Gamma:\text{rank}(\Gamma)\leq k} \|\Sigma - \Gamma\|^2_{\mathbb{R}^{p\times p}} \geq \sum_{j=k+1}^p \lambda_j^2.$$

Setting $\Gamma = \Sigma_k = H_k \Sigma H_k$ attains the lower bound on the RHS. □

7.1.1 PCA in the Gaussian Case

If $X \sim \mathcal{N}(\mu, \Sigma_{p\times p})$, it admits the Karhunen-Loève expansion:

$$X - \mu = \sum_{i=1}^{p} \xi_i u_i, \qquad \xi_i \overset{iid}{\sim} \mathcal{N}(0, \lambda_i), \tag{7.6}$$

where $\Sigma u_i = \lambda_i u_i$ for $1 \le i \le p$ gives the spectrum of Σ, and the eigenvalues are $\xi_i = \langle X, u_i \rangle$. We now notice that (i) u_i are precisely the principal components, (ii) ξ_i are precisely the scores, (iii) scores along different components are independent (not just uncorrelated), and (iv) scores along different components are Gaussian. Therefore, distinct component scores can be analysed completely separately. **So in the Gaussian case, PCA $\equiv$ Karhunen-Loève**.

From the perspective of the covariance, remember that when $X \sim \mathcal{N}(\mu, \Sigma_{p\times p})$, $\text{supp}(X - \mu) = \mathcal{R}(\Sigma)$. Thus by the Low Rank Approximation Theorem (7.1.2), PCA is equivalent to successive (and nested) dimension reductions of the support of X. Note that in the non-Gaussian case, we find the best fitting hyperplane of given dimension to the true support.

PCA basically represents a change-of-basis and in the new basis, everything is mathematically simpler. But our interest is in terms of the original basis, where coordinates in the original basis correspond to variables/features (age,weight,height,...). However, coordinates in PCA basis are linear combinations of these variables/features. Of course, ideally we would like to find combinations that are interpretable and/or sparse. However, there is no guarantee that this may be the case and this motivates l_1-penalised PCA, $u_\tau := \arg\max_{\|u\|=1}\{u^T \Sigma u - \tau \|u\|_1\}$.

7.1.2 Sample PCA and Population PCA

If we only have an i.i.d. sample $X_1, ..., X_n$, we can define the sample principal components and the sample scores in precisely the same way as before, but replacing the mean/covariance (μ, Σ) with their sample versions $\big(\overline{X}, \widehat{\Sigma}\big)$. An important note is that the observed (realised) sample scores have zero empirical correlation however they are correlated as random variables, since they are based on empirical principal components (which are approximations to the true ones). In the Gaussian case, the sample scores will not be independent, and the MLE equivariance immediately establishes the following Proposition:

Proposition 7.1.4. *Let $X_1, ..., X_n \sim \mathcal{N}(\mu, \Sigma_{p\times p})$, where Σ has spectrum $\{(\lambda_i, u_i)\}_{i=1}^p$ and assume that the MLE of (μ, Σ) exists (in which case it equals $\big(\overline{X}, \widehat{\Sigma}\big)$). Then, provided $\lambda_1 > ... > \lambda_p$, the MLE of $\{(\lambda_i, u_i)\}$ is given by the spectrum $\big\{\big(\widehat{\lambda}_i, \widehat{u}_i\big)\big\}$ of $\widehat{\Sigma}$.*

Strictly speaking, eigenvectors are unique up to a sign, thus we rather estimate the eigenprojections $u_i u_i^T$ by their sample version $\widehat{u}_i \widehat{u}_i^T$. Let us now see how the optimal dimension reduction property of PCA holds at the sample level.

Corollary 7.1.5. *Let $X_1, ..., X_n$ be i.i.d. random vectors in $\mathbb{R}^p$. The best approximating k-hyperplane to the points $\{X_1, ..., X_n\}$ is given by $\overline{X} + \mathcal{R}\big(\widehat{\Sigma}_k\big)$, where $\widehat{\Sigma}_k = \sum_{i=1}^k \widehat{\lambda}_i \widehat{u}_i \widehat{u}_i^T$ is the rank-k spectral truncation of $\widehat{\Sigma}$. Equivalently, defining $\widehat{H}_k := \sum_{i=1}^k \widehat{u}_i \widehat{u}_i^T$,*

$$\sum_{j=1}^{n} \Big\| \big(X_j - \overline{X}\big) - \widehat{H}_k \big(X_j - \overline{X}\big) \Big\|^2 \le \sum_{j=1}^{n} \|X_j - v - Q(X_j - v)\|^2, \tag{7.7}$$

for any $v \in \mathbb{R}^p$ and $n \times n$ projection operator Q of rank at most k.

Proof. $(*)$ An equivalent formulation of the statement is given by:

$$\sum_{j=1}^{p}\|X_j - H_k X_j\|^2 \leq \sum_{j=1}^{p}\|X_j - QX_j\|^2,$$

for any $n\times n$ projection operator Q of rank at most k, where $\widehat{H}_k := \sum_{i=1}^{k}\widehat{u}_i\widehat{u}_i^T$. However, this is a direct consequence of the Optimal Linear Dimension Reduction Theorem (7.1.1), taking expectations with respect to the empirical (discrete) measure $P_n := \sum_{i=1}^{p}\delta_{X_i}$. □

Viewing the spectrum $\{(\widehat{\lambda}_i, \widehat{u}_i)\}$ of $\widehat{\Sigma}$ as an estimator of the spectrum $\{(\lambda_i, u_i)\}$ of Σ, we may find some bounds involving the eigenvalues and eigenvectors.

Lemma 7.1.6 (Eigenvalue Perturbation Bound)**.**

$$\max_j \left|\widehat{\lambda}_j - \lambda_j\right| \leq \left\|\widehat{\Sigma} - \Sigma\right\|_{\mathbb{R}^{p\times p}}. \tag{7.8}$$

Proof. We expand the RHS and use the Trace Inequality (7.1.3), where for all $j = 1,...,p$:

$$\begin{aligned}\left\|\widehat{\Sigma} - \Sigma\right\|^2 =& \|\Sigma\|^2 + \left\|\widehat{\Sigma}\right\|^2 - 2\mathrm{trace}\big(\widehat{\Sigma}\Sigma\big) \geq \sum_i \lambda_i^2 + \sum_i \widehat{\lambda}_i^2 - 2\sum_i \lambda_i\widehat{\lambda}_i = \sum_i \big(\lambda_i - \widehat{\lambda}_i\big)^2 \\ &= \big(\lambda_j - \widehat{\lambda}_j\big)^2\left(1 + \sum_{i\neq j}\frac{\big(\lambda_i - \widehat{\lambda}_i\big)^2}{\big(\lambda_j - \widehat{\lambda}_j\big)^2}\right) \\ &\geq \big(\lambda_j - \widehat{\lambda}_j\big)^2.\end{aligned}$$

Thus, for all $j = 1,...,p$, we find that $\left|\lambda_j - \widehat{\lambda}_j\right| \leq \left\|\widehat{\Sigma} - \Sigma\right\|$. □

Theorem 7.1.7 (Eigenvector Perturbation Bound)**.** *Let $\Sigma \succ 0$ and $\widehat{\Sigma} \succ 0$ have spectra (λ_j, u_j) and $(\widehat{\lambda}_j, \widehat{u}_j)$ respectively, both with distinct eigenvalues. Define $u_j^* := \mathrm{sign}\big(\langle u_j, \widehat{u}_j\rangle\big)u_j$. Then,*

$$\left\|\widehat{u}_j - u_j^*\right\| \leq 2\sqrt{2}\left\|\widehat{\Sigma} - \Sigma\right\|_{\mathbb{R}^{p\times p}}, \tag{7.9}$$

where $\alpha_1 = (\lambda_1 - \lambda_2)^{-1}$ and $\alpha_j = \max\left\{\big(\lambda_{j-1} - \lambda_j\big)^{-1}, \big(\lambda_j - \lambda_{j+1}\big)^{-1}\right\}$, for $j \geq 2$.

Distinct eigenvalues allow for individual eigendirections to be identifiable, however eigenvectors are unique only up to a sign change, hence the use of u_j^*.

Proof. We will prove this by bounding the quantity $\|\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j\|$ between the two terms in the sought inequality.

- Note that,

$$\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j = (\Sigma - \widehat{\Sigma} + \widehat{\Sigma})\widehat{u}_j - (\lambda_j - \widehat{\lambda}_j + \widehat{\lambda}_j)\widehat{u}_j = (\Sigma - \widehat{\Sigma})\widehat{u}_j + (\widehat{\lambda}_j - \lambda_j)\widehat{u}_j.$$

Thus, the triangle inequality and the previously seen inequality $\|\Sigma - \Gamma\|^2_{\mathbb{R}^{p\times p}} \geq \sum_{j=1}^{p}(\lambda_j - \omega_j)^2$ imply that:

$$\|\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j\| \leq \|(\Sigma - \widehat{\Sigma})\widehat{u}_j\| + \|(\widehat{\lambda}_j - \lambda_j)\widehat{u}_j\| \leq \|\Sigma - \widehat{\Sigma}\|_\infty + \|\Sigma - \widehat{\Sigma}\|_{\mathbb{R}^{p\times p}},$$

and since $\|\Sigma - \widehat{\Sigma}\|_\infty \leq \|\Sigma - \widehat{\Sigma}\|_{\mathbb{R}^{p\times p}}$, the RHS is majorised by $2\|\Sigma - \widehat{\Sigma}\|_{\mathbb{R}^{p\times p}}$.

- Now, for all $1 \leq j \leq p$, we aim to lower bound $\|\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j\|^2$ below by $(2\alpha_j^2)^{-1}\|u_j^* - \widehat{u}_j\|^2$.

$$\begin{aligned}\|\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j\|^2 = \sum_{k=1}^{p}\langle\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j, u_k\rangle^2 = \sum_{k=1}^{p}\big(\langle\Sigma\widehat{u}_j, u_k\rangle - \langle\lambda_j\widehat{u}_j, u_k\rangle\big)^2 &= \sum_{k=1}^{p}(\lambda_k - \lambda_j)^2\langle\widehat{u}_j, u_k\rangle^2 \\ &\geq \min_{k\neq j}(\lambda_k - \lambda_j)^2\sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2 \\ &\geq \alpha_j^{-2}\sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2.\end{aligned}$$

Recalling that $u_j^* = \operatorname{sign}\big(\langle u_j, \widehat{u}_j\rangle\big)u_j$, observe that $\|u_j^* - \widehat{u}_j\|^2$ may be written as:

$$\begin{aligned}\sum_{k=1}^{p}\langle u_j^* - \widehat{u}_j, u_k\rangle^2 = \big(\operatorname{sign}\big(\langle u_j^*, u_j\rangle\big) - \langle\widehat{u}_j, u_j\rangle\big)^2 + \sum_{k\neq j}\langle u_j^* - \widehat{u}_j, u_k\rangle &= \big(1 - |\langle\widehat{u}_j, u_j\rangle|\big)^2 + \sum_{k\neq j}\big(\langle u_j^*, u_k\rangle - \langle\widehat{u}_j, u_k\rangle\big)^2 \\ &= \big(1 - |\langle\widehat{u}_j, u_j\rangle|\big)^2 + \sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2.\end{aligned}$$

Furthermore, since $\sum_{k=1}^{p}\langle\widehat{u}_j, u_k\rangle^2 = 1$, we find:

$$\big(1 - |\langle\widehat{u}_j, u_j\rangle|\big)^2 = \sum_{k=1}^{p}\langle\widehat{u}_j, u_k\rangle^2 - 2|\langle\widehat{u}_j, u_j\rangle| + |\langle\widehat{u}_j, u_j\rangle|^2 = \sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2 + 2\underbrace{\big(|\langle\widehat{u}_j, u_j\rangle|^2 - |\langle\widehat{u}_j, u_j\rangle|\big)}_{\leq 0} \leq \sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2,$$

where we have used the fact that $\langle\widehat{u}_j, u_j\rangle \leq 1$. Thus, $2\sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2 \geq \|u_j^* - \widehat{u}_j\|^2$. Combining all the inequalities and re-arranging the constant factors, we arrive at:

$$4\|\widehat{\Sigma} - \Sigma\|^2_{\mathbb{R}^{p\times p}} \geq \|\Sigma\widehat{u}_j - \lambda_j\widehat{u}_j\|^2 \geq \alpha_j^{-2}\sum_{k\neq j}\langle\widehat{u}_j, u_k\rangle^2 \geq (2\alpha_j^2)^{-1}\|u_j^* - \widehat{u}_j\|^2.$$

□

7.1.3 Sampling Distribution of the Spectrum

Notice that one way to get rid of the use of u^* is to always use the convention that U (and $\widehat{U}$) are taken so that their diagonal elements are non-negative. This eradicates the sign ambiguity from all the eigenvectors, and it will be useful in what comes next. We call this the **sign consistency**. We now seek to study the asymptotic sampling distribution of $\big\{\big(\widehat{\lambda}_i, \widehat{u}_i\big)\big\}$. In this subsection, we will develop the asymptotic law of the empirical eigenvalues, and that of the empirical eigenvectors, and then see how they simplify when dealing with a Gaussian data matrix.

Recall that, as $n \to \infty$ with p fixed, $\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big) \xrightarrow{d} Z$, where Z is a mean zero Gaussian random matrix. We can use this setup to obtain the following Theorem.

Theorem 7.1.8 (CLT for Empirical Spectrum)**.** *Let $X_1, ..., X_n$ be i.i.d. p-vectors whose covariance Σ has spectrum $\{(\lambda_i, u_i)\}_{i=1}^p$, with $\lambda_1 > ... > \lambda_p > 0$. Let $\big\{\big(\widehat{\lambda}_i, \widehat{u}_i\big)\big\}$ be the spectrum of $\widehat{\Sigma}$, and assume that $\{(\widehat{u}_i, u_i)\}$ are chosen sign-consistently. Then:*
(1) $\big\{\sqrt{n}\big(\widehat{\lambda}_j - \lambda_j\big)\big\}_{1\leq j\leq p} \xrightarrow{d} \mathcal{N}(0, \Phi)$, where $\Phi_{ij} = \mathbb{E}\big[\langle Zu_i, u_i\rangle\langle Zu_j, u_j\rangle\big]$.
(2) $\sqrt{n}\big(U^T\widehat{U} - \mathbb{I}\big) \xrightarrow{d} W$ for $W = \{W_{ij}\}$ a centred Gaussian matrix, such that,

$$\operatorname{Cov}\big(W_{ii'}, W_{jj'}\big) = \begin{cases} 0 & \text{if } i = i' \text{ or } j = j' \\ \mathbb{E}\left[\frac{\langle Zu_i, u_{i'}\rangle\langle Zu_j, u_{j'}\rangle}{(\lambda_{i'} - \lambda_i)(\lambda_{j'} - \lambda_j)}\right] & \text{otherwise} \end{cases}. \tag{7.10}$$

Note that we can easily deduce from (2) that $UW_n = \sqrt{n}\big(\widehat{U} - U\big) \xrightarrow{d} UW$ which is also a centred Gaussian limit. In fact, the proof will show that the sequences in (1) and (2) are jointly asymptotically Gaussian.

Proof. We will leverage the CLT for $\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big) \xrightarrow{d} Z$ in order to obtain the sought CLT. Assuming that U and $\widehat{U}$ are defined sign-consistently, define,

$$Q_n = U^T\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big)U = \sqrt{n}\big(U^T\widehat{\Sigma}U - U^T\Sigma U\big) =: \sqrt{n}(T_n - \Lambda)$$

$$D_n := \sqrt{n}\big(\widehat{\Lambda} - \Lambda\big)$$

$$W_n = \sqrt{n}\big(U^T\widehat{U} - \mathbb{I}\big),$$

and observe that we may write,

$$\underbrace{\Lambda + \frac{Q_n}{\sqrt{n}}}_{T_n} = \underbrace{\left(\mathbb{I} + \frac{W_n}{\sqrt{n}}\right)}_{U^T\widehat{U}}\underbrace{\left(\Lambda + \frac{D_n}{\sqrt{n}}\right)}_{\widehat{\Lambda}}\underbrace{\left(\mathbb{I} + \frac{W_n}{\sqrt{n}}\right)}_{\widehat{U}^TU},$$

or equivalently,

$$Q_n \overset{(\Delta)}{=} W_n\Lambda + \Lambda W_n^T + D_n + \frac{W_nD_n + W_n\Lambda W_n^T + D_nW_n^T}{\sqrt{n}} + \frac{W_nD_nW_n^T}{n}.$$

We furthermore note the constraint that $U^T\widehat{U}$ is orthogonal $\implies \left(\mathbb{I} + \frac{W_n}{\sqrt{n}}\right)$ is orthogonal $\implies W_n + W_n^T + \frac{W_nW_n^T}{\sqrt{n}} \overset{(\dagger)}{=} 0$. With all these definitions/relations in place, let us start to look at asymptotics.

(i) $Q_n \xrightarrow{d} Q = U^TZU$ where Z is the (centred Gaussian) weak limit of $\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big)$ and so Q is centred Gaussian itself.

(ii) All the terms in (Δ) are scaled by $\frac{1}{\sqrt{n}}$ or $\frac{1}{n}$ converges to zero in probability, by submultiplicativity of the matrix norm $\|\cdot\|_\infty$ and our permutation bounds:

$$\|W_n\|_\infty \leq \big\|\sqrt{n}\big(U^T\widehat{U} - \mathbb{I}\big)\big\|_{\mathbb{R}^{p\times p}} = \big\|\sqrt{n}\big(\widehat{U} - U\big)\|_{\mathbb{R}^{p\times p}} \leq c_1\big\|\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big)\|_{\mathbb{R}^{p\times p}} \xrightarrow{d} c_1\xi$$

$$\|D_n\|_\infty \leq \big\|\sqrt{n}\big(\widehat{\Lambda} - \Lambda\big)\big\|_{\mathbb{R}^{p\times p}} \leq \sqrt{p}\sup_{1\leq j\leq p}\big|\widehat{\lambda}_j - \lambda_j\big| \leq c_2\big\|\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big)\|_{\mathbb{R}^{p\times p}} \xrightarrow{d} c_2\xi,$$

where $c_1, c_2 \in (0,\infty)$ and ξ is a scalar random variable, so dividing by a negative power of n kills off the last two terms of (Δ) in the limit.

(iii) Thus by (i) and (ii) combined with (Δ), Slutsky's Theorem (5.3.13) we have that,

$$W_n\Lambda + \Lambda W_n^T + D_n \xrightarrow{d} Q.$$

(iv) Additionally, (ii) combined with $(\dagger)$ implies that $W_n + W_n^T \xrightarrow{d} 0$, which means that the diagonal of W_n vanishes asymptotically, and consequently so does the diagonal of $W_n\Lambda + \Lambda W_n^T$ seeing as Λ is a diagonal matrix.

(v) On the other hand, D_n is (by definition) diagonal for all $n \geq 1$.

Consequently, letting $\mathcal{G} : \mathbb{R}^{p\times p} \to \mathbb{R}^{p\times p}$ be the projection onto diagonal matrices,

$$\begin{aligned} D_n - \mathcal{G}Q_n = \mathcal{G}D_n - \mathcal{G}Q_n &= \mathcal{G}D_n + \mathcal{G}\big(W_n\Lambda + \Lambda W_n^T\big) - \mathcal{G}Q_n - \mathcal{G}\big(W_n\Lambda + \Lambda W_n^T\big) \\ &= \mathcal{G}\big(W_n\Lambda + \Lambda W_n^T + D_n - Q_n\big) - \mathcal{G}\big(W_n\Lambda + \Lambda W_n^T\big) \xrightarrow{d} 0. \end{aligned}$$

This proves that $D_n = \sqrt{n}\big(\widehat{\Lambda} - \Lambda\big) \xrightarrow{d} \mathcal{G}Q = \lim_{n\to\infty}\{\text{law}(\mathcal{G}Q_n)\}$, and so $\big\{\sqrt{n}\big(\widehat{\lambda}_j - \lambda_j\big)\big\}_{1\leq j\leq p}$ has a centred Gaussian limit in distribution.

As for the limiting covariance of $\{\sqrt{n}(\widehat{\lambda}_j-\lambda_j)\}_{1\le j\le p}$, this is simply the covariance of the diagonal elements of $Q=U^TZU$ (which coincide with the diagonal elements of $\mathcal{G}Q$). Noting that the latter is,

$$\mathrm{Cov}(e_i^TQe_i,e_j^TQe_j)=\mathrm{Cov}(e_i^TU^TZUe_i,e_j^TU^TZUe_j)=\mathrm{Cov}(u_i^TZu_i,u_j^TZu_j),$$

and since $\mathbb{E}(Z)=0$ the latter is $\mathbb{E}\big[\langle Zu_i,u_i\rangle\langle Zu_j,u_j\rangle\big]$, as claimed. This settles the eigenvalues, and now we turn our attention to the eigenvectors.
Letting $\mathcal{G}^\perp$ be the projection onto matrices with zeroes on the diagonal,

$$W_n\Lambda-\Lambda W_n-\frac{W_nW_n^T}{\sqrt{n}}\overset{(\dagger)}{\equiv}W_n\Lambda+\Lambda W_n^T=\mathcal{G}^\perp(W_n\Lambda+\Lambda W_n^T+D_n)\xrightarrow{d}\mathcal{G}\perp Q,$$

is asymptotically mean zero Gaussian. But $\frac{W_nW_n^T}{\sqrt{n}}\xrightarrow{\mathbb{P}}0$ and we notice that the elements of $W_n\Lambda-\Lambda W_n$ are simply $w_{ij}(n)\lambda_j-\lambda_iw_{ij}(n)=(\lambda_j-\lambda_i)w_{ij}(n)$, so,

$$(W_n)_{ij}=\frac{1}{\lambda_j-\lambda_i}(W_n\Lambda-\Lambda W_n)_{ij}.$$

Hence W_n itself has a centred Gaussian limit W, by Slutsky, with,

$$W_{ij}=\frac{\mathbb{1}_{\{i\neq j\}}}{\lambda_j-\lambda_i}Q_{ij}.$$

And we can get the covariance between pairs of entries W by suitably rescaling the covariance of the corresponding pair of entries of $Q=U^TZU$:

$$\mathrm{Cov}(W_{ii'},W_{jj'})=\begin{cases}0 & \text{if } i=i' \text{ or } j=j'\\ \mathbb{E}\left[\frac{\langle Zu_i,u_{i'}\rangle\langle Zu_j,u_{j'}\rangle}{(\lambda_{i'}-\lambda_i)(\lambda_{j'}-\lambda_j)}\right] & \text{otherwise}\end{cases}. \tag{7.11}$$

□

Corollary 7.1.9 (Asymptotic Law of Wishart Spectrum)**.** *Let $X_1,...,X_n\sim\mathcal{N}(\mu,\Sigma)$ be i.i.d. p-vectors whose covariance Σ has spectrum $\{(\lambda_i,u_i)\}_{i=1}^p$, with $\lambda_1>...>\lambda_p>0$. Let $\{(\widehat{\lambda}_i,\widehat{u}_i)\}$ be the spectrum of $\widehat{\Sigma}$, and assume that $\{(\widehat{u}_i,u_i)\}$ are chosen sign-consistently. Then:*
(1) $\{\sqrt{n}(\widehat{\lambda}_j-\lambda_j)\}_{1\le j\le p}\xrightarrow{d}\mathcal{N}(0,\Phi)$, where $\Phi=\mathrm{diag}(2\lambda_1^2,...,2\lambda_p^2)$.
(2) $\sqrt{n}(U^T\widehat{U}-\mathbb{I})\xrightarrow{d}W$ for $W=\{W_{ij}\}$ a centred Gaussian random matrix.
(3) $W\perp D$.
(4) Writing $W=(W_1,...,W_p)$ columnwise, we have:

$$\mathrm{Cov}(W_i,W_i)=\sum_{k\neq i}\frac{\lambda_i\lambda_ke_ke_k^T}{(\lambda_i-\lambda_k)^2}\quad\text{and,}\quad\mathrm{Cov}(W_i,W_j)=-\frac{\lambda_i\lambda_je_je_i^T}{(\lambda_i-\lambda_j)^2},\ i<j. \tag{7.12}$$

Proof. • By our Gaussian assumption, we know that $n\widehat{\Sigma}\sim W(\Sigma,n-1)$ by Theorem (5.1.2) and therefore, $nU^T\widehat{\Sigma}U\equiv nT_n\sim W(\Lambda,n-1)$. Consequently, $Q_n=\sqrt{n}(T_n-\Lambda)\xrightarrow{d}Q\sim\mathcal{N}(0,C)$ where $C\equiv$ covariance of $W(\Lambda,1)$ and so the entries of Q are uncorrelated by Proposition (5.3.11) (Since Λ is diagonal). Thus, we have independence as they are jointly Gaussian. It then follows that $\mathcal{G}Q$ is independent of $\mathcal{G}^\perp Q$ for any projection $\mathcal{G}:\mathbb{R}^{p\times p}\to\mathbb{R}^{p\times p}$ that zeroes elements and its complementary projection $\mathcal{G}^\perp$, in particular for $\mathcal{G}$ being the projection onto diagonal matrices. Recalling from our previous proof that $\sqrt{n}(\widehat{\Lambda}-\Lambda)\xrightarrow{d}\mathcal{G}Q$ and $\sqrt{n}(U^T\widehat{U}-\mathbb{I})\xrightarrow{d}\mathcal{G}^\perp Q$ we establish (3) and the centred Gaussian limits claimed. For the covariance in

(4), we recall from our last proof that $W_{ij} = \frac{\mathbb{1}_{\{i\neq j\}}}{\lambda_j - \lambda_i} Q_{ij}$ and after a suitable rescaling we obtain the result. We now present the proof of part (1), where the key step is to use the vectorization equation (5.20). Recall that $n\widehat{\Sigma} \sim W(\Sigma, n-1)$ and thus we have the identity in law $n\widehat{\Sigma} \overset{d}{=} \sum_{i=1}^{n-1} W_i$ for $W_i \overset{iid}{\sim} W(\Sigma, 1)$. As a result we know that,

$$\mathrm{Cov}\left(\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big)\right)_{(ij)(kl)} = \frac{n-1}{n}\big(\Sigma_{ik}\Sigma_{jl} - \Sigma_{il}\Sigma_{jk}\big).$$

In particular, this gives us the covariance (taking the limit) of Z:

$$\mathrm{Cov}(Z)_{(ij)(kl)} = \big(\Sigma_{ik}\Sigma_{jl} - \Sigma_{il}\Sigma_{jk}\big).$$

This may be succinctly written with equation (5.20) as $\mathrm{Cov}\big(\mathrm{vec}(Z)\big) = (\Sigma \otimes \Sigma)(\mathbb{I} + K)$ where $\mathbb{I}$ is the identity matrix and K the commutator, i.e. $K\mathrm{vec}(A) = \mathrm{vec}(A^T)$. We now have:

$$\begin{aligned}
\mathbb{E}\big[\langle Zu_i, u_i\rangle\langle Zu_j, u_j\rangle\big] = \mathbb{E}\big[\langle Z, u_i u_i^T\rangle\langle Z, u_j u_j^T\rangle\big] &= \Big\langle \mathrm{Cov}(Z)u_i u_i^T, u_j u_j^T \Big\rangle \\
&= \Big\langle \mathrm{Cov}\big(\mathrm{vec}(Z)\big)\mathrm{vec}(u_i u_i^T), \mathrm{vec}(u_j u_j^T)\Big\rangle \\
&= \Big\langle (\Sigma\otimes\Sigma)(\mathbb{I}+K)\mathrm{vec}(u_i u_i^T), \mathrm{vec}(u_j u_j^T)\Big\rangle \\
&= 2\Big\langle (\Sigma\otimes\Sigma)\mathrm{vec}(u_i u_i^T), \mathrm{vec}(u_j u_j^T)\Big\rangle.
\end{aligned}$$

Remark that the inner product in the first two equations is not the same as we go from vectors to matrices. We now use the identity $(A\otimes B)\mathrm{vec}(V) = \mathrm{vec}(BVA^T)$ to find:

$$\begin{aligned}
\mathbb{E}\big[\langle Zu_i, u_i\rangle\langle Zu_j, u_j\rangle\big] = 2\Big\langle \mathrm{vec}(\Sigma u_i u_i^T\Sigma), \mathrm{vec}(u_j u_j^T)\Big\rangle &= 2\lambda_i\Big\langle \mathrm{vec}(u_i u_i^T\Sigma), \mathrm{vec}(u_j u_j^T)\Big\rangle \\
&= 2\lambda_i \mathrm{trace}\big(u_i u_i^T \Sigma u_j u_j^T\big) \\
&= 2\lambda_i\lambda_j \mathrm{trace}\big(u_i u_i^T u_j u_j^T\big) \\
&= 2\lambda_i\lambda_j\delta_{ij}\mathrm{trace}\big(u_i u_j^T\big) = 2\lambda_i\lambda_j\delta_{ij}.
\end{aligned}$$

□

We have some very interesting results in the Gaussian case:

- Eigenvalues are asymptotically independent between them.
- Eigenvectors are asymptotically independent of the eigenvalues, however the eigenvectors are not asymptotically independent between them (this makes sense since they are orthogonal).
- Judging from the (Gaussian) asymptotic standard deviation of $\sqrt{2}\lambda_j$ we see that crossings will happen often even for well-spaces eigenvalues.
- We can easily deduce that $\Delta_n = UW_n = \sqrt{n}\big(\widehat{U} - U\big) \xrightarrow{d} UW = \Delta$ with,

$$\mathrm{Cov}(\Delta_i, \Delta_i) = \sum_{k\neq i} \frac{\lambda_i\lambda_k u_k u_k^T}{(\lambda_i - \lambda_k)^2} \quad \text{and,} \quad \mathrm{Cov}(\Delta_i, \Delta_j) = -\frac{\lambda_i\lambda_j u_j u_i^T}{(\lambda_i - \lambda_j)^2},\ \ i \neq j. \tag{7.13}$$

Note that we only used the Gaussian assumption to specify the asymptotic covariance. In the non-Gaussian settings, we can still specify the asymptotic covariance, but it depends on comprehensive (mixed) fourth moment structure. However, we previously saw that dependence on fourth moments is *minimal* in elliptical families. Indeed, we can straightforwardly deduce the extension below:

Theorem 7.1.10 (Asymptotic Law of Elliptical Spectrum)**.** *Let $X_1, ..., X_n$ be centred i.i.d. elliptical p-vectors whose covariance Σ has spectrum $\{(\lambda_i, u_i)\}_{i=1}^p$, with $\lambda_1 > ... > \lambda_p > 0$. Let $\big\{\big(\widehat{\lambda}_i, \widehat{u}_i\big)\big\}$ be the spectrum of $\widehat{\Sigma}$, and assume that $\{(\widehat{u}_i, u_i)\}$ are chosen sign-consistently. Letting κ as in equation (5.22) we have:*

(1) $D_n = \sqrt{n}\big(\widehat{\Lambda} - \Lambda\big) \xrightarrow{d} D$ for D a diagonal and centred Gaussian random matrix.
(2) $W_n = \sqrt{n}\big(U^T\widehat{U} - \mathbb{I}\big) \xrightarrow{d} W$ for $W = \{W_{ij}\}$ a centred Gaussian random matrix.
(3) $W \perp D$.
(4) Writing $W = (W_1, ..., W_p)$ columnwise, we have:

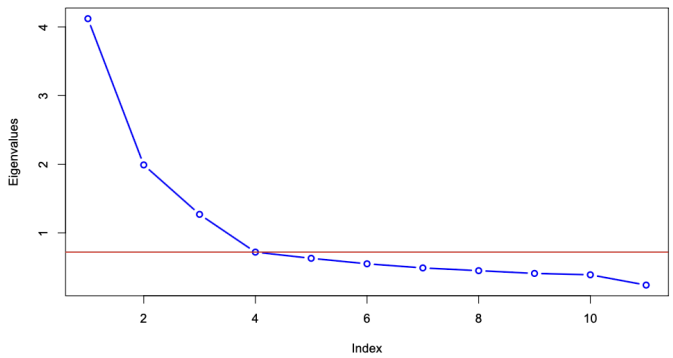
$$\mathrm{Cov}(W_i, W_i) = (1+\kappa)\sum_{k\neq i}\frac{\lambda_i\lambda_k e_k e_k^T}{(\lambda_i - \lambda_k)^2} \quad \text{and,} \quad \mathrm{Cov}(W_i, W_j) = -(1+\kappa)\frac{\lambda_i\lambda_j e_j e_i^T}{(\lambda_i - \lambda_j)^2}, \;\; i < j. \tag{7.14}$$

As a high level summary, assuming $\lambda_1 > ... > \lambda_p > 0$ and sign-consistency we have that in any case: (i) sample eigenvectors and eigenvalues are jointly asymptotically Gaussian, and (ii) sample eigenvectors of different index remain dependent, even asymptotically. In the **Gaussian case**, the covariance structure for sample eigenvalues and eigenvectors is tractable and depends only on second moments. This structure shows that (i) sample eigenvalues of different index are asymptotically mutually independent, and (ii) sample eigenvectors are asymptotically independent of sample eigenvalues. In the **Elliptical case**, the covariance structure for sample eigenvalues and eigenvectors is tractable, depends on second moments and the fourth moment parameter κ. The structure shows that (i) sample eigenvalues of different index are asymptotically mutually independent, and (ii) sample eigenvectors are asymptotically independent of sample eigenvalues. Finally in the **non-Elliptical case**, the covariance structure for sample eigenvalues and eigenvectors is possible intractable, depends on comprehensive mixed fourth moments. Furthermore, even asymptotically, sample eigenvalues may be dependent for different indices and may be dependent with sample eigenvectors.

7.1.4 Selecting the Number of Components

A priori, there is no unequivocal way to choose a truncation level k. In fact, we can interpret k in multiple ways: (i) as a tuning parameter in an approximation problem (% of variance explained), (ii) as a tuning parameter in an inverse problem (condition numbers), (iii) as a model parameter to be inferred (testing/estimation), or (iv) as a model index to be selected over (model selection). Conversely, approaches to choosing k implicitly or explicitly represent a choice of interpretation. Sometimes, different approaches give essentially the same criterion, but not always. They often boil down to *eigenvalue decay* criteria. No single approach is superior in all circumstances, and the choice of method is often guided by the specific data and problem at hand. Combinations of methods may be employed (with careful calibration of significance levels if the testing approach is among them, to avoid data snooping bias).

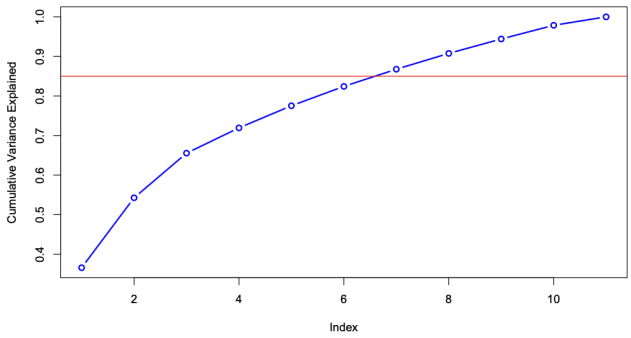
• No matter the method, the **scree plot** (see figure below) often shows up. It represents *derivative* of approximation error function, and levelling off suggests diminishing returns in terms of approximations. We often seek for **elbows** if such are present. The rationale is that past the elbow, the scree plot is essentially constant. No point in truncating to dimensions past elbow point, we might as well not reduce at all since all the remaining dimensions are virtually exchangeable. We can also add approximate error bars in the Gaussian case.



• The % of variance explained is simple enough in principle:

$$k = \min\left\{1 \le j \le p : \frac{\widehat{\lambda}_1 + ... + \widehat{\lambda}_k}{\text{trace}(\widehat{\Sigma})} \ge 1 - \beta\right\}, \tag{7.15}$$

where β may be chosen to some standard level (e.g. 0.15 or 0.1 - no gold standard), but more often β is chosen to depend on the empirical eigenvalues, e.g. via simultaneous inspection of the scree plot and **cumulative variance plot** (see figure below - line is drawn at $\beta = 0.15$ corresponding to 85% variance explained):



• Often, the sample covariance is used as a device for a downstream task, usually through its (generalised) inverse, or that of its square root: for linear prediction, testing and classification. In this case, if the sample covariance is ill-conditioned $CN := \widehat{\lambda}_1/\widehat{\lambda}_p \gg 1$, it can lead to wildly fluctuating outcomes even under small sampling variation. Look at condition indices $CI_j := \widehat{\lambda}_1/\widehat{\lambda}_j, j = 2, ..., p$, and truncate at first j where $CI_j > c^*$ for some threshold c^*. Intuitively, we can try to choose the maximal rank k truncation that still leads to a well-conditioned matrix (according to the threshold). In the case of prediction, one can also use Cross Validation (CV).

• For testing, the possible model for random vectors with *nearly rank k* covariance is {lower dimensional signal} + {isotropic noise}. Specifically we model X as $X = L + \epsilon$, i.e. X is a noisy measurement of the latent vector of interest L where:
† L is a random vector in $\mathbb{R}^p$ with rank $k < p$ and covariance $\Phi = \sum_{j=1}^k \phi_j u_j u_j^T$.
† ϵ is a random vector in $\mathbb{R}^p$ with diagonal covariance $\theta \mathbb{I}_{p\times p}$, $\theta > 0$.
† $\text{Cov}(L, \epsilon) = 0$ and [{ϕ distinct} OR {$\phi_k > \theta$}] (for identifiability).
This is also known as the **spiked covariance model**. The covariance of X then becomes,

$$\text{Cov}(X) = \Phi + \theta\mathbb{I} = \sum_{i=1}^{k}(\phi_i + \theta)u_i u_i^T + \sum_{i=k+1}^{p} \theta u_i u_i^T. \tag{7.16}$$

And in this setting, choosing k is equivalent to inferring as of where population scree plot becomes flat. Now, given a candidate $k \in \{0, ..., k_{\max}\} = K$, where $k_{\max} \le p - 2$, we can test the last $(p-k)$ principal components of X for sphericity. Equivalently, we test the sphericity of $\Delta_k = \text{Cov}(Q_k U^T X)$ with Q_k being the $(p-k) \times p$ matrix obtained when deleting the first k rows of $\mathbb{I}_{p\times p}$:

$$\begin{cases} H_{k,0} : \Delta_k = \theta \mathbb{I}_{(p-k)\times(p-k)} & \text{for some } \theta > 0 \\ H_{k,1} : \Delta_k \ne \theta \mathbb{I}_{(p-k)\times(p-k)} & \text{for all } \theta > 0 \end{cases}. \tag{7.17}$$

Whenever the hypothesized value k is chosen by scree plot inspection (data snooping) over the set K, we will need to adjust for multiple testing: let p_k be the p-value corresponding to $H_{k,0}$, and let $p_{(1)} \le p_{(2)} \le ...$ be the ordered p-values from smallest to largest. Starting at $j = 1$ and going up, (i) if $p_{(j)} \le \frac{\alpha}{(|K|-j+1)}$ we reject the hypothesis corresponding to $p_{(j)}$ and go to $j+1$, and (ii) if $p_{(j)} > \frac{\alpha}{(|K|-j+1)}$ we accept the hypothesis

corresponding to $p_{(j)}$ and also all hypotheses corresponding to $p_{(j')}$ with $j' \geq j$, and terminate.
This corresponds to the **Bonferroni-Holm Adjustment**, ensuring that the probability of falsely rejecting $H_K = \bigcup_{k \in K} H_{0,k}$ is at most α. Note that $H_K = \bigcup_{k \in K} H_{0,k} = \{H_0 : \text{rank}(\Phi) \leq k_{\max}\}$, since the least j (if any) for which $H_{j,0}$ is accepted is the *de facto* estimate of $\text{rank}(\Phi)$.

• If we assume a specific distribution (e.g. Gaussian) we can employ **model selection** in the context of the low rank plus noise model. Then, the low rank plus noise covariance testing scheme may be seen as a sort of analysis of deviance for covariance: (i) each such model is a restricted version of the general (unrestricted model) when $k = p$, (ii) for $k_1 \leq k_2$ the corresponding models are nested. Thus, the test at step k may be seen as a likelihood ration test for a sub-model. More generally (and for different models) small k yields parsimonious models, and at the same time smaller k usually yields worse fit (i.e. lowers the maximum likelihood). Thus we can use an information criterion (AIC/BIC), and depending on the precise model these will lead to threshold criteria.

Proposition 7.1.11. *Considering the Gaussian low rank plus noise model $X_i = L_i + \epsilon_i$ for $X_i \overset{iid}{\sim} \mathcal{N}(0, \Sigma)$, $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2 \mathbb{I}_p)$ where $i = 1, ..., n$, σ is known, Σ is unknown, and X_i, ϵ_i are totally independent across indices. Then, the model selection takes the form $\widehat{\lambda}_k > \text{threshold}(n)$ [which resembles a (sample-size dependent) condition number criterion].*

Proof. $(*)$ Suppose that we have a statistical model M_k of some data, with k the number of estimated parameters in the model. Let $l_k = l(\widehat{M}_k)$ be the maximized value of the log-likelihood function for the k-th model. Then the AIC value of the model is,

$$\text{AIC}_{(M_k)} = 2k - 2l_k.$$

Given a set of candidate models for the data, the preferred model is the one with the minimum AIC value. Thus, AIC rewards goodness of fit (as assessed by the likelihood function), but it also includes a penalty that is an increasing function of the number of estimated parameter, which discourages over-fitting. Given the setting, the observed data satisfies $X_i \overset{iid}{\sim} \mathcal{N}(0, \Sigma + \sigma^2 \mathbb{I}_p)$. Note that the log-likelihood for a postulated Σ is:

$$l(\Sigma) \approx -\frac{n}{2} \log|\Sigma + \sigma^2 \mathbb{I}_p| - \frac{n}{2} \text{trace}\left((\Sigma + \sigma^2 \mathbb{I}_p)^{-1} \widehat{C} \right),$$

where $\widehat{C}$ is the empirical covariance of the perturbed i.i.d. observations $X_1, ..., X_n$:

$$\widehat{C} = \frac{1}{n} \sum_{i=1}^{n} X_i X_i^T = \sum_{i=1}^{p} \widehat{\lambda}_i u_i u_i^T,$$

which is the maximum likelihood estimator of $C = \Sigma + \sigma^2 \mathbb{I}$. In turn, conditioned on Σ being of rank k, the restricted MLE is given by,

$$\widehat{C}_k = \sum_{i=1}^{k} \widehat{\lambda}_i u_i u_i^T + \sum_{i=k+1}^{p} \sigma^2 u_i u_i^T.$$

Hence, the log-likelihood for the model prescribing that Σ has rank k is:

$$\begin{aligned} l_k \approx -\frac{n}{2} \log|\widehat{C}_k| - \frac{n}{2} \text{trace}(\widehat{C}_k^{-1} \widehat{C}) &= -\frac{n}{2} \left(\sum_{i=1}^{k} \log\left(\widehat{\lambda}_i\right) + \sum_{i=k+1}^{p} \log(\sigma^2) \right) - \frac{nk}{2} - \frac{n}{2} \sigma^{-2} \left(\sum_{i=k+1}^{p} \widehat{\lambda}_i \right) \\ &= -\frac{n}{2} \sum_{i=1}^{k} \log\left(\widehat{\lambda}_i\right) - \frac{n}{2}(p-k) \log(\sigma^2) - \frac{nk}{2} - \frac{n}{2} \sigma^{-2} \left(\sum_{i=k+1}^{p} \widehat{\lambda}_i \right) \end{aligned}$$

$$\approx -\frac{n}{2}\left(\sum_{i=1}^{k}\log\left(\widehat{\lambda}_i\right) - k\log\left(\sigma^2\right) + k + \sigma^{-2}\left(\sum_{i=k+1}^{p}\widehat{\lambda}_i\right)\right)$$
$$\approx -\frac{n}{2}\left(\sum_{i=1}^{k}\log\left(\widehat{\lambda}_i\sigma^{-2}\right) + k + \sigma^{-2}\left(\sum_{i=k+1}^{p}\widehat{\lambda}_i\sigma^{-2}\right)\right),$$

up to additive constants independent of k. Hence:

$$\mathrm{AIC}_k = 2k + \frac{n}{2}\left(\sum_{i=1}^{k}\log\left(\widehat{\lambda}_i\sigma^{-2}\right) + k + \sum_{i=k+1}^{p}\left(\widehat{\lambda}_i\sigma^{-2}\right)\right).$$

Inclusion of an additional principal component (i.e. increasing the number of parameters of the model) is justified if the criterion value decreases. That is, the model M_{k+1} is preferred to the model M_k if $\mathrm{AIC}_{k+1} < \mathrm{AIC}_k$. That is, if we have:

$$\mathrm{AIC}_{k+1} - \mathrm{AIC}_k = n\left(\log\left(\frac{\widehat{\lambda}_{k+1}}{\sigma^2}\right) + 1 - \frac{\widehat{\lambda}_{k+1}}{\sigma^2}\right) + 2 < 0.$$

Finally, this occurs when:

$$\widehat{\lambda}_{k+1} > \sigma^2\left(\frac{2}{n} + \log\left(\frac{\widehat{\lambda}_{k+1}}{\sigma^2}\right) + 1\right) \equiv \mathrm{threshold}(n).$$

□

7.2 Canonical Correlation Analysis (CCA)

Suppose that our variables can be naturally assigned into two groups: inputs (e.g. lifestyle/exercise variables) and outputs (e.g. health indicators). We seek to understand associations between the inputs and outputs, and naturally we would investigate the pairwise correlations between all input/output pairs. However, such an approach is arguably inefficient and ineffective since if both groups have cardinality p then there are $p(p-1)/2$ such pairs (i.e. a large number of pairs). Also, possibly no single pair may be too correlated, but it may be that the group is as a whole. **Canonical Correlation Analysis** seeks to approximate/summarize the associations with relatively few statistical summaries. Each summary is the correlation between some linear combination of input variables with some other linear combination of output variables. It is in this sense that CCA can be thought of as an extension of regression (or regression may be thought of CCA with a singleton output group). Another way of thinking about it is that CCA is to cross-covariance matrices what PCA is to covariance matrices (*between* versus *within* dependencies).

CCA

- Let X and Y be random vectors in $\mathbb{R}^p$ and $\mathbb{R}^q$ respectively with covariance matrix $\Sigma = \mathrm{Cov}(X,Y) = \begin{pmatrix}\Sigma_X & \Sigma_{XY}\\ \Sigma_{XY}^T & \Sigma_Y\end{pmatrix}$. Assume wlog that $p \le q$.
- We seek $\alpha \in \mathbb{R}^p$ and $\beta \in \mathbb{R}^q$ to maximize $\mathrm{Cov}\left(\alpha^T X, \beta^T Y\right) = \alpha^T \Sigma_{XY}\beta$.
- For this, we need a constraint or else the objective can grow without bound: $\mathrm{Var}\left(\alpha^T X\right) = \alpha^T\Sigma_X\alpha = 1$ and $\mathrm{Var}\left(\beta^T Y\right) = \beta^T\Sigma_Y\beta = 1$.

Such a pair (α_1, β_1) is called the **first pair of canonical variables**, and its covariance $\mathrm{Cov}\left(\alpha_1^T X, \beta_1^T Y\right) = \mathrm{corr}\left(\alpha_1^T X, \beta_1^T Y\right) = \omega_1$ is called the **first canonical correlation**. The **second pair of canonical variables**

(α_2, β_2) and the **second canonical correlation** is defined in a similar way, but with the additional constraints: $\mathrm{Cov}\big(\alpha_1^T X, \alpha_2^T X\big) = \alpha_1^T \Sigma_X \alpha_2 = 0$ and $\mathrm{Cov}\big(\beta_1^T Y, \beta_2^T Y\big) = \beta_1^T \Sigma_Y \beta_2 = 0$.

Solution to CCA. Provided $\Sigma_X, \Sigma_Y \succ 0$, we will show in the next Theorem that $k = \mathrm{rank}(\Sigma_{XY})$ canonical pairs exist. Moreover, we will also show that the canonical pairs are given by $\alpha_j = \Sigma_X^{-\frac{1}{2}} u_j$ and $\beta_j = \Sigma_Y^{-\frac{1}{2}} v_j$ for $j \leq k$, where (u_j, v_j) are the singular vectors of the *canonical correlation matrix* $\Phi = \Sigma_X^{-\frac{1}{2}} \Sigma_{XY} \Sigma_Y^{-\frac{1}{2}} = U \Omega V^T$, and canonical correlations given by the corresponding singular values of Φ. □

Theorem 7.2.1 (Canonical Correlation Analysis - CCA)**.** *Let X and Y be random vectors in $\mathbb{R}^p$ and $\mathbb{R}^q$ respectively, with $p \leq q$, with covariance matrix*

$$\Sigma = \mathrm{Cov}(X, Y) = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{XY}^T & \Sigma_Y \end{pmatrix}, \tag{7.18}$$

where $\Sigma_X, \Sigma_Y \succ 0$. Let $k = \mathrm{rank}(\Sigma_{XY})$ and let $U \Omega V^T$ be the SVD of,

$$\Phi = \Sigma_X^{-\frac{1}{2}} \Sigma_{XY} \Sigma_Y^{-\frac{1}{2}}, \tag{7.19}$$

where $U = \begin{pmatrix} u_1 & \dots & u_p \end{pmatrix}$ and $V = \begin{pmatrix} v_1 & \dots & v_q \end{pmatrix}$. Then, for $j = 1, ..., k$,

$$\sup_{\begin{pmatrix} \alpha & \beta \end{pmatrix}^T \in C_j(X,Y)} \mathrm{Cov}\big(\alpha^T X, \beta^T Y\big) = \mathrm{corr}\left(\left(\Sigma_X^{-\frac{1}{2}} u_j \right)^T X, \left(\Sigma_Y^{-\frac{1}{2}} v_j \right)^T Y \right) = \omega_j, \tag{7.20}$$

where the constraint sets $C_j(X,Y) \subset \mathbb{R}^{p+q}$ are defined as:

$$C_1(X,Y) = \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{R}^{p+q} : Var\big(\alpha^T X\big) = Var\big(\beta^T Y\big) = 1 \right\}, \tag{7.21}$$

$$C_j(X,Y) = \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in C_1(X,Y) : Cov\big(\alpha^T X, \alpha_i^T X\big) = Cov\big(\beta^T Y, \beta_i^T Y\big) = 0, \ \ \forall i < j \right\}, \ \ j \geq 2. \tag{7.22}$$

Proof. The $\alpha_j = \Sigma_X^{-\frac{1}{2}} u_j$ and $\beta_j = \Sigma_Y^{-\frac{1}{2}} v_j$ are feasible (i.e. they satisfy the constraints). To establish that the spectrum is equal to the two quantities as stated, we proceed in two steps. First we notice that:

$$\mathrm{corr}\left(\left(\Sigma_X^{-\frac{1}{2}} u_j \right)^T X, \left(\Sigma_Y^{-\frac{1}{2}} v_j \right)^T Y \right) = u_j^T \Sigma_X^{-\frac{1}{2}} \Sigma_{XY} \Sigma_Y^{-\frac{1}{2}} v_j = u_j^T U \Omega V^T v_j = \omega_j.$$

Thus the second equality is immediately true by the SVD. As for the first equality:

$$\mathrm{Cov}\big(\alpha^T X, \beta^T Y\big) = \alpha^T \Sigma_{XY} \beta = \left(\Sigma_X^{\frac{1}{2}} \alpha \right)^T \Sigma_X^{-\frac{1}{2}} \Sigma_{XY} \Sigma_Y^{-\frac{1}{2}} \left(\Sigma_Y^{\frac{1}{2}} \beta \right) = \gamma^T \Phi \theta.$$

So we have the equivalences (with the analogous implications for β's and θ's),

$$\alpha^T \Sigma_X \alpha = 1 \iff \gamma^T \gamma = 1 \ \text{ and } \ \mathrm{Cov}\big(\alpha^T X, \alpha_i^T X\big) = 0 \iff \underbrace{\left(\Sigma_X^{\frac{1}{2}} \alpha_i \right)^T}_{:= \gamma_i^T} \gamma = 0.$$

Hence as (α,β) ranges over the constraint sets $C_j(X,Y)$, (γ,θ) range over,

$$C_1' = \left\{ \left(\alpha^T,\beta^T\right)^T \in \mathbb{R}^{p+q} : \gamma^T\gamma = \theta^T\theta = 1 \right\},$$

$$C_j' = \left\{ \left(\alpha^T,\beta^T\right)^T \in C_1' : \gamma^T\gamma_i = \theta^T\theta_i = 0,\ \ \forall i < j \right\},\ \ j = 2,...,k.$$

The result will now follow from Cauchy-Schwarz and the SVD of Φ. Now, by the Cauchy-Schwarz Inequality, we have:

$$\left|\gamma^T\Phi\theta\right| \leq \|\gamma\|\|\Phi\theta\| \leq \|\gamma\|\sqrt{\theta^T\Phi^T\Phi\theta}.$$

With this in mind we note:

- The upper-bound is attained when γ is collinear with $\Phi\theta$. So to maximize the expression, we seek feasible γ and θ such that γ is collinear with $\Phi\theta$.
- $\Phi^T\Phi \succeq 0$. Thus by PCA, the second term of the upper-bound is maximized over the constraint sets C_1' and C_j' at the first k eigenvectors of $\Phi^T\Phi$ respectively. Equivalently, over the first k right singular vectors $\{v_j\}_{j\leq k}$ of Φ.
- Once this choice is made, we note that the choice of unit vectors γ in the constraint sets do not affect the value of the objective, so long as they are collinear to the corresponding $\Phi v_j \overset{SVD}{=} \omega_j u_j$. This forces us to choose the γ from the constraint sets C_1' and C_j' as the left singular vectors $\{u_j\}_{j\leq k}$ of Φ.

Back-transforming from γ's and θ's to α's and β's now completes the proof. □

Contrary to PCA, the nature of CCA constraints make it equivariant under standardisation, and invertible affine transformation more generally:

Theorem 7.2.2 (Invariance/Equivariance of Canonical Correlations/Pairs)**.** *In the same context as the previous Theorem on CCA, let $f(x) = Fx + \phi$ and $g(y) = Gy + \gamma$ be invertible affine transformations on $\mathbb{R}^p$ and $\mathbb{R}^q$ respectively. Then:*
(i) The canonical correlations of $\{f(X), g(Y)\}$ are the same as those of $\{X,Y\}$.
(ii) The canonical pairs of $\{f(X), g(Y)\}$ are the inversely transformed canonical pairs of $\{X,Y\}$, via f^{-1} and g^{-1} respectively.

Proof. Covariance is invariant to translations, so we may assume $\phi = \gamma = 0$. Let $\kappa_{X,Y}(\alpha,\beta) = \mathrm{Cov}\left(\alpha^T X, \beta^T Y\right)$ and $\kappa_{f,g}(\alpha,\beta) = \mathrm{Cov}\left(\alpha^T FX, \beta^T GY\right)$ be the original and transformed objectives. Consider the (bijective) change of variables,

$$\begin{pmatrix}\widetilde{\alpha}\\ \widetilde{\beta}\end{pmatrix} = D\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}F^T\alpha\\ G^T\beta\end{pmatrix}, \qquad \text{where } D = \begin{pmatrix}F^T & 0\\ 0 & G^T\end{pmatrix}.$$

In these variables:

$$\kappa_{f,g}(\alpha,\beta) = \mathrm{Cov}\left(\alpha^T FX, \beta^T GY\right) = \mathrm{Cov}\left(\widetilde{\alpha}^T X, \widetilde{\beta}^T Y\right) = \kappa_{X,Y}\left(\widetilde{\alpha},\widetilde{\beta}\right),$$

and,

$$(\alpha,\beta) \in C_1(f,g) \iff \mathrm{Var}\underbrace{\left(\alpha^T FX\right)}_{\widetilde{\alpha}^T X} = \mathrm{Var}\underbrace{\left(\beta^T GY\right)}_{\widetilde{\beta}^T Y} = 1 \iff \left(\widetilde{\alpha},\widetilde{\beta}\right) \in C_1(X,Y).$$

Now, given this equivalence, we have for $j = 2,...,k$:

$$\begin{aligned}(\alpha,\beta) \in C_j(f,g) &\iff (\alpha,\beta) \in C_1(f,g) \ \text{ and } \ \mathrm{Cov}\left(\alpha^T FX, \alpha_i^T FX\right) = \mathrm{Cov}\left(\beta^T GY, \beta_i^T GX\right) = 0 \ \ \forall i < j\\ &\iff \left(\widetilde{\alpha},\widetilde{\beta}\right) \in C_1(X,Y) \ \text{ and } \ \mathrm{Cov}\left(\widetilde{\alpha}^T X, \widetilde{\alpha}_i^T X\right) = \mathrm{Cov}\left(\widetilde{\beta}^T Y, \widetilde{\beta}_i^T X\right) = 0 \ \ \forall i < j\\ &\iff \left(\widetilde{\alpha},\widetilde{\beta}\right) \in C_j(X,Y).\end{aligned}$$

Letting $(\widetilde{\alpha}_j^*, \widetilde{\beta}_j^*)$ be the maximizer of $\kappa_{X,Y}$ over $C_j(X,Y)$ and $\begin{pmatrix}\alpha^*\\ \beta^*\end{pmatrix} = D^{-1}\begin{pmatrix}\widetilde{\alpha}_j^*\\ \widetilde{\beta}_j^*\end{pmatrix}$,

$$\kappa_{X,Y}(\widetilde{\alpha}_j^*, \widetilde{\beta}_j^*) \geq \kappa_{X,Y}(\widetilde{\alpha}_j, \widetilde{\beta}_j), \quad \forall (\widetilde{\alpha}_j, \widetilde{\beta}_j) \in C_j(X,Y) \Longrightarrow \kappa_{f,g}(\alpha_j^*, \beta_j^*) \geq \kappa_{f,g}(\alpha_j, \beta_j), \quad \forall (\alpha_j, \beta_j) \in C_j(f,g).$$

□

At the level of the sample, CCA may be carried out by replacing the covariance matrix by the sample covariance matrix. Since $(\widehat{\Sigma}_X, \widehat{\Sigma}_Y)$ are consistent for (Σ_X, Σ_Y), and we have $\mathbb{P}(\widehat{\Sigma}_X, \widehat{\Sigma}_Y \succ 0) = 1$ for all n sufficiently large.

Proposition 7.2.3. *Let $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$ be random vectors with strictly positive covariances Σ_X and Σ_Y, with absolutely continuous law. Then, for all n sufficiently large:*

$$\mathbb{P}\left(\widehat{\Sigma}_X \succ 0, \widehat{\Sigma}_Y \succ 0\right) = 1. \tag{7.23}$$

Proof. $(*)$ Let $X_1, ..., X_n$ be i.i.d. copies of the random vector $X \in \mathbb{R}^p$, with $n \geq p$. Suppose that $\widehat{\Sigma}_n$ is not full rank. Then, there exists $v \in \mathbb{R}^p$ such that:

$$0 = v^T \widehat{\Sigma}_n v = \frac{1}{n-1}\sum_{j=1}^n v^T X_i X_i^T v = \sum_{j=1}^n \langle v, X_i \rangle^2,$$

which yields that v is orthogonal to all X_i. But this occurs with probability zero since $\mathbb{P}(\text{span}(X_1, ..., X_p) = \mathbb{R}^p) = 1$. Indeed, the probability of the complementary is upper-bounded by that of drawing a vector in a $(p-1)$-dimensional subspace, which is 0 for all absolutely continuous measures. To conclude, we observe that a finite intersection of a.s. events is a.s. . □

Thus for n sufficiently large, when the singular values of the cross-covariance Σ_{XY} are distinct, the sample canonical pairs and canonical correlations are the MLE of their population versions. As for their asymptotic properties, notice that the sample canonical pairs/correlations can be related to the eigenvectors/eigenvalues of $\widehat{\Phi}\widehat{\Phi}^T$ and $\widehat{\Phi}^T\widehat{\Phi}$, where $\widehat{\Phi} := \widehat{\Sigma}_X^{-\frac{1}{2}}\widehat{\Sigma}_{XY}\widehat{\Sigma}_Y^{-\frac{1}{2}}$. Thus we can use our perturbation bounds once we can control the deviations $\left\|\widehat{\Phi}\widehat{\Phi}^T - \Phi\Phi^T\right\|_{\mathbb{R}^{q\times q}}$ and $\left\|\widehat{\Phi}^T\widehat{\Phi} - \Phi^T\Phi\right\|_{\mathbb{R}^{p\times p}}$. In the presence of invertibility, $\Sigma \mapsto \Phi$ and $\widehat{\Sigma} \mapsto \widehat{\Phi}$ are Lipschitz continuous, and so we may obtain such bounds. Similar arguments involving the differentiability of these maps can be used to obtain $\sqrt{n}$ asymptotic Gaussian limits (allowing inference) via the Delta Method.

Proposition 7.2.4. *We have that $\Sigma \mapsto \Sigma^2$, $\Sigma \mapsto \Sigma^{\frac{1}{2}}$ and $\Sigma \mapsto \Sigma^{-1}$ are all C^1 for $\Sigma \succ 0$.*

Proof. $(*)$ To show that all these matrix functions are continuously differentiable (C^1) for $\Sigma \succ 0$, we will use the fact that these functions are smooth functions of the eigenvalues of the matrix. The argument is very similar for all these three cases. Wlog, consider the function $\Sigma \mapsto \Sigma^{-1}$. For $\Sigma = U\Lambda U^T$, the inverse is given by $\Sigma^{-1} = U\Lambda^{-1}U^T$ where Λ^{-1} is the diagonal matrix with entries λ_i^{-1}. The function $x \mapsto x^{-1}$ is smooth and continuously differentiable on $\mathbb{R}_{>0}$. This, with the smoothness of the *SVD mapping* $\Sigma \mapsto U\Lambda U^T$ with respect to Σ, ensures that $\Sigma \mapsto \Sigma^{-1}$ is C^1. □

Chapter 8

Gaussian Graphical Models

In this final Chapter, we briefly apply Statistical Theory to the study of (Gaussian) Graphical Models. We first start with a warm-up by presenting Markov chains, before rapidly moving on to the AR(1) model. We then move on to general Gaussian graphical models, along with the Markov properties. We finally end with the notions of density factorization with respect to a graph, and statistical inference for Gaussian graphical models.

As an important special case, we consider a stationary Markov chain. Set a sequence of identically distributed random scalars $\xi_1, \xi_2, \xi_3, ...$, satisfying the Markov property, i.e. the past is independent of the future given the present: $\{\xi_i\}_{i<k} \perp \{\xi_j\}_{j>k} | \xi_k$ for all k. By stationarity, the transition density $f_{\xi_{k+1}|\xi_k} = g$ is time-invariant.



8.1 Stationary Gaussian AR(1) Model

We first start with a characterization of the stationary Gaussian AR(1) Model.

Proposition 8.1.1 (AR(1) Model)**.** *Assuming that $X = (\xi_1, ..., \xi_p)^T$ is jointly centred Gaussian, this implies that $\xi_1 \sim \mathcal{N}\left(0, \frac{\sigma^2}{1-\rho^2}\right)$ and $\xi_{k+1} = \rho\xi_k + \epsilon_{k+1}$, where $\rho := \text{corr}(\xi_k, \xi_{k+1})$, $|\rho| < 1$, $\epsilon_i \perp \xi_i$ and $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.*

Proof. $(*)$ Let $\rho := \text{corr}(\xi_k, \xi_{k+1})$, which is independent of k by stationarity. Assume $\rho \notin \{-1, 1\}$, otherwise the claim is trivial. Then, by the regression representation [Theorem (3.3.1)], for all k there exists a centred random variable ϵ_k, independent of ξ_k with variance $\sigma^2 > 0$ given by,

$$\epsilon_k := \xi_{k+1} - \rho\xi_k.$$

Note that ϵ_k need be Gaussian, since Gaussians are closed under linear modifications and that the sequence of ϵ_k need be i.i.d. by stationarity. By assumption, all ξ_k are centred and Gaussian, say $\mathcal{N}(0, \nu^2)$. By stationarity:

$$\xi_1 \overset{d}{=} \xi_2 \overset{d}{=} \rho\xi_1 + \epsilon_k,$$

so that we must have $\nu^2 = \rho^2\nu^2 + \sigma^2$. Solving for ν yields $\nu^2 = \frac{\sigma^2}{1-\rho^2}$, as required. □

Now, the Markov property stipulates that the density factorizes:

$$f(\xi_1,...,\xi_p;\Theta) = f_{\xi_1}(\xi_1;\Theta)\prod_{j=1}^{p-1} f_{\xi_{j+1}|\xi_j}(\xi_{j+1}|\xi_j;\Theta), \tag{8.1}$$

where the conditional densities $f_{\xi_{j+1}|\xi_j}(\cdot|y;\Theta)$ are $\mathcal{N}(\rho y,\sigma^2)$ PDFs. For a **single realisation** of the vector $X=(\xi_1,...,\xi_p)^T$, this yields a log-likelihood (up to constants):

$$l_1(\rho,\sigma^2) = -\frac{1}{2}\log\left(\frac{\sigma^2}{1-\rho^2}\right) - \frac{(p-1)}{2}\log(\sigma^2) - \frac{(1-\rho^2)}{2\sigma^2}\xi_1^2 - \sum_{j=1}^{p-1}\frac{(\xi_{j+1}-\rho\xi_j)^2}{2\sigma^2}. \tag{8.2}$$

For n **independent realisations** $X_i=(\xi_{i,1},...,\xi_{i,p})^T$ are available, we get:

$$l_1(\rho,\sigma^2) = -\frac{n}{2}\log\left(\frac{\sigma^2}{1-\rho^2}\right) - \frac{n(p-1)}{2}\log(\sigma^2) - \frac{(1-\rho^2)}{2\sigma^2}\sum_{i=1}^{n}\xi_{i,1}^2 - \sum_{i=1}^{n}\sum_{j=1}^{p-1}\frac{(\xi_{i,j+1}-\rho\xi_{i,j})^2}{2\sigma^2}. \tag{8.3}$$

Let us now look at the precision matrix of the AR(1) Model. The only pairs (ξ_i,ξ_j) that are not conditionally independent given $\{\xi_k : k\notin\{i,j\}\}$ are adjacent pairs,i.e. $|i-j|=1$. As a result, the precision matrix Θ of a stationary Gaussian AR(1) model is **tridiagonal**. Now notice that Σ is determined element-wise via,

$$\mathrm{Cov}(\xi_i,\xi_{i+k}) = \sqrt{\mathrm{Var}(\xi_i)\mathrm{Var}(\xi_{i+k})}\cdot\mathrm{corr}(\xi_i,\xi_{i+k}) = \frac{\sigma^2}{1-\rho^2}\rho^k. \tag{8.4}$$

Thus, we can directly verify that $\Theta=\Sigma^{-1}=L^TL$ where,

$$L = \sigma^{-1}\begin{pmatrix} \sqrt{1-\rho^2} & 0 & \dots & \dots & 0 \\ -\rho & 1 & 0 & \dots & 0 \\ 0 & -\rho & 1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & -\rho & 1 \end{pmatrix}. \tag{8.5}$$

With this decomposition, we may calculate the log-likelihood of L based on a single realisation of X using the expression of the multivariate Gaussian density:

$$l_1(L) = \frac{1}{2}\log|L^TL| - \frac{1}{2}X^TL^TLX. \tag{8.6}$$

Thus in summary, the Markov property yields a factorisation of the joint density by suitable conditioning. It also leads to a sparse precision matrix and it substantially increases the statistical efficiency. Now, let us see if there is a more general structure underlying all this. After all, we recall that for jointly Gaussian $(X^T,Y^T,Z^T)^T$ with $\Sigma\succ 0$, $\{X\perp Y\}|Z \iff f_{XZY}=f_{X|Z}f_{Y|Z}f_Z \iff \Theta_{XY}=0$. Let us now see the equivalent formulation of the AR(1) Model as a graphical model. To this end, define a graph $G=(V,E)$ with vertex/edge set as $V=[p]=\{1,...,p\}$ and $E=\{(i,j)\in[p]^2 : |i-j|=1\}\subset V^2$. We do not allow for self-loops, and therefore:

$$\{\xi_i\perp\xi_j\}\big|\{\xi_k : k\in[p]\setminus\{i,j\}\} \iff (i,j)\notin E. \tag{8.7}$$

In other words: (i) absence of an edge (i,j) indicates conditional independence given all these, (ii) absence of an edge (i,j) indicates that dependence between ξ_i and ξ_j is indirect, and (iii) presence of an edge (i,j) indicates direct dependence between ξ_i and ξ_j.

A *graphical model* generalises the chain-like dependence structure to more general combinatorial dependence structures encoded by a more general graph.

8.2 Markov Properties in Gaussian Graphical Models

Definition 8.2.1. *A **Gaussian Graphical Model** encodes the conditional independencies amongst the coordinates $V := \{1, ..., p\}$ of a Gaussian vector via the edges in a graph on V.*

Thus, there are three ways a graph $G = (V, E)$ could define a Markov property.

Definition 8.2.2. *The **Pairwise Markov Property**: No edge between X_i and X_j implies their conditional independence given remaining variables, i.e. $(i,j) \notin E \Longrightarrow \{X_i \perp X_j\} \big| \{X_k\}_{k \neq i,j}$.*

Definition 8.2.3. *The **Local Markov Property**: Conditional on its graph neighbours, i-th variable is independent of all other variables, i.e. $\big\{X_i \perp \{X_j : (i,j) \notin E\}\big\} \big| \{X_k : (i,k) \in E\}$.*

Definition 8.2.4. *The **Global Markov Property**: Two subvectors are conditionally independent given a subvector that separates them in G ($S \subset V$ **separates** $A, B \subset V$ if removing S from V disconnects A from B), i.e. $S \subset V$ separates $A \subset V$ from $B \subset V$ in $G \Longrightarrow \{X_A \perp X_B\} \big| X_S$.*

In full generality, regardless of Gaussianity, it is not hard to see that:

$$\text{Global Markov} \Longrightarrow \text{Local Markov} \Longrightarrow \text{Pairwise Markov}.$$

To get the converse, we need to exclude *perfect dependence*.

Theorem 8.2.5 (Equivalence of Markov Properties - Gaussian case)**.** *The three Markov properties are equivalent for $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^p$ with $\Sigma \succ 0$.*

Proof. Write $V = \{1, ..., p\}$ and $\Theta = \Sigma^{-1}$. Given the list of implications, it suffices to prove that when $\Sigma \succ 0$, the pairwise Markov property with respect to some graph $G = (V, E)$ implies the Global Markov property with respect to G. Thus assume that G encodes the pairwise Markov property. In the Gaussian case, this happens if and only if $E = \{i \neq j : \Theta_{ij} \neq 0\}$, by the Gaussian conditional independence Theorem (3.3.3). Now, we need to show that this graph structure also yields the Global Markov property. To this aim, assume $S \subset V$ separates $A, B \subset V$ in the graph G. If (A, B, S) partition V, i.e. $A \cup B \cup S = V$, then we are done. To see this, note that since A and B are separated, there is no edge from $i \in A$ to $j \in B$. It follows that $\Theta_{ij} = 0$ for all $(i,j) \in A \times B$. Block-wise, this says that $\Theta_{AB} = 0$. Thus, by Theorem (3.3.3), we have $X_A \perp X_B$ given all other variables. However all other variables coincide with X_S, since $V = A \cup B \cup S$.
Now consider the general case. Let $R = V \setminus [A \cup B \cup S]$ be the remaining vertices. Partition $R = R_A \cup R_B \cup R_0$ where:

- R_A contains all the vertices in R that are path-connected with A.
- R_B contains all the vertices in R that are path-connected with B.
- $R_0 = R \setminus [R_A \cup R_B]$ are the remaining vertices in R.

We highlight that this is indeed a partition of R: it must be that $R_A \cap R_B = \varnothing$ or else S would not separate A and B. For the same reason, S necessarily separates R_A from R_B, R_A from B and R_B from A. Finally, any $v \in R_0$ is disconnected from both B and R_B, or else v would be contained in R_B. Now, the trick will be to augment the sets A and B, use the first part and finally marginalise. Thus define $\widetilde{A} = A \cup R_A \cup R_0$ and $\widetilde{B} = B \cup R_B$. Then, from our preceding discussion, S separates $\left(\widetilde{A} \cup \widetilde{B}\right)$ and furthermore $\left(\widetilde{A}, S, \widetilde{B}\right)$ is

a partition of V, and so by the first part of the proof,

$$\{X_{\widetilde{A}} \perp X_{\widetilde{B}}\}|X_S.$$

But since $A \subset \widetilde{A}$ and $B \subset \widetilde{B}$, this implies that $\{X_A \perp X_B\}|X_S$, as required. □

In this proof, we relied on Gaussianity to go from pairwise conditional independence to blockwise conditional independence. This is in fact a remarkable feature of Gaussians, where no interactions involve more than pairs of variables. This may already be seen at the level of the density:

$$\log(f(x)) = \text{constant} - \frac{1}{2}\sum_{v\in V}\Theta_{vv}x_v^2 - \frac{1}{2}\sum_{(v,v')\in E}\Theta_{vv'}x_v x_{v'}, \qquad x=(x_1,...,x_p)^T \in \mathbb{R}^p. \tag{8.8}$$

Remark 8.2.6. *In the non-Gaussian case, the above Theorem is still valid provided we replace non-singularity of the covariance with the everywhere positivity of the joint density.*

8.3 Density Factorisation with Respect to a Graph

Now that we have clarified how the Markov property with respect to a graph relates to precision matrix sparsity, let us turn to the factorisation of the density. For this, let us revisit the AR(1) model example momentarily. The factorisation,

$$f(u_1,...,u_n) = f_{\xi_1}(u_1)\prod_{j=1}^{p-1} f_{\xi_{j+1}|\xi_j}(u_{j+1}|u_j), \tag{8.9}$$

used the fact that the graph was well-ordered, to arrive at a form $f(u_1,...,u_n) \propto \prod_{j=1}^{p-1}\psi_j(u_j,u_{j+1})$, for $\psi_i : \mathbb{R}^2 \to (0,\infty)$. To see the general picture, where there is usually no ordering, we need some definitions related to graphs.

Definition 8.3.1. *A **clique** of $G=(V,E)$ is a fully connected subset of V.*

Definition 8.3.2. *A **maximal clique** of $G=(V,E)$ is a clique that is not a strict subset of another clique.*

The important fact is that the **AR(1) model joint density factorises over its maximal cliques**.

Definition 8.3.3. *Let $V=\{1,...,p\}$, $G=(V,E)$ a graph and $f(x_1,...,x_p)>0$ be an everywhere positive density on $\mathbb{R}^p$. We say that **$f>0$ factorises with respect to G** if,*

$$f_{X_1,...,X_p}(x_1,...,x_p) = \prod_{C\subseteq V}\psi_C(x_C), \tag{8.10}$$

for 2^p interaction functions $\psi_C>0$ such that $\psi_C=1$ unless C is a clique.

We use the shorthand notation $x_C=(x_{i_1},...,x_{i_k})$ for $C=\{i_1,...,i_k\}\subseteq V$. Said differently, $f>0$ factorises as a product of positive functions ψ_C with C ranging over the collection $\mathcal{C}(G)$ of cliques of G. The reason we give the definition the way we do, is to give a parsimony/reductive intuition, removing terms from a larger product corresponding to non-cliques. This factorization implies that the global distribution may be understood in terms of local interactions. Conversely, this provides a way to construct complex distributions from simpler building blocks, modelling local interactions.

In statistical physics terminology, one calls a density of the form,

$$f_{X_1,...,X_p}(x_1,...,x_p) \propto \exp\left(\sum_{C\in\mathcal{C}(G)} \phi_C(x_C)\right), \tag{8.11}$$

with real-valued and non-identically vanishing valued potential functions $\phi_C : \mathbb{R}^{|C|} \to \mathbb{R}$, a **Gibbs distribution with respect to G**. Now, the fundamental result linking conditional independence and factorisation is the following:

Theorem 8.3.4 (Hammersley-Clifford)**.** *Let $f : \mathbb{R}^p \to (0,\infty)$ be an everywhere positive probability density function. Then:*

f factorises with respect to $G \iff f$ satisfies the local Markov property with respect to G.

Proof. ($\Longrightarrow$) Assume $f > 0$ factorises with respect to G. Consider the conditional of X_i given all else:

$$f\big(x_i|x_{V\setminus\{i\}}\big) = \frac{\prod_{C\subseteq V}\psi_C(x_C)}{\int_{\mathbb{R}}\prod_{C\subseteq V}\psi_C(x_C)dx_i} = \frac{\prod_{i\notin C}\psi_C(x_C)\prod_{i\in C}\psi_C(x_C)}{\prod_{i\notin C}\psi_C(x_C)\int_{\mathbb{R}}\prod_{i\in C}\psi_C(x_C)dx_i} = \frac{\prod_{i\in C}\psi_C(x_C)}{\int_{\mathbb{R}}\prod_{i\in C}\psi_C(x_C)dx_i}.$$

Therefore, given any $j \neq i$, the RHS depends on x_j only if $(i,j) \in E$:
- if $i \in C$ and C is not a clique, then $\psi_C = 1$, so there is no dependence on x_j.
- if $i \in C$ and C is a clique, and then $j \in C$ if and only if $(i,j) \in E$. So, in turn, ψ_C (and hence the RHS) depends on x_j if and only if $(i,j) \in E$.

In other words, f satisfies the local Markov property with respect to G.

($\Longleftarrow$) Assume $f > 0$ satisfies the local Markov property with respect to G. Define $\psi_\varnothing \equiv \psi_\varnothing(x_\varnothing) = f(\mu)$ for some fixed reference point $\mu \in \mathbb{R}^p$ (e.g. take $\mu = 0$). The argument $x_\varnothing$ simply corresponds to the function $\psi_\varnothing$ being a constant (does not depend on any coordinates). Define the remaining ψ's recursively via:

$$\psi_C(x_C) = \begin{cases} 1 & \text{if } C \text{ is not a clique} \\ 1 & \text{if } x_j = \mu_j \text{ for some } j \in C \\ \frac{f(x_C,\mu_{V\setminus C})}{\prod_{B\subset C}\psi_B(x_B)} & \text{otherwise} \end{cases}.$$

Clearly the collection $\{\psi_C\}$ has the form we seek. It only remains to show that f factorises as $\prod_{C\subseteq V}\psi_C$ with these specific interaction functions. Given any $x \in \mathbb{R}^p$, let $A = A(x)$ be the set of all coordinates where x and μ disagree: $A = \{j \in V : x_j \neq \mu_j\}$. Decompose $x = (x_A, x_{V\setminus A})^T \equiv (x_A, \mu_{V\setminus A})^T$. Suppose $A = \varnothing$. In this case $x = \mu$ and we need to show that $f(\mu)$ factorises as stipulated. Checking the definition of $\psi_C(\mu_C) \neq 1$ except $C = \varnothing$, for which we defined $\psi_\varnothing(x_\varnothing) \equiv f(\mu)$. Therefore, the factorisation holds in the form,

$$\prod_{C\subseteq V}\psi_C(\mu_C) = \psi_\varnothing = f(\mu).$$

Now suppose that $A \neq \varnothing$ and is a clique. Then,

$$\prod_{C\subseteq V}\psi_C(x_C) = \psi_A(x_A)\prod_{C\subset A}\psi_C(x_C)\prod_{C\not\subset A}\psi_C(x_C) = \frac{f(x_A,\mu_{V\setminus A})}{\prod_{C\subset A}\psi_C(x_C)}\prod_{C\subset A}\psi_C(x_C)\prod_{C\not\subset A}\psi_C(x_C),$$

and the terms in the last product equals 1 since $C \not\subseteq A$ means that $x_j = \mu_j$ for some $j \in C$, in which case our construction yields $\psi_C = 1$. Noting that $f(x_A,\mu_{V\setminus A}) = f(x_A,x_{V\setminus A})$, we once again get the sought factorisation.

Finally, suppose that $A \neq \varnothing$ and is not a clique. In this case, we will establish the factorisation by induction on the size of A. The base case is $|A| = 0$, where we have already established the factorisation. For $k \leq p$, assume the factorisation holds for $|A| = k-1$, and let us show that it also holds for $|A| = k$. Since A is not a clique, there exists $i, j \in A$ with $(i,j) \notin E$. We thus have:

$$\begin{aligned} f(x_A, x_{V\setminus A}) \equiv f(x_A, \mu_{V\setminus A}) &= f\left(x_i | x_{A\setminus\{i\}}, \mu_{V\setminus A}\right) f\left(x_{A\setminus\{i\}}, \mu_{V\setminus A}\right) \\ &= f\left(x_i | x_{A\setminus\{i\}}, \mu_{V\setminus A}\right) \frac{f\left(x_{A\setminus\{i\}}, \mu_{V\setminus A}, \mu_i\right)}{f\left(\mu_i | x_{A\setminus\{i\}}, \mu_{V\setminus A}\right)} \\ &= \underbrace{\frac{f\left(x_i | x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A}\right)}{f\left(\mu_i | x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A}\right)}}_{(\dagger)} f\left(x_{A\setminus\{i\}}, \mu_{V\setminus A}, \mu_i\right). \end{aligned}$$

Note that we have used the local Markov Property to go from the third to the last step. Now, the term (†) doesn't depend on x_j since $(i,j) \notin E$, so we can fix x_j to whichever value we want without changing the expression. Thus we may fix $x_j = \mu_j$. Using the inductive hypothesis [IH] (i.e. the factorisation):

$$\begin{aligned} f(x_A, x_{V\setminus A}) \equiv f(x_A, \mu_{V\setminus A}) &= \frac{\left(\frac{f(x_i, x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A})}{f(x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A})}\right)}{\left(\frac{f(\mu_i, x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A})}{f(x_{A\setminus\{i,j\}}, \mu_j, \mu_{V\setminus A})}\right)} f\left(x_{A\setminus\{i\}}, \mu_{V\setminus A}, \mu_i\right) \\ &= \frac{\prod_{C\subseteq A\setminus\{j\}} \psi_C(x_C)}{\prod_{C\subseteq A\setminus\{i,j\}} \psi_C(x_C)} \underbrace{\prod_{C\subseteq A\setminus\{i\}} \psi_C(x_C)}_{\text{IH}} = \prod_{C\subseteq A} \psi_C(x_C) = \prod_{C\subseteq V} \psi_C(x_C), \end{aligned}$$

by definition of ψ_C and the fact that $(A\setminus\{j\})\setminus(A\setminus\{i,j\}) = \{i\}$. □

We finally investigate the density factorisation with respect to a graph in the Gaussian case.

Corollary 8.3.5 (Hammersley-Clifford Theorem - Gaussian case)**.** *Let $V = \{1,...,p\}$, $G = (V,E)$ a graph and $X \sim \mathcal{N}(0,\Sigma)$ on $\mathbb{R}^p$ with $\Sigma \succ 0$. Then, the following statements are equivalent:*
(i) The density of X factorises with respect to G.
(ii) X satisfies a Markov property (since all three are equivalent for Gaussians when $\Sigma \succ 0$) with respect to G.
(iii) Given $i \neq j$, the (i,j) entry of $\Theta = \Sigma^{-1}$ is zero if and only if $(i,j) \notin E$.

When the graph is geometric, then one can make use of factorisation cleverly in order to carry out likelihood estimation conforming to a graphical model. The key here is that the graph is quite sparse. However in general, the list of cliques is difficult to obtain, where in fact it is NP-hard.

8.4 Statistical Inference for Gaussian Graphical Models

There are two problems that one might want to consider in the context of statistical inference, given $X_1, ..., X_n \sim \mathcal{N}(\mu, \Sigma)$:

- Fitting a Gaussian distribution (estimating μ and Σ) subject to the constraint that $\mathcal{N}(\mu,\Sigma)$ factorises with respect to a given graph G. This is known as **covariance selection**.
- Estimating the graph G with respect to which $\mathcal{N}(\mu,\Sigma)$ factorises when the parameters are unknown. This is known as **structure estimation**.

In light of the Hammersley-Clifford Theorem, when $\Sigma \succ 0$, these two problems reduce to:

- Estimating the model parameters under the constraint that $\Theta_{ij} = 0$ for known set E of pairs (i,j).
- Estimating the location of zeroes of Θ_{ij} amongst pairs (i,j).

We recall some models for covariance selection, where given a normal data matrix X, a (Gaussian) **covariance selection model** consists in the family $\{\mathcal{N}(\mu,\Sigma) : \mu \in \mathbb{R}^p, \Sigma \in \mathcal{S}\}$, for some subset $\mathcal{S}$ of the cone of $p \times p$ non-negative definite matrices. Depending on our choice of $\mathcal{S}$, an MLE may or may not exist, where a standard choice for $\mathcal{S}$ is the set of strictly positive definite matrices. Even then, we saw that the MLE does not exist when $\widehat{\Sigma}$ is singular, so we expect that existence/uniqueness of the MLE is subtle for general $\mathcal{S}$. Some reasonable choices could be that we choose to impose linear constraints on Σ or on Σ^{-1} (provided it exists). Focussing on the special case where constraints fix some elements to be zero, (i) imposing this on Σ leads to straightforward estimation where when the MLE exists we can annihilate the corresponding entries by equivariance, or (ii) imposing this on Σ^{-1} *leads to estimation under a graphical model.*

Since the constraints apply to the covariance, we can take $\mu = 0$. Now consider a graph $G = (V, E)$ on the index set $V = \{1, ..., p\}$. The model we wish to fit is $\{\mathcal{N}(0,\Sigma) : \Sigma \succ 0 \text{ and } e_i^T \Sigma^{-1} e_j = 0 \text{ whenever } (i,j) \notin E\}$. When $\Sigma \succ 0$, the log-likelihood is (up to constants),

$$l(\Sigma) = -\log|\Sigma| - \operatorname{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big), \tag{8.12}$$

which can be equivalently expressed via $\Theta = \Sigma^{-1}$ as:

$$l(\Theta) = \log|\Theta| - \operatorname{trace}\big(\Theta\widehat{\Sigma}\big). \tag{8.13}$$

The crucial observations are that (i) the objective $l(\Theta)$ is strictly concave over the set $\Theta \succ 0$, and (ii) the constraint set $\{\Theta \succ 0 : e_i^T \Theta e_j = 0 \text{ whenever } (i,j) \notin E\}$ is convex. In conclusion, maximising the log-likelihood under a graphical model constraint is equivalent to a strictly convex optimisation problem and thus provided a maximiser exists, it will also be unique. It turns out that log-likelihood maximisation under a graphical model constraint is equivalent to the entropy maximisation under second moment constraints.

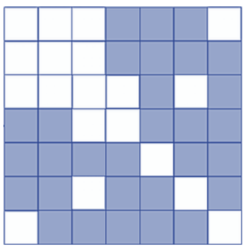
Theorem 8.4.1 (Graphical Modelling as Matrix Completion)**.** *Maximising the log-likelihood* $l(\Sigma) = -\log|\Sigma| - \operatorname{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big)$ *over the set,*

$$\{\Sigma \succ 0 : e_i^T(\Sigma^{-1})e_j = 0 \ \textit{whenever}\ (i,j) \notin E\}, \tag{8.14}$$

is equivalent to maximising the entropy $H(\Sigma) \propto \log|\Sigma| + \textit{constant over the set,}$

$$\{\Sigma \succ 0 : e_i^T \Sigma e_j = e_i^T \widehat{\Sigma} e_j \ \ \textit{whenever}\ (i,j) \in E \ \textit{or}\ i = j\}. \tag{8.15}$$

Intuitively: delete all non-adjacent (with respect to G) off-diagonal entries of $\widehat{\Sigma}$. Then, complete the missing entries to maximise the determinant.



Proof. The objective is negatively infinite at singular matrices, so we focus on interior points of the non-negative definite cone. We then have a differentiable strictly concave objective in Θ with equality con-

straints. We thus resort to the method of Lagrange multipliers. For this, define the Lagrangian,

$$\mathcal{L}(\Theta,\epsilon) = \log|\Theta| - \operatorname{trace}\big(\Theta\widehat{\Sigma}\big) - \operatorname{trace}(\Theta\epsilon) = \log|\Theta| - \operatorname{trace}\big(\Theta(\widehat{\Sigma}+\epsilon)\big),$$

for a symmetric $\epsilon = \{\epsilon_{ij}\}$ with $\epsilon_{ij} = 0$ when $i = j$ or $(i,j) \in E$ (ϵ_{ij} are Lagrange multipliers corresponding to the equality constraints). Constrained optima must be saddle points of the Lagrangian. So if Θ_* is a constrained maximum of l, then Θ_* is a critical point of $\widetilde{l}(\Theta) = \log|\Theta| - \operatorname{trace}\big(\Theta\widehat{\Sigma}\big) = \log|\Theta| - \operatorname{trace}\big(\Theta(\widehat{\Sigma}+\epsilon)\big)$. This is just a Gaussian likelihood corresponding to an empirical covariance $\widetilde{\Sigma} = \widehat{\Sigma} + \epsilon$ instead of $\widehat{\Sigma}$. When $\Theta \succ 0$, this can only have a unique critical point when $\widetilde{\Sigma} \succ 0$, and the critical point is $\Theta_* = \widetilde{\Sigma}^{-1} = \big(\widehat{\Sigma}+\epsilon\big)^{-1}$.
Now, it remains to plug such a Θ_* back into the Lagrangian, and minimise over ϵ with $\epsilon_{ij} = 0$ when $i = j$ or $(i,j) \in E$. Equivalently, it remains to choose ϵ to minimise $l\big((\widehat{\Sigma}+\epsilon)^{-1}\big) = -\log\big|\widehat{\Sigma}+\epsilon\big| - p$. Evidently, minimisation occurs at ϵ such that $\widehat{\Sigma}+\epsilon \succ 0$, compatibly with the requirement of the critical point Θ_*. Finally, recall that entropy is minus the expected log-likelihood to find:

$$-\mathbb{E}\Big(-\log|\Sigma| - \operatorname{trace}\big(\Sigma^{-1}\widehat{\Sigma}\big) + \text{constant}\Big) = \log|\Sigma| + p + \text{constant}.$$

□

8.5 Partial Covariance Modelling and Estimating the Graph Structure from Data

Suppose we are interested in a Gaussian random vector $X = (X_1, X_2, X_3, X_4)^T$, however we can only model the pairs (X_i, X_j) for $(i,j) \in \{(1,2),(2,3),(3,4),(1,4)\}$. For example, we prescribe the corresponding covariances as follows:

$$\begin{pmatrix} 1 & \rho & ? & -\rho \\ \rho & 1 & \rho & ? \\ ? & \rho & 1 & \rho \\ -\rho & ? & \rho & 1 \end{pmatrix}. \tag{8.16}$$

Of course, we must take care that all specified marginal covariances of all orders are positive definite (otherwise, we cannot do anything from the start). This is valid in the matrix (8.16), however there is no valid positive-definite completion in this case. The missing entry pattern implicitly specifies a graph G: $i \neq j$ are adjacent if and only if their covariance is specified. It turns out that so long as sub-covariances are $\succ 0$, the problem of completion is entirely contingent on the structure of the graph.

Theorem 8.5.1 (Grone, Johnson, Sá & Wolkowicz)**.** *Let $\Sigma_{partial}$ be a partial covariance with missing entry pattern graph G. Provided all specified sub-covariances of $\Sigma_{partial}$ are positive-definite:*

$$\Sigma_{partial} \text{ admits a positive-definite completion} \iff G \text{ is chordal,}$$

*where **chordal** means that every cycle of length 4 has a cord.*

Remarkably, the proof is constructive as it manifests a completion. It makes use of the fact that any chordal graph may be turned into a complete graph by adding one edge at a time in such a way that the resulting graph remains chordal at each step. Following this ordering of edge additions, the partial matrix is completed entry by entry in such a way as to maximise the determinant of the largest complete sub-matrix that contains the missing entry. Thus, the construction yields the maximum entropy completion, not just any completion!

We now consider the opposite direction: *Given* $X_1, ..., X_n \sim \mathcal{N}(\mu, \Sigma)$ *with* $\Sigma \succ 0$*, estimate the corresponding Markov structure (equivalently the edge set of its graphical model).* In light of our results, the problem reduces to estimating the zero pattern of Θ. When $n > p$ we have a natural approach based on thresholding: we can test for the presence of each possible edge, by testing for the corresponding partial correlation. The test statistic has the same null distribution with same parameters in each case. Selecting a significance level is in 1-1 correspondence with selecting a threshold for hard thresholding. The former can be chosen according to FDR considerations, and the latter via asymptotic considerations. A weakness of this approach is that it is unclear if the resulting matrix is $\succ 0$.

A different approach is to use a **penalised log-likelihood** to promote graph sparsity:

$$l_{\text{pen}}(\Theta) = \log|\Theta| - \text{trace}\big(\Theta\widehat{\Sigma}\big) + \tau \sum_{i<j} |\Theta_{ij}|, \tag{8.17}$$

where this method requires tuning $\tau > 0$; and we actually use the partial correlation matrix instead of Θ to balance scales.